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### SYSTEMS AND METHODS FOR NAVIGATING A HOST VEHICLE ON A ROAD SEGMENT WITH OBJECTS ON THE ROAD EDGES

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#### Abstract

A system for navigating a host vehicle relative to a road segment may comprise at least one processor comprising circuitry and a memory. the memory may include instructions that when executed by the circuitry cause the at least one processor to: receive at least one image captured by at least one camera from an environment of the host vehicle, wherein the at least one image includes representations of at least two objects with offset edges in the environment of the host vehicle and located on one side of the road segment; analyze the at least one image to determine a free space boundary relative to the at least two objects, wherein the free space boundary represents an edge of at least one region of free space and follows a path different from a path represented by the offset edges of the at least two objects; determine at least one navigational action based on the free space boundary; and cause the host vehicle to implement the at least one navigational action.

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## **Background/Summary**

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application claims the benefit of priority of U.S. Provisional Application No. 63/561,961, filed on Mar. 6, 2024. The foregoing application is incorporated herein by reference in its entirety.

### **TECHNICAL FIELD**

[0002] The present disclosure relates generally to autonomous vehicle navigation.

### **BACKGROUND**

[0003] As technology continues to advance, the goal of a fully autonomous vehicle that is capable of navigating on roadways is on the horizon. Autonomous vehicles may need to take into account a variety of factors and make appropriate decisions based on those factors to safely and accurately reach an intended destination. For example, an autonomous vehicle may need to process and interpret visual information (e.g., information captured from a camera) and may also use information obtained from other sources (e.g., from a GPS device, a speed sensor, an accelerometer, a suspension sensor, a RADAR, a LIDAR etc.).

[0004] To make real-time decisions regarding navigation, speed control, and/or steering, an autonomous vehicle relying on visual information must extract or derive useful data from the captured images. This process involves obtaining valuable information, such as the 3D position and/or relative velocity of various features within a scene in the host vehicle environment. For example, the vehicle's system might need may need to detect objects positioned along the edges of road segments to accurately identify a safe, navigable area in which the vehicle can travel without incident. Additionally, the system may recognize and track dynamic elements such as pedestrians, cyclists, and other vehicles, as well as interpret traffic signs, lane markings, and obstacles. These details are essential for the vehicle to understand its environment accurately, predict the movements of other objects, and make informed decisions to navigate safely. Therefore, there is a need for an autonomous vehicle to have the capability to extract and derive valuable information from captured images.

[0005] The present disclosure describes solutions that enable improved autonomous navigation relative to a road segment. The disclosed embodiments include innovative systems, methods, and non-transitory computer-readable media for deriving valuable information from one or more captured images.

### **SUMMARY**

[0006] Embodiments consistent with the present disclosure provide systems and methods for autonomous vehicle navigation. The disclosed embodiments may use cameras to provide autonomous vehicle navigation features. For example, consistent with the disclosed embodiments, the disclosed systems may include one, two, or more cameras that monitor the environment of a vehicle. The disclosed systems may provide a navigational response based on, for example, an analysis of images captured by one or more of the cameras.

[0007] In an embodiment, a system for navigating a host vehicle relative to a road segment is disclosed. The system may comprise at least one processor comprising circuitry and a memory, wherein the memory includes instructions that when executed by the circuitry cause the at least one processor to: receive at least one image captured by at least one camera from an environment of the

host vehicle, wherein the at least one image includes representations of at least two objects with offset edges in the environment of the host vehicle and located on one side of the road segment; analyze the at least one image to determine a free space boundary relative to the at least two objects, wherein the free space boundary represents an edge of at least one region of free space and follows a path different from a path represented by the offset edges of the at least two objects; determine at least one navigational action based on the free space boundary; and cause the host vehicle to implement the at least one navigational action.

[0008] In another embodiment, a method for navigating a host vehicle relative to a road segment is disclosed. The method may comprise: receiving at least one image captured by at least one camera from an environment of the host vehicle, wherein the at least one image includes representations of at least two objects with offset edges in the environment of the host vehicle and located on one side of the road segment; analyzing the at least one image to determine a free space boundary relative to the at least two objects, wherein the free space boundary represents an edge of at least one region of free space and follows a path different from a path represented by the offset edges of the at least two objects; determining at least one navigational action based on the free space boundary; and causing the host vehicle to implement the at least one navigational action.

[0009] Consistent with other disclosed embodiments, non-transitory computer-readable storage media may store program instructions, which are executed by at least one processing device and perform any of the methods described herein.

[0010] The foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the claims.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The accompanying drawings, which are incorporated in and constitute a part of this disclosure, illustrate various disclosed embodiments. In the drawings:

[0012] FIG. 1 is a diagrammatic representation of an exemplary system consistent with the disclosed embodiments.

[0013] FIG. 2A is a diagrammatic side view representation of an exemplary vehicle including a system consistent with the disclosed embodiments.

[0014] FIG. 2B is a diagrammatic top view representation of the vehicle and system shown in FIG. 2A consistent with the disclosed embodiments.

[0015] FIG. 2C is a diagrammatic top view representation of another embodiment of a vehicle including a system consistent with the disclosed embodiments.

[0016] FIG. 2D is a diagrammatic top view representation of yet another embodiment of a vehicle including a system consistent with the disclosed embodiments.

[0017] FIG. 2E is a diagrammatic top view representation of yet another embodiment of a vehicle including a system consistent with the disclosed embodiments.

[0018] FIG. 2F is a diagrammatic representation of exemplary vehicle control systems consistent with the disclosed embodiments.

[0019] FIG. 3A is a diagrammatic representation of an interior of a vehicle including a rearview mirror and a user interface for a vehicle imaging system consistent with the disclosed embodiments.

[0020] FIG. 3B is an illustration of an example of a camera mount that is configured to be positioned behind a rearview mirror and against a vehicle windshield consistent with the disclosed embodiments.

[0021] FIG. 3C is an illustration of the camera mount shown in FIG. 3B from a different perspective consistent with the disclosed embodiments.

[0022] FIG. 3D is an illustration of an example of a camera mount that is configured to be positioned behind a rearview mirror and against a vehicle windshield consistent with the disclosed embodiments.

[0023] FIG. 4 is an exemplary block diagram of a memory configured to store instructions for performing one or more operations consistent with the disclosed embodiments.

[0024] FIG. 5A is a flowchart showing an exemplary process for causing one or more navigational responses based on monocular image analysis consistent with disclosed embodiments.

[0025] FIG. 5B is a flowchart showing an exemplary process for detecting one or more vehicles and/or pedestrians in a set of images consistent with the disclosed embodiments.

[0026] FIG. 5C is a flowchart showing an exemplary process for detecting road marks and/or lane geometry information in a set of images consistent with the disclosed embodiments.

[0027] FIG. 5D is a flowchart showing an exemplary process for detecting traffic lights in a set of images consistent with the disclosed embodiments.

[0028] FIG. 5E is a flowchart showing an exemplary process for causing one or more navigational responses based on a vehicle path consistent with the disclosed embodiments.

[0029] FIG. 5F is a flowchart showing an exemplary process for determining whether a leading vehicle is changing lanes consistent with the disclosed embodiments.

[0030] FIG. 6 is a flowchart showing an exemplary process for causing one or more navigational responses based on stereo image analysis consistent with the disclosed embodiments.

[0031] FIG. 7 is a flowchart showing an exemplary process for causing one or more navigational responses based on an analysis of three sets of images consistent with the disclosed embodiments.

[0032] FIG. 8 shows a sparse map for providing autonomous vehicle navigation, consistent with the disclosed embodiments.

[0033] FIG. 9A illustrates a polynomial representation of a portions of a road segment consistent with the disclosed embodiments.

[0034] FIG. 9B illustrates a curve in three-dimensional space representing a target trajectory of a vehicle, for a particular road segment, included in a sparse map consistent with the disclosed embodiments.

[0035] FIG. 10 illustrates example landmarks that may be included in sparse map consistent with the disclosed embodiments.

[0036] FIG. 11A shows polynomial representations of trajectories consistent with the disclosed embodiments.

[0037] FIGS. 11B and 11C show target trajectories along a multi-lane road consistent with disclosed embodiments.

[0038] FIG. 11D shows an example road signature profile consistent with disclosed embodiments.

[0039] FIG. 12 is a schematic illustration of a system that uses crowd sourcing data received from a plurality of vehicles for autonomous vehicle navigation, consistent with the disclosed embodiments.

[0040] FIG. 13 illustrates an example autonomous vehicle road navigation model represented by a plurality of three-dimensional splines, consistent with the disclosed embodiments.

[0041] FIG. 14 shows a map skeleton generated from combining location information from many drives, consistent with the disclosed embodiments.

[0042] FIG. 15 shows an example of a longitudinal alignment of two drives with example signs as landmarks, consistent with the disclosed embodiments.

[0043] FIG. 16 shows an example of a longitudinal alignment of many drives with an example sign as a landmark, consistent with the disclosed embodiments.

[0044] FIG. 17 is a schematic illustration of a system for generating drive data using a camera, a vehicle, and a server, consistent with the disclosed embodiments.

[0045] FIG. 18 is a schematic illustration of a system for crowdsourcing a sparse map, consistent with the disclosed embodiments.

[0046] FIG. **19** is a flowchart showing an exemplary process for generating a sparse map for autonomous vehicle navigation along a road segment, consistent with the disclosed embodiments.

[0047] FIG. **20** illustrates a block diagram of a server consistent with the disclosed embodiments.

[0048] FIG. **21** illustrates a block diagram of a memory consistent with the disclosed embodiments.

[0049] FIG. **22** illustrates a process of clustering vehicle trajectories associated with vehicles, consistent with the disclosed embodiments.

[0050] FIG. **23** illustrates a navigation system for a vehicle, which may be used for autonomous navigation, consistent with the disclosed embodiments.

[0051] FIGS. **24A**, **24B**, **24C**, and **24D** illustrate exemplary lane marks that may be detected consistent with the disclosed embodiments.

[0052] FIG. **24E** shows exemplary mapped lane marks consistent with the disclosed embodiments.

[0053] FIG. **24F** shows an exemplary anomaly associated with detecting a lane mark consistent with the disclosed embodiments.

[0054] FIG. **25A** shows an exemplary image of a vehicle's surrounding environment for navigation based on the mapped lane marks consistent with the disclosed embodiments.

[0055] FIG. **25B** illustrates a lateral localization correction of a vehicle based on mapped lane marks in a road navigation model consistent with the disclosed embodiments.

[0056] FIGS. **25C** and **25D** provide conceptual representations of a localization technique for locating a host vehicle along a target trajectory using mapped features included in a sparse map.

[0057] FIG. **26A** is a flowchart showing an exemplary process for mapping a lane mark for use in autonomous vehicle navigation consistent with disclosed embodiments.

[0058] FIG. **26B** is a flowchart showing an exemplary process for autonomously navigating a host vehicle along a road segment using mapped lane marks consistent with disclosed embodiments.

[0059] FIG. **27** is a flowchart showing an exemplary process for navigating a host vehicle relative to a road segment, consistent with the disclosed embodiments.

[0060] FIGS. **28A**, **28B**, **28C**, and **28D** represent exemplary images captured by an onboard camera in a host vehicle traveling on a road segment with objects located in the environment of the host vehicle, consistent with the disclosed embodiments.

[0061] FIG. **29** represents an exemplary image captured by an onboard camera in a host vehicle traveling on a road segment with at least one object overhanging the surface of the road segment, consistent with the disclosed embodiments.

#### DETAILED DESCRIPTION

[0062] The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar parts. While several illustrative embodiments are described herein, modifications, adaptations and other implementations are possible. For example, substitutions, additions or modifications may be made to the components illustrated in the drawings, and the illustrative methods described herein may be modified by substituting, reordering, removing, or adding steps to the disclosed methods. Accordingly, the following detailed description is not limited to the disclosed embodiments and examples. Instead, the proper scope is defined by the appended claims.

#### Autonomous Vehicle (AV) Overview

[0063] As used throughout this disclosure, the term “autonomous vehicle” refers to a vehicle capable of implementing at least one navigational change without driver input. A “navigational change” refers to a change in one or more of steering, braking, or acceleration of the vehicle. To be autonomous, a vehicle need not be fully automatic (e.g., fully operational without a driver or without driver input). Rather, an autonomous vehicle includes those that can operate under driver control during certain time periods and without driver control during other time periods. Autonomous vehicles may also include vehicles that control only some aspects of vehicle navigation, such as steering (e.g., to maintain a vehicle course between vehicle lane constraints), but may leave other aspects to the driver (e.g., braking). In some cases, autonomous vehicles may

handle some or all aspects of braking, speed control, and/or steering of the vehicle.

[0064] As human drivers typically rely on visual cues and observations to control a vehicle, transportation infrastructures are built accordingly, with lane markings, traffic signs, and traffic lights, all designed to provide visual information to drivers. In view of these design characteristics of transportation infrastructures, an autonomous vehicle may include a camera and a processing unit that analyzes visual information captured from the environment of the vehicle. The visual information may include, for example, components of the transportation infrastructure (e.g., lane markings, traffic signs, traffic lights, etc.) that are observable by drivers and other obstacles (e.g., other vehicles, pedestrians, debris, etc.). Additionally, an autonomous vehicle may also use stored information, such as information that provides a model of the vehicle's environment when navigating. For example, the vehicle may use GPS data, sensor data (e.g., from an accelerometer, a speed sensor, a suspension sensor, etc.), and/or other map data to provide information related to its environment while the vehicle is traveling, and the vehicle (as well as other vehicles) may use the information to localize itself on the model.

[0065] In some embodiments in this disclosure, an autonomous vehicle may use information obtained while navigating (e.g., from a camera, GPS device, an accelerometer, a speed sensor, a suspension sensor, etc.). In other embodiments, an autonomous vehicle may use information obtained from past navigations by the vehicle (or by other vehicles) while navigating. In yet other embodiments, an autonomous vehicle may use a combination of information obtained while navigating and information obtained from past navigations. The following sections provide an overview of a system consistent with the disclosed embodiments, followed by an overview of a forward-facing imaging system and methods consistent with the system. The sections that follow disclose systems and methods for constructing, using, and updating a sparse map for autonomous vehicle navigation.

#### System Overview

[0066] FIG. 1 is a block diagram representation of a system **100** consistent with the exemplary disclosed embodiments. System **100** may include various components depending on the requirements of a particular implementation. In some embodiments, system **100** may include a processing unit **110**, an image acquisition unit **120**, a position sensor **130**, one or more memory units **140**, **150**, a map database **160**, a user interface **170**, and a wireless transceiver **172**. Processing unit **110** may include one or more processing devices. In some embodiments, processing unit **110** may include an applications processor **180**, an image processor **190**, or any other suitable processing device. Similarly, image acquisition unit **120** may include any number of image acquisition devices and components depending on the requirements of a particular application. In some embodiments, image acquisition unit **120** may include one or more image capture devices (e.g., cameras), such as image capture device **122**, image capture device **124**, and image capture device **126**. System **100** may also include a data interface **128** communicatively connecting processing unit **110** to image acquisition unit **120**. For example, data interface **128** may include any wired and/or wireless link or links for transmitting image data acquired by image acquisition unit **120** to processing unit **110**.

[0067] Wireless transceiver **172** may include one or more devices configured to exchange transmissions over an air interface to one or more networks (e.g., cellular, the Internet, etc.) by use of a radio frequency, infrared frequency, magnetic field, or an electric field. Wireless transceiver **172** may use any known standard to transmit and/or receive data (e.g., Wi-Fi, Bluetooth®, Bluetooth Smart, 802.15.4, ZigBee, etc.). Such transmissions can include communications from the host vehicle to one or more remotely located servers. Such transmissions may also include communications (one-way or two-way) between the host vehicle and one or more target vehicles in an environment of the host vehicle (e.g., to facilitate coordination of navigation of the host vehicle in view of or together with target vehicles in the environment of the host vehicle), or even a broadcast transmission to unspecified recipients in a vicinity of the transmitting vehicle.

[0068] Both applications processor **180** and image processor **190** may include various types of processing devices. For example, either or both of applications processor **180** and image processor **190** may include a microprocessor, preprocessors (such as an image preprocessor), a graphics processing unit (GPU), a central processing unit (CPU), support circuits, digital signal processors, integrated circuits, memory, or any other types of devices suitable for running applications and for image processing and analysis. In some embodiments, applications processor **180** and/or image processor **190** may include any type of single or multi-core processor, mobile device microcontroller, central processing unit, etc. Various processing devices may be used, including, for example, processors available from manufacturers such as Intel®, AMD®, etc., or GPUs available from manufacturers such as NVIDIA®, ATI®, etc. and may include various architectures (e.g., x86 processor, ARM®, etc.).

[0069] In some embodiments, applications processor **180** and/or image processor **190** may include any of the EyeQ series of processor chips available from Mobileye®. These processor designs each include multiple processing units with local memory and instruction sets. Such processors may include video inputs for receiving image data from multiple image sensors and may also include video out capabilities. In one example, the EyeQ2® uses 90 nm-micron technology operating at 332 Mhz. The EyeQ2® architecture consists of two floating point, hyper-thread 32-bit RISC CPUs (MIPS32® 34K® cores), five Vision Computing Engines (VCE), three Vector Microcode Processors (VMP®), Denali 64-bit Mobile DDR Controller, 128-bit internal Sonics Interconnect, dual 16-bit Video input and 18-bit Video output controllers, 16 channels DMA and several peripherals. The MIPS34K CPU manages the five VCEs, three VMP™ and the DMA, the second MIPS34K CPU and the multi-channel DMA as well as the other peripherals. The five VCEs, three VMP® and the MIPS34K CPU can perform intensive vision computations required by multi-function bundle applications. In another example, the EyeQ3®, which is a third generation processor and is six times more powerful than the EyeQ2®, may be used in the disclosed embodiments. In other examples, the EyeQ4® and/or the EyeQ5® may be used in the disclosed embodiments. Of course, any newer or future EyeQ processing devices may also be used together with the disclosed embodiments.

[0070] Any of the processing devices disclosed herein may be configured to perform certain functions. Configuring a processing device, such as any of the described EyeQ processors or other controller or microprocessor, to perform certain functions may include programming of computer executable instructions and making those instructions available to the processing device for execution during operation of the processing device. In some embodiments, configuring a processing device may include programming the processing device directly with architectural instructions. For example, processing devices such as field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), and the like may be configured using, for example, one or more hardware description languages (HDLs).

[0071] In other embodiments, configuring a processing device may include storing executable instructions on a memory that is accessible to the processing device during operation. For example, the processing device may access the memory to obtain and execute the stored instructions during operation. In either case, the processing device configured to perform the sensing, image analysis, and/or navigational functions disclosed herein represents a specialized hardware-based system in control of multiple hardware based components of a host vehicle.

[0072] While FIG. **1** depicts two separate processing devices included in processing unit **110**, more or fewer processing devices may be used. For example, in some embodiments, a single processing device may be used to accomplish the tasks of applications processor **180** and image processor **190**. In other embodiments, these tasks may be performed by more than two processing devices. Further, in some embodiments, system **100** may include one or more of processing unit **110** without including other components, such as image acquisition unit **120**.

[0073] Processing unit **110** may comprise various types of devices. For example, processing unit

**110** may include various devices, such as a controller, an image preprocessor, a central processing unit (CPU), a graphics processing unit (GPU), support circuits, digital signal processors, integrated circuits, memory, or any other types of devices for image processing and analysis. The image preprocessor may include a video processor for capturing, digitizing and processing the imagery from the image sensors. The CPU may comprise any number of microcontrollers or microprocessors. The GPU may also comprise any number of microcontrollers or microprocessors. The support circuits may be any number of circuits generally well known in the art, including cache, power supply, clock and input-output circuits. The memory may store software that, when executed by the processor, controls the operation of the system. The memory may include databases and image processing software. The memory may comprise any number of random access memories, read only memories, flash memories, disk drives, optical storage, tape storage, removable storage and other types of storage. In one instance, the memory may be separate from the processing unit **110**. In another instance, the memory may be integrated into the processing unit **110**.

[0074] Each memory **140, 150** may include software instructions that when executed by a processor (e.g., applications processor **180** and/or image processor **190**), may control operation of various aspects of system **100**. These memory units may include various databases and image processing software, as well as a trained system, such as a neural network, or a deep neural network, for example. The memory units may include random access memory (RAM), read only memory (ROM), flash memory, disk drives, optical storage, tape storage, removable storage and/or any other types of storage. In some embodiments, memory units **140, 150** may be separate from the applications processor **180** and/or image processor **190**. In other embodiments, these memory units may be integrated into applications processor **180** and/or image processor **190**.

[0075] Position sensor **130** may include any type of device suitable for determining a location associated with at least one component of system **100**. In some embodiments, position sensor **130** may include a GPS receiver. Such receivers can determine a user position and velocity by processing signals broadcasted by global positioning system satellites. Position information from position sensor **130** may be made available to applications processor **180** and/or image processor **190**.

[0076] In some embodiments, system **100** may include components such as a speed sensor (e.g., a tachometer, a speedometer) for measuring a speed of vehicle **200** and/or an accelerometer (either single axis or multi-axis) for measuring acceleration of vehicle **200**.

[0077] User interface **170** may include any device suitable for providing information to or for receiving inputs from one or more users of system **100**. In some embodiments, user interface **170** may include user input devices, including, for example, a touchscreen, microphone, keyboard, pointer devices, track wheels, cameras, knobs, buttons, etc. With such input devices, a user may be able to provide information inputs or commands to system **100** by typing instructions or information, providing voice commands, selecting menu options on a screen using buttons, pointers, or eye-tracking capabilities, or through any other suitable techniques for communicating information to system **100**.

[0078] User interface **170** may be equipped with one or more processing devices configured to provide and receive information to or from a user and process that information for use by, for example, applications processor **180**. In some embodiments, such processing devices may execute instructions for recognizing and tracking eye movements, receiving and interpreting voice commands, recognizing and interpreting touches and/or gestures made on a touchscreen, responding to keyboard entries or menu selections, etc. In some embodiments, user interface **170** may include a display, speaker, tactile device, and/or any other devices for providing output information to a user.

[0079] Map database **160** may include any type of database for storing map data useful to system **100**. In some embodiments, map database **160** may include data relating to the position, in a



reference coordinate system, of various items, including roads, water features, geographic features, businesses, points of interest, restaurants, gas stations, etc. Map database **160** may store not only the locations of such items, but also descriptors relating to those items, including, for example, names associated with any of the stored features. In some embodiments, map database **160** may be physically located with other components of system **100**. Alternatively or additionally, map database **160** or a portion thereof may be located remotely with respect to other components of system **100** (e.g., processing unit **110**). In such embodiments, information from map database **160** may be downloaded over a wired or wireless data connection to a network (e.g., over a cellular network and/or the Internet, etc.). In some cases, map database **160** may store a sparse data model including polynomial representations of certain road features (e.g., lane markings) or target trajectories for the host vehicle. Systems and methods of generating such a map are discussed below with references to FIGS. **8-19**.

[0080] Image capture devices **122**, **124**, and **126** may each include any type of device suitable for capturing at least one image from an environment. Moreover, any number of image capture devices may be used to acquire images for input to the image processor. Some embodiments may include only a single image capture device, while other embodiments may include two, three, or even four or more image capture devices. Image capture devices **122**, **124**, and **126** will be further described with reference to FIGS. **2B-2E**, below.

[0081] System **100**, or various components thereof, may be incorporated into various different platforms. In some embodiments, system **100** may be included on a vehicle **200**, as shown in FIG. **2A**. For example, vehicle **200** may be equipped with a processing unit **110** and any of the other components of system **100**, as described above relative to FIG. **1**. While in some embodiments vehicle **200** may be equipped with only a single image capture device (e.g., camera), in other embodiments, such as those discussed in connection with FIGS. **2B-2E**, multiple image capture devices may be used. For example, either of image capture devices **122** and **124** of vehicle **200**, as shown in FIG. **2A**, may be part of an ADAS (Advanced Driver Assistance Systems) imaging set.

[0082] The image capture devices included on vehicle **200** as part of the image acquisition unit **120** may be positioned at any suitable location. In some embodiments, as shown in FIGS. **2A-2E** and **3A-3C**, image capture device **122** may be located in the vicinity of the rearview mirror. This position may provide a line of sight similar to that of the driver of vehicle **200**, which may aid in determining what is and is not visible to the driver. Image capture device **122** may be positioned at any location near the rearview mirror, but placing image capture device **122** on the driver side of the mirror may further aid in obtaining images representative of the driver's field of view and/or line of sight.

[0083] Other locations for the image capture devices of image acquisition unit **120** may also be used. For example, image capture device **124** may be located on or in a bumper of vehicle **200**. Such a location may be especially suitable for image capture devices having a wide field of view. The line of sight of bumper-located image capture devices can be different from that of the driver and, therefore, the bumper image capture device and driver may not always see the same objects. The image capture devices (e.g., image capture devices **122**, **124**, and **126**) may also be located in other locations. For example, the image capture devices may be located on or in one or both of the side mirrors of vehicle **200**, on the roof of vehicle **200**, on the hood of vehicle **200**, on the trunk of vehicle **200**, on the sides of vehicle **200**, mounted on, positioned behind, or positioned in front of any of the windows of vehicle **200**, and mounted in or near light figures on the front and/or back of vehicle **200**, etc.

[0084] In addition to image capture devices, vehicle **200** may include various other components of system **100**. For example, processing unit **110** may be included on vehicle **200** either integrated with or separate from an engine control unit (ECU) of the vehicle. Vehicle **200** may also be equipped with a position sensor **130**, such as a GPS receiver and may also include a map database **160** and memory units **140** and **150**.

[0085] As discussed earlier, wireless transceiver **172** may and/or receive data over one or more networks (e.g., cellular networks, the Internet, etc.). For example, wireless transceiver **172** may upload data collected by system **100** to one or more servers, and download data from the one or more servers. Via wireless transceiver **172**, system **100** may receive, for example, periodic or on demand updates to data stored in map database **160**, memory **140**, and/or memory **150**. Similarly, wireless transceiver **172** may upload any data (e.g., images captured by image acquisition unit **120**, data received by position sensor **130** or other sensors, vehicle control systems, etc.) from by system **100** and/or any data processed by processing unit **110** to the one or more servers.

[0086] System **100** may upload data to a server (e.g., to the cloud) based on a privacy level setting. For example, system **100** may implement privacy level settings to regulate or limit the types of data (including metadata) sent to the server that may uniquely identify a vehicle and or driver/owner of a vehicle. Such settings may be set by user via, for example, wireless transceiver **172**, be initialized by factory default settings, or by data received by wireless transceiver **172**.

[0087] In some embodiments, system **100** may upload data according to a “high” privacy level, and under setting a setting, system **100** may transmit data (e.g., location information related to a route, captured images, etc.) without any details about the specific vehicle and/or driver/owner. For example, when uploading data according to a “high” privacy setting, system **100** may not include a vehicle identification number (VIN) or a name of a driver or owner of the vehicle, and may instead of transmit data, such as captured images and/or limited location information related to a route.

[0088] Other privacy levels are contemplated. For example, system **100** may transmit data to a server according to an “intermediate” privacy level and include additional information not included under a “high” privacy level, such as a make and/or model of a vehicle and/or a vehicle type (e.g., a passenger vehicle, sport utility vehicle, truck, etc.). In some embodiments, system **100** may upload data according to a “low” privacy level. Under a “low” privacy level setting, system **100** may upload data and include information sufficient to uniquely identify a specific vehicle, owner/driver, and/or a portion or entirely of a route traveled by the vehicle. Such “low” privacy level data may include one or more of, for example, a VIN, a driver/owner name, an origination point of a vehicle prior to departure, an intended destination of the vehicle, a make and/or model of the vehicle, a type of the vehicle, etc.

[0089] FIG. 2A is a diagrammatic side view representation of an exemplary vehicle imaging system consistent with the disclosed embodiments. FIG. 2B is a diagrammatic top view illustration of the embodiment shown in FIG. 2A. As illustrated in FIG. 2B, the disclosed embodiments may include a vehicle **200** including in its body a system **100** with a first image capture device **122** positioned in the vicinity of the rearview mirror and/or near the driver of vehicle **200**, a second image capture device **124** positioned on or in a bumper region (e.g., one of bumper regions **210**) of vehicle **200**, and a processing unit **110**.

[0090] As illustrated in FIG. 2C, image capture devices **122** and **124** may both be positioned in the vicinity of the rearview mirror and/or near the driver of vehicle **200**. Additionally, while two image capture devices **122** and **124** are shown in FIGS. 2B and 2C, it should be understood that other embodiments may include more than two image capture devices. For example, in the embodiments shown in FIGS. 2D and 2E, first, second, and third image capture devices **122**, **124**, and **126**, are included in the system **100** of vehicle **200**.

[0091] As illustrated in FIG. 2D, image capture device **122** may be positioned in the vicinity of the rearview mirror and/or near the driver of vehicle **200**, and image capture devices **124** and **126** may be positioned on or in a bumper region (e.g., one of bumper regions **210**) of vehicle **200**. And as shown in FIG. 2E, image capture devices **122**, **124**, and **126** may be positioned in the vicinity of the rearview mirror and/or near the driver seat of vehicle **200**. The disclosed embodiments are not limited to any particular number and configuration of the image capture devices, and the image capture devices may be positioned in any appropriate location within and/or on vehicle **200**.

[0092] It is to be understood that the disclosed embodiments are not limited to vehicles and could

be applied in other contexts. It is also to be understood that disclosed embodiments are not limited to a particular type of vehicle **200** and may be applicable to all types of vehicles including automobiles, trucks, trailers, and other types of vehicles.

[0093] The first image capture device **122** may include any suitable type of image capture device. Image capture device **122** may include an optical axis. In one instance, the image capture device **122** may include an Aptina M9V024 WVGA sensor with a global shutter. In other embodiments, image capture device **122** may provide a resolution of 1280×960 pixels and may include a rolling shutter. Image capture device **122** may include various optical elements. In some embodiments one or more lenses may be included, for example, to provide a desired focal length and field of view for the image capture device. In some embodiments, image capture device **122** may be associated with a 6 mm lens or a 12 mm lens. In some embodiments, image capture device **122** may be configured to capture images having a desired field-of-view (FOV) **202**, as illustrated in FIG. 2D. For example, image capture device **122** may be configured to have a regular FOV, such as within a range of 40 degrees to 56 degrees, including a 46 degree FOV, 50 degree FOV, 52 degree FOV, or greater. Alternatively, image capture device **122** may be configured to have a narrow FOV in the range of 23 to 40 degrees, such as a 28 degree FOV or 36 degree FOV. In addition, image capture device **122** may be configured to have a wide FOV in the range of 100 to 180 degrees. In some embodiments, image capture device **122** may include a wide angle bumper camera or one with up to a 180 degree FOV. In some embodiments, image capture device **122** may be a 7.2 M pixel image capture device with an aspect ratio of about 2:1 (e.g., H×V=3800×1900 pixels) with about 100 degree horizontal FOV. Such an image capture device may be used in place of a three image capture device configuration. Due to significant lens distortion, the vertical FOV of such an image capture device may be significantly less than 50 degrees in implementations in which the image capture device uses a radially symmetric lens. For example, such a lens may not be radially symmetric which would allow for a vertical FOV greater than 50 degrees with 100 degree horizontal FOV.

[0094] The first image capture device **122** may acquire a plurality of first images relative to a scene associated with the vehicle **200**. Each of the plurality of first images may be acquired as a series of image scan lines, which may be captured using a rolling shutter. Each scan line may include a plurality of pixels.

[0095] The first image capture device **122** may have a scan rate associated with acquisition of each of the first series of image scan lines. The scan rate may refer to a rate at which an image sensor can acquire image data associated with each pixel included in a particular scan line.

[0096] Image capture devices **122**, **124**, and **126** may contain any suitable type and number of image sensors, including CCD sensors or CMOS sensors, for example. In one embodiment, a CMOS image sensor may be employed along with a rolling shutter, such that each pixel in a row is read one at a time, and scanning of the rows proceeds on a row-by-row basis until an entire image frame has been captured. In some embodiments, the rows may be captured sequentially from top to bottom relative to the frame.

[0097] In some embodiments, one or more of the image capture devices (e.g., image capture devices **122**, **124**, and **126**) disclosed herein may constitute a high resolution imager and may have a resolution greater than 5 M pixel, 7 M pixel, 10 M pixel, or greater.

[0098] The use of a rolling shutter may result in pixels in different rows being exposed and captured at different times, which may cause skew and other image artifacts in the captured image frame. On the other hand, when the image capture device **122** is configured to operate with a global or synchronous shutter, all of the pixels may be exposed for the same amount of time and during a common exposure period. As a result, the image data in a frame collected from a system employing a global shutter represents a snapshot of the entire FOV (such as FOV **202**) at a particular time. In contrast, in a rolling shutter application, each row in a frame is exposed and data is captured at different times. Thus, moving objects may appear distorted in an image capture device having a

rolling shutter. This phenomenon will be described in greater detail below.

[0099] The second image capture device **124** and the third image capturing device **126** may be any type of image capture device. Like the first image capture device **122**, each of image capture devices **124** and **126** may include an optical axis. In one embodiment, each of image capture devices **124** and **126** may include an Aptina M9V024 WVGA sensor with a global shutter. Alternatively, each of image capture devices **124** and **126** may include a rolling shutter. Like image capture device **122**, image capture devices **124** and **126** may be configured to include various lenses and optical elements. In some embodiments, lenses associated with image capture devices **124** and **126** may provide FOVs (such as FOVs **204** and **206**) that are the same as, or narrower than, a FOV (such as FOV **202**) associated with image capture device **122**. For example, image capture devices **124** and **126** may have FOVs of 40 degrees, 30 degrees, 26 degrees, 23 degrees, 20 degrees, or less.

[0100] Image capture devices **124** and **126** may acquire a plurality of second and third images relative to a scene associated with the vehicle **200**. Each of the plurality of second and third images may be acquired as a second and third series of image scan lines, which may be captured using a rolling shutter. Each scan line or row may have a plurality of pixels. Image capture devices **124** and **126** may have second and third scan rates associated with acquisition of each of image scan lines included in the second and third series.

[0101] Each image capture device **122**, **124**, and **126** may be positioned at any suitable position and orientation relative to vehicle **200**. The relative positioning of the image capture devices **122**, **124**, and **126** may be selected to aid in fusing together the information acquired from the image capture devices. For example, in some embodiments, a FOV (such as FOV **204**) associated with image capture device **124** may overlap partially or fully with a FOV (such as FOV **202**) associated with image capture device **122** and a FOV (such as FOV **206**) associated with image capture device **126**.

[0102] Image capture devices **122**, **124**, and **126** may be located on vehicle **200** at any suitable relative heights. In one instance, there may be a height difference between the image capture devices **122**, **124**, and **126**, which may provide sufficient parallax information to enable stereo analysis. For example, as shown in FIG. 2A, the two image capture devices **122** and **124** are at different heights. There may also be a lateral displacement difference between image capture devices **122**, **124**, and **126**, giving additional parallax information for stereo analysis by processing unit **110**, for example. The difference in the lateral displacement may be denoted by  $dx$ , as shown in FIGS. 2C and 2D. In some embodiments, fore or aft displacement (e.g., range displacement) may exist between image capture devices **122**, **124**, and **126**. For example, image capture device **122** may be located 0.5 to 2 meters or more behind image capture device **124** and/or image capture device **126**. This type of displacement may enable one of the image capture devices to cover potential blind spots of the other image capture device(s).

[0103] Image capture devices **122** may have any suitable resolution capability (e.g., number of pixels associated with the image sensor), and the resolution of the image sensor(s) associated with the image capture device **122** may be higher, lower, or the same as the resolution of the image sensor(s) associated with image capture devices **124** and **126**. In some embodiments, the image sensor(s) associated with image capture device **122** and/or image capture devices **124** and **126** may have a resolution of 640×480, 1024×768, 1280×960, or any other suitable resolution.

[0104] The frame rate (e.g., the rate at which an image capture device acquires a set of pixel data of one image frame before moving on to capture pixel data associated with the next image frame) may be controllable. The frame rate associated with image capture device **122** may be higher, lower, or the same as the frame rate associated with image capture devices **124** and **126**. The frame rate associated with image capture devices **122**, **124**, and **126** may depend on a variety of factors that may affect the timing of the frame rate. For example, one or more of image capture devices **122**, **124**, and **126** may include a selectable pixel delay period imposed before or after acquisition of image data associated with one or more pixels of an image sensor in image capture device **122**,

**124**, and/or **126**. Generally, image data corresponding to each pixel may be acquired according to a clock rate for the device (e.g., one pixel per clock cycle). Additionally, in embodiments including a rolling shutter, one or more of image capture devices **122**, **124**, and **126** may include a selectable horizontal blanking period imposed before or after acquisition of image data associated with a row of pixels of an image sensor in image capture device **122**, **124**, and/or **126**. Further, one or more of image capture devices **122**, **124**, and/or **126** may include a selectable vertical blanking period imposed before or after acquisition of image data associated with an image frame of image capture device **122**, **124**, and **126**.

[0105] These timing controls may enable synchronization of frame rates associated with image capture devices **122**, **124**, and **126**, even where the line scan rates of each are different. Additionally, as will be discussed in greater detail below, these selectable timing controls, among other factors (e.g., image sensor resolution, maximum line scan rates, etc.) may enable synchronization of image capture from an area where the FOV of image capture device **122** overlaps with one or more FOVs of image capture devices **124** and **126**, even where the field of view of image capture device **122** is different from the FOVs of image capture devices **124** and **126**.

[0106] Frame rate timing in image capture device **122**, **124**, and **126** may depend on the resolution of the associated image sensors. For example, assuming similar line scan rates for both devices, if one device includes an image sensor having a resolution of 640×480 and another device includes an image sensor with a resolution of 1280×960, then more time will be required to acquire a frame of image data from the sensor having the higher resolution.

[0107] Another factor that may affect the timing of image data acquisition in image capture devices **122**, **124**, and **126** is the maximum line scan rate. For example, acquisition of a row of image data from an image sensor included in image capture device **122**, **124**, and **126** will require some minimum amount of time. Assuming no pixel delay periods are added, this minimum amount of time for acquisition of a row of image data will be related to the maximum line scan rate for a particular device. Devices that offer higher maximum line scan rates have the potential to provide higher frame rates than devices with lower maximum line scan rates. In some embodiments, one or more of image capture devices **124** and **126** may have a maximum line scan rate that is higher than a maximum line scan rate associated with image capture device **122**. In some embodiments, the maximum line scan rate of image capture device **124** and/or **126** may be 1.25, 1.5, 1.75, or 2 times or more than a maximum line scan rate of image capture device **122**.

[0108] In another embodiment, image capture devices **122**, **124**, and **126** may have the same maximum line scan rate, but image capture device **122** may be operated at a scan rate less than or equal to its maximum scan rate. The system may be configured such that one or more of image capture devices **124** and **126** operate at a line scan rate that is equal to the line scan rate of image capture device **122**. In other instances, the system may be configured such that the line scan rate of image capture device **124** and/or image capture device **126** may be 1.25, 1.5, 1.75, or 2 times or more than the line scan rate of image capture device **122**.

[0109] In some embodiments, image capture devices **122**, **124**, and **126** may be asymmetric. That is, they may include cameras having different fields of view (FOV) and focal lengths. The fields of view of image capture devices **122**, **124**, and **126** may include any desired area relative to an environment of vehicle **200**, for example. In some embodiments, one or more of image capture devices **122**, **124**, and **126** may be configured to acquire image data from an environment in front of vehicle **200**, behind vehicle **200**, to the sides of vehicle **200**, or combinations thereof.

[0110] Further, the focal length associated with each image capture device **122**, **124**, and/or **126** may be selectable (e.g., by inclusion of appropriate lenses etc.) such that each device acquires images of objects at a desired distance range relative to vehicle **200**. For example, in some embodiments image capture devices **122**, **124**, and **126** may acquire images of close-up objects within a few meters from the vehicle. Image capture devices **122**, **124**, and **126** may also be

configured to acquire images of objects at ranges more distant from the vehicle (e.g., 25 m, 50 m, 100 m, 150 m, or more). Further, the focal lengths of image capture devices **122**, **124**, and **126** may be selected such that one image capture device (e.g., image capture device **122**) can acquire images of objects relatively close to the vehicle (e.g., within 10 m or within 20 m) while the other image capture devices (e.g., image capture devices **124** and **126**) can acquire images of more distant objects (e.g., greater than 20 m, 50 m, 100 m, 150 m, etc.) from vehicle **200**.

[0111] According to some embodiments, the FOV of one or more image capture devices **122**, **124**, and **126** may have a wide angle. For example, it may be advantageous to have a FOV of 140 degrees, especially for image capture devices **122**, **124**, and **126** that may be used to capture images of the area in the vicinity of vehicle **200**. For example, image capture device **122** may be used to capture images of the area to the right or left of vehicle **200** and, in such embodiments, it may be desirable for image capture device **122** to have a wide FOV (e.g., at least 140 degrees).

[0112] The field of view associated with each of image capture devices **122**, **124**, and **126** may depend on the respective focal lengths. For example, as the focal length increases, the corresponding field of view decreases.

[0113] Image capture devices **122**, **124**, and **126** may be configured to have any suitable fields of view. In one particular example, image capture device **122** may have a horizontal FOV of 46 degrees, image capture device **124** may have a horizontal FOV of 23 degrees, and image capture device **126** may have a horizontal FOV in between 23 and 46 degrees. In another instance, image capture device **122** may have a horizontal FOV of 52 degrees, image capture device **124** may have a horizontal FOV of 26 degrees, and image capture device **126** may have a horizontal FOV in between 26 and 52 degrees. In some embodiments, a ratio of the FOV of image capture device **122** to the FOVs of image capture device **124** and/or image capture device **126** may vary from 1.5 to 2.0. In other embodiments, this ratio may vary between 1.25 and 2.25.

[0114] System **100** may be configured so that a field of view of image capture device **122** overlaps, at least partially or fully, with a field of view of image capture device **124** and/or image capture device **126**. In some embodiments, system **100** may be configured such that the fields of view of image capture devices **124** and **126**, for example, fall within (e.g., are narrower than) and share a common center with the field of view of image capture device **122**. In other embodiments, the image capture devices **122**, **124**, and **126** may capture adjacent FOVs or may have partial overlap in their FOVs. In some embodiments, the fields of view of image capture devices **122**, **124**, and **126** may be aligned such that a center of the narrower FOV image capture devices **124** and/or **126** may be located in a lower half of the field of view of the wider FOV image capture device **122**.

[0115] FIG. 2F is a diagrammatic representation of exemplary vehicle control systems, consistent with the disclosed embodiments. As indicated in FIG. 2F, vehicle **200** may include throttling system **220**, braking system **230**, and steering system **240**. System **100** may provide inputs (e.g., control signals) to one or more of throttling system **220**, braking system **230**, and steering system **240** over one or more data links (e.g., any wired and/or wireless link or links for transmitting data). For example, based on analysis of images acquired by image capture devices **122**, **124**, and/or **126**, system **100** may provide control signals to one or more of throttling system **220**, braking system **230**, and steering system **240** to navigate vehicle **200** (e.g., by causing an acceleration, a turn, a lane shift, etc.). Further, system **100** may receive inputs from one or more of throttling system **220**, braking system **230**, and steering system **24** indicating operating conditions of vehicle **200** (e.g., speed, whether vehicle **200** is braking and/or turning, etc.). Further details are provided in connection with FIGS. 4-7, below.

[0116] As shown in FIG. 3A, vehicle **200** may also include a user interface **170** for interacting with a driver or a passenger of vehicle **200**. For example, user interface **170** in a vehicle application may include a touch screen **320**, knobs **330**, buttons **340**, and a microphone **350**. A driver or passenger of vehicle **200** may also use handles (e.g., located on or near the steering column of vehicle **200** including, for example, turn signal handles), buttons (e.g., located on the steering wheel of vehicle

**200**), and the like, to interact with system **100**. In some embodiments, microphone **350** may be positioned adjacent to a rearview mirror **310**. Similarly, in some embodiments, image capture device **122** may be located near rearview mirror **310**. In some embodiments, user interface **170** may also include one or more speakers **360** (e.g., speakers of a vehicle audio system). For example, system **100** may provide various notifications (e.g., alerts) via speakers **360**.

[0117] FIGS. 3B-3D are illustrations of an exemplary camera mount **370** configured to be positioned behind a rearview mirror (e.g., rearview mirror **310**) and against a vehicle windshield, consistent with disclosed embodiments. As shown in FIG. 3B, camera mount **370** may include image capture devices **122**, **124**, and **126**. Image capture devices **124** and **126** may be positioned behind a glare shield **380**, which may be flush against the vehicle windshield and include a composition of film and/or anti-reflective materials. For example, glare shield **380** may be positioned such that the shield aligns against a vehicle windshield having a matching slope. In some embodiments, each of image capture devices **122**, **124**, and **126** may be positioned behind glare shield **380**, as depicted, for example, in FIG. 3D. The disclosed embodiments are not limited to any particular configuration of image capture devices **122**, **124**, and **126**, camera mount **370**, and glare shield **380**. FIG. 3C is an illustration of camera mount **370** shown in FIG. 3B from a front perspective.

[0118] As will be appreciated by a person skilled in the art having the benefit of this disclosure, numerous variations and/or modifications may be made to the foregoing disclosed embodiments. For example, not all components are essential for the operation of system **100**. Further, any component may be located in any appropriate part of system **100** and the components may be rearranged into a variety of configurations while providing the functionality of the disclosed embodiments. Therefore, the foregoing configurations are examples and, regardless of the configurations discussed above, system **100** can provide a wide range of functionality to analyze the surroundings of vehicle **200** and navigate vehicle **200** in response to the analysis.

[0119] As discussed below in further detail and consistent with various disclosed embodiments, system **100** may provide a variety of features related to autonomous driving and/or driver assist technology. For example, system **100** may analyze image data, position data (e.g., GPS location information), map data, speed data, and/or data from sensors included in vehicle **200**. System **100** may collect the data for analysis from, for example, image acquisition unit **120**, position sensor **130**, and other sensors. Further, system **100** may analyze the collected data to determine whether or not vehicle **200** should take a certain action, and then automatically take the determined action without human intervention. For example, when vehicle **200** navigates without human intervention, system **100** may automatically control the braking, acceleration, and/or steering of vehicle **200** (e.g., by sending control signals to one or more of throttling system **220**, braking system **230**, and steering system **240**). Further, system **100** may analyze the collected data and issue warnings and/or alerts to vehicle occupants based on the analysis of the collected data. Additional details regarding the various embodiments that are provided by system **100** are provided below.

#### Forward-Facing Multi-Imaging System

[0120] As discussed above, system **100** may provide drive assist functionality that uses a multi-camera system. The multi-camera system may use one or more cameras facing in the forward direction of a vehicle. In other embodiments, the multi-camera system may include one or more cameras facing to the side of a vehicle or to the rear of the vehicle. In one embodiment, for example, system **100** may use a two-camera imaging system, where a first camera and a second camera (e.g., image capture devices **122** and **124**) may be positioned at the front and/or the sides of a vehicle (e.g., vehicle **200**). The first camera may have a field of view that is greater than, less than, or partially overlapping with, the field of view of the second camera. In addition, the first camera may be connected to a first image processor to perform monocular image analysis of images provided by the first camera, and the second camera may be connected to a second image processor to perform monocular image analysis of images provided by the second camera. The

outputs (e.g., processed information) of the first and second image processors may be combined. In some embodiments, the second image processor may receive images from both the first camera and second camera to perform stereo analysis. In another embodiment, system **100** may use a three-camera imaging system where each of the cameras has a different field of view. Such a system may, therefore, make decisions based on information derived from objects located at varying distances both forward and to the sides of the vehicle. References to monocular image analysis may refer to instances where image analysis is performed based on images captured from a single point of view (e.g., from a single camera). Stereo image analysis may refer to instances where image analysis is performed based on two or more images captured with one or more variations of an image capture parameter. For example, captured images suitable for performing stereo image analysis may include images captured: from two or more different positions, from different fields of view, using different focal lengths, along with parallax information, etc.

[0121] For example, in one embodiment, system **100** may implement a three camera configuration using image capture devices **122**, **124**, and **126**. In such a configuration, image capture device **122** may provide a narrow field of view (e.g., 34 degrees, or other values selected from a range of about 20 to 45 degrees, etc.), image capture device **124** may provide a wide field of view (e.g., 150 degrees or other values selected from a range of about 100 to about 180 degrees), and image capture device **126** may provide an intermediate field of view (e.g., 46 degrees or other values selected from a range of about 35 to about 60 degrees). In some embodiments, image capture device **126** may act as a main or primary camera. Image capture devices **122**, **124**, and **126** may be positioned behind rearview mirror **310** and positioned substantially side-by-side (e.g., 6 cm apart). Further, in some embodiments, as discussed above, one or more of image capture devices **122**, **124**, and **126** may be mounted behind glare shield **380** that is flush with the windshield of vehicle **200**. Such shielding may act to minimize the impact of any reflections from inside the car on image capture devices **122**, **124**, and **126**.

[0122] In another embodiment, as discussed above in connection with FIGS. 3B and 3C, the wide field of view camera (e.g., image capture device **124** in the above example) may be mounted lower than the narrow and main field of view cameras (e.g., image capture devices **122** and **126** in the above example). This configuration may provide a free line of sight from the wide field of view camera. To reduce reflections, the cameras may be mounted close to the windshield of vehicle **200**, and may include polarizers on the cameras to damp reflected light.

[0123] A three camera system may provide certain performance characteristics. For example, some embodiments may include an ability to validate the detection of objects by one camera based on detection results from another camera. In the three camera configuration discussed above, processing unit **110** may include, for example, three processing devices (e.g., three EyeQ series of processor chips, as discussed above), with each processing device dedicated to processing images captured by one or more of image capture devices **122**, **124**, and **126**.

[0124] In a three camera system, a first processing device may receive images from both the main camera and the narrow field of view camera, and perform vision processing of the narrow FOV camera to, for example, detect other vehicles, pedestrians, lane marks, traffic signs, traffic lights, and other road objects. Further, the first processing device may calculate a disparity of pixels between the images from the main camera and the narrow camera and create a 3D reconstruction of the environment of vehicle **200**. The first processing device may then combine the 3D reconstruction with 3D map data or with 3D information calculated based on information from another camera.

[0125] The second processing device may receive images from main camera and perform vision processing to detect other vehicles, pedestrians, lane marks, traffic signs, traffic lights, and other road objects. Additionally, the second processing device may calculate a camera displacement and, based on the displacement, calculate a disparity of pixels between successive images and create a 3D reconstruction of the scene (e.g., a structure from motion). The second processing device may



send the structure from motion based 3D reconstruction to the first processing device to be combined with the stereo 3D images.

[0126] The third processing device may receive images from the wide FOV camera and process the images to detect vehicles, pedestrians, lane marks, traffic signs, traffic lights, and other road objects. The third processing device may further execute additional processing instructions to analyze images to identify objects moving in the image, such as vehicles changing lanes, pedestrians, etc.

[0127] In some embodiments, having streams of image-based information captured and processed independently may provide an opportunity for providing redundancy in the system. Such redundancy may include, for example, using a first image capture device and the images processed from that device to validate and/or supplement information obtained by capturing and processing image information from at least a second image capture device.

[0128] In some embodiments, system **100** may use two image capture devices (e.g., image capture devices **122** and **124**) in providing navigation assistance for vehicle **200** and use a third image capture device (e.g., image capture device **126**) to provide redundancy and validate the analysis of data received from the other two image capture devices. For example, in such a configuration, image capture devices **122** and **124** may provide images for stereo analysis by system **100** for navigating vehicle **200**, while image capture device **126** may provide images for monocular analysis by system **100** to provide redundancy and validation of information obtained based on images captured from image capture device **122** and/or image capture device **124**. That is, image capture device **126** (and a corresponding processing device) may be considered to provide a redundant sub-system for providing a check on the analysis derived from image capture devices **122** and **124** (e.g., to provide an automatic emergency braking (AEB) system). Furthermore, in some embodiments, redundancy and validation of received data may be supplemented based on information received from one more sensors (e.g., radar, lidar, acoustic sensors, information received from one or more transceivers outside of a vehicle, etc.).

[0129] One of skill in the art will recognize that the above camera configurations, camera placements, number of cameras, camera locations, etc., are examples only. These components and others described relative to the overall system may be assembled and used in a variety of different configurations without departing from the scope of the disclosed embodiments. Further details regarding usage of a multi-camera system to provide driver assist and/or autonomous vehicle functionality follow below.

[0130] FIG. **4** is an exemplary functional block diagram of memory **140** and/or **150**, which may be stored/programmed with instructions for performing one or more operations consistent with the disclosed embodiments. Although the following refers to memory **140**, one of skill in the art will recognize that instructions may be stored in memory **140** and/or **150**.

[0131] As shown in FIG. **4**, memory **140** may store a monocular image analysis module **402**, a stereo image analysis module **404**, a velocity and acceleration module **406**, and a navigational response module **408**. The disclosed embodiments are not limited to any particular configuration of memory **140**. Further, application processor **180** and/or image processor **190** may execute the instructions stored in any of modules **402**, **404**, **406**, and **408** included in memory **140**. One of skill in the art will understand that references in the following discussions to processing unit **110** may refer to application processor **180** and image processor **190** individually or collectively.

Accordingly, steps of any of the following processes may be performed by one or more processing devices.

[0132] In one embodiment, monocular image analysis module **402** may store instructions (such as computer vision software) which, when executed by processing unit **110**, performs monocular image analysis of a set of images acquired by one of image capture devices **122**, **124**, and **126**. In some embodiments, processing unit **110** may combine information from a set of images with additional sensory information (e.g., information from radar, lidar, etc.) to perform the monocular

image analysis. As described in connection with FIGS. 5A-5D below, monocular image analysis module **402** may include instructions for detecting a set of features within the set of images, such as lane markings, vehicles, pedestrians, road signs, highway exit ramps, traffic lights, hazardous objects, and any other feature associated with an environment of a vehicle. Based on the analysis, system **100** (e.g., via processing unit **110**) may cause one or more navigational responses in vehicle **200**, such as a turn, a lane shift, a change in acceleration, and the like, as discussed below in connection with navigational response module **408**.

[0133] In one embodiment, stereo image analysis module **404** may store instructions (such as computer vision software) which, when executed by processing unit **110**, performs stereo image analysis of first and second sets of images acquired by a combination of image capture devices selected from any of image capture devices **122**, **124**, and **126**. In some embodiments, processing unit **110** may combine information from the first and second sets of images with additional sensory information (e.g., information from radar) to perform the stereo image analysis. For example, stereo image analysis module **404** may include instructions for performing stereo image analysis based on a first set of images acquired by image capture device **124** and a second set of images acquired by image capture device **126**. As described in connection with FIG. 6 below, stereo image analysis module **404** may include instructions for detecting a set of features within the first and second sets of images, such as lane markings, vehicles, pedestrians, road signs, highway exit ramps, traffic lights, hazardous objects, and the like. Based on the analysis, processing unit **110** may cause one or more navigational responses in vehicle **200**, such as a turn, a lane shift, a change in acceleration, and the like, as discussed below in connection with navigational response module **408**. Furthermore, in some embodiments, stereo image analysis module **404** may implement techniques associated with a trained system (such as a neural network or a deep neural network) or an untrained system, such as a system that may be configured to use computer vision algorithms to detect and/or label objects in an environment from which sensory information was captured and processed. In one embodiment, stereo image analysis module **404** and/or other image processing modules may be configured to use a combination of a trained and untrained system.

[0134] In one embodiment, velocity and acceleration module **406** may store software configured to analyze data received from one or more computing and electromechanical devices in vehicle **200** that are configured to cause a change in velocity and/or acceleration of vehicle **200**. For example, processing unit **110** may execute instructions associated with velocity and acceleration module **406** to calculate a target speed for vehicle **200** based on data derived from execution of monocular image analysis module **402** and/or stereo image analysis module **404**. Such data may include, for example, a target position, velocity, and/or acceleration, the position and/or speed of vehicle **200** relative to a nearby vehicle, pedestrian, or road object, position information for vehicle **200** relative to lane markings of the road, and the like. In addition, processing unit **110** may calculate a target speed for vehicle **200** based on sensory input (e.g., information from radar) and input from other systems of vehicle **200**, such as throttling system **220**, braking system **230**, and/or steering system **240** of vehicle **200**. Based on the calculated target speed, processing unit **110** may transmit electronic signals to throttling system **220**, braking system **230**, and/or steering system **240** of vehicle **200** to trigger a change in velocity and/or acceleration by, for example, physically depressing the brake or easing up off the accelerator of vehicle **200**.

[0135] In one embodiment, navigational response module **408** may store software executable by processing unit **110** to determine a desired navigational response based on data derived from execution of monocular image analysis module **402** and/or stereo image analysis module **404**. Such data may include position and speed information associated with nearby vehicles, pedestrians, and road objects, target position information for vehicle **200**, and the like. Additionally, in some embodiments, the navigational response may be based (partially or fully) on map data, a predetermined position of vehicle **200**, and/or a relative velocity or a relative acceleration between vehicle **200** and one or more objects detected from execution of monocular image analysis module

**402** and/or stereo image analysis module **404**. Navigational response module **408** may also determine a desired navigational response based on sensory input (e.g., information from radar) and inputs from other systems of vehicle **200**, such as throttling system **220**, braking system **230**, and steering system **240** of vehicle **200**. Based on the desired navigational response, processing unit **110** may transmit electronic signals to throttling system **220**, braking system **230**, and steering system **240** of vehicle **200** to trigger a desired navigational response by, for example, turning the steering wheel of vehicle **200** to achieve a rotation of a predetermined angle. In some embodiments, processing unit **110** may use the output of navigational response module **408** (e.g., the desired navigational response) as an input to execution of velocity and acceleration module **406** for calculating a change in speed of vehicle **200**.

[0136] Furthermore, any of the modules (e.g., modules **402**, **404**, and **406**) disclosed herein may implement techniques associated with a trained system (such as a neural network or a deep neural network) or an untrained system.

[0137] FIG. 5A is a flowchart showing an exemplary process **500A** for causing one or more navigational responses based on monocular image analysis, consistent with disclosed embodiments. At step **510**, processing unit **110** may receive a plurality of images via data interface **128** between processing unit **110** and image acquisition unit **120**. For instance, a camera included in image acquisition unit **120** (such as image capture device **122** having field of view **202**) may capture a plurality of images of an area forward of vehicle **200** (or to the sides or rear of a vehicle, for example) and transmit them over a data connection (e.g., digital, wired, USB, wireless, Bluetooth, etc.) to processing unit **110**. Processing unit **110** may execute monocular image analysis module **402** to analyze the plurality of images at step **520**, as described in further detail in connection with FIGS. 5B-5D below. By performing the analysis, processing unit **110** may detect a set of features within the set of images, such as lane markings, vehicles, pedestrians, road signs, highway exit ramps, traffic lights, and the like.

[0138] Processing unit **110** may also execute monocular image analysis module **402** to detect various road hazards at step **520**, such as, for example, parts of a truck tire, fallen road signs, loose cargo, small animals, and the like. Road hazards may vary in structure, shape, size, and color, which may make detection of such hazards more challenging. In some embodiments, processing unit **110** may execute monocular image analysis module **402** to perform multi-frame analysis on the plurality of images to detect road hazards. For example, processing unit **110** may estimate camera motion between consecutive image frames and calculate the disparities in pixels between the frames to construct a 3D-map of the road. Processing unit **110** may then use the 3D-map to detect the road surface, as well as hazards existing above the road surface.

[0139] At step **530**, processing unit **110** may execute navigational response module **408** to cause one or more navigational responses in vehicle **200** based on the analysis performed at step **520** and the techniques as described above in connection with FIG. 4. Navigational responses may include, for example, a turn, a lane shift, a change in acceleration, and the like. In some embodiments, processing unit **110** may use data derived from execution of velocity and acceleration module **406** to cause the one or more navigational responses. Additionally, multiple navigational responses may occur simultaneously, in sequence, or any combination thereof. For instance, processing unit **110** may cause vehicle **200** to shift one lane over and then accelerate by, for example, sequentially transmitting control signals to steering system **240** and throttling system **220** of vehicle **200**. Alternatively, processing unit **110** may cause vehicle **200** to brake while at the same time shifting lanes by, for example, simultaneously transmitting control signals to braking system **230** and steering system **240** of vehicle **200**.

[0140] FIG. 5B is a flowchart showing an exemplary process **500B** for detecting one or more vehicles and/or pedestrians in a set of images, consistent with disclosed embodiments. Processing unit **110** may execute monocular image analysis module **402** to implement process **500B**. At step **540**, processing unit **110** may determine a set of candidate objects representing possible vehicles

and/or pedestrians. For example, processing unit **110** may scan one or more images, compare the images to one or more predetermined patterns, and identify within each image possible locations that may contain objects of interest (e.g., vehicles, pedestrians, or portions thereof). The predetermined patterns may be designed in such a way to achieve a high rate of “false hits” and a low rate of “misses.” For example, processing unit **110** may use a low threshold of similarity to predetermined patterns for identifying candidate objects as possible vehicles or pedestrians. Doing so may allow processing unit **110** to reduce the probability of missing (e.g., not identifying) a candidate object representing a vehicle or pedestrian.

[0141] At step **542**, processing unit **110** may filter the set of candidate objects to exclude certain candidates (e.g., irrelevant or less relevant objects) based on classification criteria. Such criteria may be derived from various properties associated with object types stored in a database (e.g., a database stored in memory **140**). Properties may include object shape, dimensions, texture, position (e.g., relative to vehicle **200**), and the like. Thus, processing unit **110** may use one or more sets of criteria to reject false candidates from the set of candidate objects.

[0142] At step **544**, processing unit **110** may analyze multiple frames of images to determine whether objects in the set of candidate objects represent vehicles and/or pedestrians. For example, processing unit **110** may track a detected candidate object across consecutive frames and accumulate frame-by-frame data associated with the detected object (e.g., size, position relative to vehicle **200**, etc.). Additionally, processing unit **110** may estimate parameters for the detected object and compare the object's frame-by-frame position data to a predicted position.

[0143] At step **546**, processing unit **110** may construct a set of measurements for the detected objects. Such measurements may include, for example, position, velocity, and acceleration values (relative to vehicle **200**) associated with the detected objects. In some embodiments, processing unit **110** may construct the measurements based on estimation techniques using a series of time-based observations such as Kalman filters or linear quadratic estimation (LQE), and/or based on available modeling data for different object types (e.g., cars, trucks, pedestrians, bicycles, road signs, etc.). The Kalman filters may be based on a measurement of an object's scale, where the scale measurement is proportional to a time to collision (e.g., the amount of time for vehicle **200** to reach the object). Thus, by performing steps **540-546**, processing unit **110** may identify vehicles and pedestrians appearing within the set of captured images and derive information (e.g., position, speed, size) associated with the vehicles and pedestrians. Based on the identification and the derived information, processing unit **110** may cause one or more navigational responses in vehicle **200**, as described in connection with FIG. 5A, above.

[0144] At step **548**, processing unit **110** may perform an optical flow analysis of one or more images to reduce the probabilities of detecting a “false hit” and missing a candidate object that represents a vehicle or pedestrian. The optical flow analysis may refer to, for example, analyzing motion patterns relative to vehicle **200** in the one or more images associated with other vehicles and pedestrians, and that are distinct from road surface motion. Processing unit **110** may calculate the motion of candidate objects by observing the different positions of the objects across multiple image frames, which are captured at different times. Processing unit **110** may use the position and time values as inputs into mathematical models for calculating the motion of the candidate objects. Thus, optical flow analysis may provide another method of detecting vehicles and pedestrians that are nearby vehicle **200**. Processing unit **110** may perform optical flow analysis in combination with steps **540-546** to provide redundancy for detecting vehicles and pedestrians and increase the reliability of system **100**.

[0145] FIG. 5C is a flowchart showing an exemplary process **500C** for detecting road marks and/or lane geometry information in a set of images, consistent with disclosed embodiments. Processing unit **110** may execute monocular image analysis module **402** to implement process **500C**. At step **550**, processing unit **110** may detect a set of objects by scanning one or more images. To detect segments of lane markings, lane geometry information, and other pertinent road marks, processing

unit **110** may filter the set of objects to exclude those determined to be irrelevant (e.g., minor potholes, small rocks, etc.). At step **552**, processing unit **110** may group together the segments detected in step **550** belonging to the same road mark or lane mark. Based on the grouping, processing unit **110** may develop a model to represent the detected segments, such as a mathematical model.

[0146] At step **554**, processing unit **110** may construct a set of measurements associated with the detected segments. In some embodiments, processing unit **110** may create a projection of the detected segments from the image plane onto the real-world plane. The projection may be characterized using a 3rd-degree polynomial having coefficients corresponding to physical properties such as the position, slope, curvature, and curvature derivative of the detected road. In generating the projection, processing unit **110** may take into account changes in the road surface, as well as pitch and roll rates associated with vehicle **200**. In addition, processing unit **110** may model the road elevation by analyzing position and motion cues present on the road surface. Further, processing unit **110** may estimate the pitch and roll rates associated with vehicle **200** by tracking a set of feature points in the one or more images.

[0147] At step **556**, processing unit **110** may perform multi-frame analysis by, for example, tracking the detected segments across consecutive image frames and accumulating frame-by-frame data associated with detected segments. As processing unit **110** performs multi-frame analysis, the set of measurements constructed at step **554** may become more reliable and associated with an increasingly higher confidence level. Thus, by performing steps **550**, **552**, **554**, and **556**, processing unit **110** may identify road marks appearing within the set of captured images and derive lane geometry information. Based on the identification and the derived information, processing unit **110** may cause one or more navigational responses in vehicle **200**, as described in connection with FIG. 5A, above.

[0148] At step **558**, processing unit **110** may consider additional sources of information to further develop a safety model for vehicle **200** in the context of its surroundings. Processing unit **110** may use the safety model to define a context in which system **100** may execute autonomous control of vehicle **200** in a safe manner. To develop the safety model, in some embodiments, processing unit **110** may consider the position and motion of other vehicles, the detected road edges and barriers, and/or general road shape descriptions extracted from map data (such as data from map database **160**). By considering additional sources of information, processing unit **110** may provide redundancy for detecting road marks and lane geometry and increase the reliability of system **100**.

[0149] FIG. 5D is a flowchart showing an exemplary process **500D** for detecting traffic lights in a set of images, consistent with disclosed embodiments. Processing unit **110** may execute monocular image analysis module **402** to implement process **500D**. At step **560**, processing unit **110** may scan the set of images and identify objects appearing at locations in the images likely to contain traffic lights. For example, processing unit **110** may filter the identified objects to construct a set of candidate objects, excluding those objects unlikely to correspond to traffic lights. The filtering may be done based on various properties associated with traffic lights, such as shape, dimensions, texture, position (e.g., relative to vehicle **200**), and the like. Such properties may be based on multiple examples of traffic lights and traffic control signals and stored in a database. In some embodiments, processing unit **110** may perform multi-frame analysis on the set of candidate objects reflecting possible traffic lights. For example, processing unit **110** may track the candidate objects across consecutive image frames, estimate the real-world position of the candidate objects, and filter out those objects that are moving (which are unlikely to be traffic lights). In some embodiments, processing unit **110** may perform color analysis on the candidate objects and identify the relative position of the detected colors appearing inside possible traffic lights.

[0150] At step **562**, processing unit **110** may analyze the geometry of a junction. The analysis may be based on any combination of: (i) the number of lanes detected on either side of vehicle **200**, (ii) markings (such as arrow marks) detected on the road, and (iii) descriptions of the junction extracted

from map data (such as data from map database **160**). Processing unit **110** may conduct the analysis using information derived from execution of monocular analysis module **402**. In addition, Processing unit **110** may determine a correspondence between the traffic lights detected at step **560** and the lanes appearing near vehicle **200**.

[0151] As vehicle **200** approaches the junction, at step **564**, processing unit **110** may update the confidence level associated with the analyzed junction geometry and the detected traffic lights. For instance, the number of traffic lights estimated to appear at the junction as compared with the number actually appearing at the junction may impact the confidence level. Thus, based on the confidence level, processing unit **110** may delegate control to the driver of vehicle **200** in order to improve safety conditions. By performing steps **560**, **562**, and **564**, processing unit **110** may identify traffic lights appearing within the set of captured images and analyze junction geometry information. Based on the identification and the analysis, processing unit **110** may cause one or more navigational responses in vehicle **200**, as described in connection with FIG. 5A, above.

[0152] FIG. 5E is a flowchart showing an exemplary process **500E** for causing one or more navigational responses in vehicle **200** based on a vehicle path, consistent with the disclosed embodiments. At step **570**, processing unit **110** may construct an initial vehicle path associated with vehicle **200**. The vehicle path may be represented using a set of points expressed in coordinates ( $x$ ,  $z$ ), and the distance  $d_{sub.i}$  between two points in the set of points may fall in the range of 1 to 5 meters. In one embodiment, processing unit **110** may construct the initial vehicle path using two polynomials, such as left and right road polynomials. Processing unit **110** may calculate the geometric midpoint between the two polynomials and offset each point included in the resultant vehicle path by a predetermined offset (e.g., a smart lane offset), if any (an offset of zero may correspond to travel in the middle of a lane). The offset may be in a direction perpendicular to a segment between any two points in the vehicle path. In another embodiment, processing unit **110** may use one polynomial and an estimated lane width to offset each point of the vehicle path by half the estimated lane width plus a predetermined offset (e.g., a smart lane offset).

[0153] At step **572**, processing unit **110** may update the vehicle path constructed at step **570**. Processing unit **110** may reconstruct the vehicle path constructed at step **570** using a higher resolution, such that the distance  $d_{sub.k}$  between two points in the set of points representing the vehicle path is less than the distance  $d_{sub.i}$  described above. For example, the distance  $d_{sub.k}$  may fall in the range of 0.1 to 0.3 meters. Processing unit **110** may reconstruct the vehicle path using a parabolic spline algorithm, which may yield a cumulative distance vector  $S$  corresponding to the total length of the vehicle path (i.e., based on the set of points representing the vehicle path).

[0154] At step **574**, processing unit **110** may determine a look-ahead point (expressed in coordinates as ( $x_{sub.l}$ ,  $z_{sub.l}$ )) based on the updated vehicle path constructed at step **572**. Processing unit **110** may extract the look-ahead point from the cumulative distance vector  $S$ , and the look-ahead point may be associated with a look-ahead distance and look-ahead time. The look-ahead distance, which may have a lower bound ranging from 10 to 20 meters, may be calculated as the product of the speed of vehicle **200** and the look-ahead time. For example, as the speed of vehicle **200** decreases, the look-ahead distance may also decrease (e.g., until it reaches the lower bound). The look-ahead time, which may range from 0.5 to 1.5 seconds, may be inversely proportional to the gain of one or more control loops associated with causing a navigational response in vehicle **200**, such as the heading error tracking control loop. For example, the gain of the heading error tracking control loop may depend on the bandwidth of a yaw rate loop, a steering actuator loop, car lateral dynamics, and the like. Thus, the higher the gain of the heading error tracking control loop, the lower the look-ahead time.

[0155] At step **576**, processing unit **110** may determine a heading error and yaw rate command based on the look-ahead point determined at step **574**. Processing unit **110** may determine the heading error by calculating the arctangent of the look-ahead point, e.g.,  $\arctan(x_{sub.l}/z_{sub.l})$ . Processing unit **110** may determine the yaw rate command as the product of the heading error and a

high-level control gain. The high-level control gain may be equal to:  $(2/\text{look-ahead time})$ , if the look-ahead distance is not at the lower bound. Otherwise, the high-level control gain may be equal to:  $(2 \times \text{speed of vehicle } 200 / \text{look-ahead distance})$ .

[0156] FIG. 5F is a flowchart showing an exemplary process 500F for determining whether a leading vehicle is changing lanes, consistent with the disclosed embodiments. At step 580, processing unit 110 may determine navigation information associated with a leading vehicle (e.g., a vehicle traveling ahead of vehicle 200). For example, processing unit 110 may determine the position, velocity (e.g., direction and speed), and/or acceleration of the leading vehicle, using the techniques described in connection with FIGS. 5A and 5B, above. Processing unit 110 may also determine one or more road polynomials, a look-ahead point (associated with vehicle 200), and/or a snail trail (e.g., a set of points describing a path taken by the leading vehicle), using the techniques described in connection with FIG. 5E, above.

[0157] At step 582, processing unit 110 may analyze the navigation information determined at step 580. In one embodiment, processing unit 110 may calculate the distance between a snail trail and a road polynomial (e.g., along the trail). If the variance of this distance along the trail exceeds a predetermined threshold (for example, 0.1 to 0.2 meters on a straight road, 0.3 to 0.4 meters on a moderately curvy road, and 0.5 to 0.6 meters on a road with sharp curves), processing unit 110 may determine that the leading vehicle is likely changing lanes. In the case where multiple vehicles are detected traveling ahead of vehicle 200, processing unit 110 may compare the snail trails associated with each vehicle. Based on the comparison, processing unit 110 may determine that a vehicle whose snail trail does not match with the snail trails of the other vehicles is likely changing lanes. Processing unit 110 may additionally compare the curvature of the snail trail (associated with the leading vehicle) with the expected curvature of the road segment in which the leading vehicle is traveling. The expected curvature may be extracted from map data (e.g., data from map database 160), from road polynomials, from other vehicles' snail trails, from prior knowledge about the road, and the like. If the difference in curvature of the snail trail and the expected curvature of the road segment exceeds a predetermined threshold, processing unit 110 may determine that the leading vehicle is likely changing lanes.

[0158] In another embodiment, processing unit 110 may compare the leading vehicle's instantaneous position with the look-ahead point (associated with vehicle 200) over a specific period of time (e.g., 0.5 to 1.5 seconds). If the distance between the leading vehicle's instantaneous position and the look-ahead point varies during the specific period of time, and the cumulative sum of variation exceeds a predetermined threshold (for example, 0.3 to 0.4 meters on a straight road, 0.7 to 0.8 meters on a moderately curvy road, and 1.3 to 1.7 meters on a road with sharp curves), processing unit 110 may determine that the leading vehicle is likely changing lanes. In another embodiment, processing unit 110 may analyze the geometry of the snail trail by comparing the lateral distance traveled along the trail with the expected curvature of the snail trail. The expected radius of curvature may be determined according to the calculation:

$(\delta_{\text{sub.z}}^2 + \delta_{\text{sub.x}}^2) / (2 \times \delta_{\text{sub.x}})$ , where  $\delta_{\text{sub.x}}$  represents the lateral distance traveled and  $\delta_{\text{sub.z}}$  represents the longitudinal distance traveled. If the difference between the lateral distance traveled and the expected curvature exceeds a predetermined threshold (e.g., 500 to 700 meters), processing unit 110 may determine that the leading vehicle is likely changing lanes. In another embodiment, processing unit 110 may analyze the position of the leading vehicle. If the position of the leading vehicle obscures a road polynomial (e.g., the leading vehicle is overlaid on top of the road polynomial), then processing unit 110 may determine that the leading vehicle is likely changing lanes. In the case where the position of the leading vehicle is such that, another vehicle is detected ahead of the leading vehicle and the snail trails of the two vehicles are not parallel, processing unit 110 may determine that the (closer) leading vehicle is likely changing lanes.

[0159] At step 584, processing unit 110 may determine whether or not leading vehicle 200 is changing lanes based on the analysis performed at step 582. For example, processing unit 110 may

make the determination based on a weighted average of the individual analyses performed at step **582**. Under such a scheme, for example, a decision by processing unit **110** that the leading vehicle is likely changing lanes based on a particular type of analysis may be assigned a value of “1” (and “0” to represent a determination that the leading vehicle is not likely changing lanes). Different analyses performed at step **582** may be assigned different weights, and the disclosed embodiments are not limited to any particular combination of analyses and weights.

[0160] FIG. **6** is a flowchart showing an exemplary process **600** for causing one or more navigational responses based on stereo image analysis, consistent with disclosed embodiments. At step **610**, processing unit **110** may receive a first and second plurality of images via data interface **128**. For example, cameras included in image acquisition unit **120** (such as image capture devices **122** and **124** having fields of view **202** and **204**) may capture a first and second plurality of images of an area forward of vehicle **200** and transmit them over a digital connection (e.g., USB, wireless, Bluetooth, etc.) to processing unit **110**. In some embodiments, processing unit **110** may receive the first and second plurality of images via two or more data interfaces. The disclosed embodiments are not limited to any particular data interface configurations or protocols.

[0161] At step **620**, processing unit **110** may execute stereo image analysis module **404** to perform stereo image analysis of the first and second plurality of images to create a 3D map of the road in front of the vehicle and detect features within the images, such as lane markings, vehicles, pedestrians, road signs, highway exit ramps, traffic lights, road hazards, and the like. Stereo image analysis may be performed in a manner similar to the steps described in connection with FIGS. **5A-5D**, above. For example, processing unit **110** may execute stereo image analysis module **404** to detect candidate objects (e.g., vehicles, pedestrians, road marks, traffic lights, road hazards, etc.) within the first and second plurality of images, filter out a subset of the candidate objects based on various criteria, and perform multi-frame analysis, construct measurements, and determine a confidence level for the remaining candidate objects. In performing the steps above, processing unit **110** may consider information from both the first and second plurality of images, rather than information from one set of images alone. For example, processing unit **110** may analyze the differences in pixel-level data (or other data subsets from among the two streams of captured images) for a candidate object appearing in both the first and second plurality of images. As another example, processing unit **110** may estimate a position and/or velocity of a candidate object (e.g., relative to vehicle **200**) by observing that the object appears in one of the plurality of images but not the other or relative to other differences that may exist relative to objects appearing in the two image streams. For example, position, velocity, and/or acceleration relative to vehicle **200** may be determined based on trajectories, positions, movement characteristics, etc. of features associated with an object appearing in one or both of the image streams.

[0162] At step **630**, processing unit **110** may execute navigational response module **408** to cause one or more navigational responses in vehicle **200** based on the analysis performed at step **620** and the techniques as described above in connection with FIG. **4**. Navigational responses may include, for example, a turn, a lane shift, a change in acceleration, a change in velocity, braking, and the like. In some embodiments, processing unit **110** may use data derived from execution of velocity and acceleration module **406** to cause the one or more navigational responses. Additionally, multiple navigational responses may occur simultaneously, in sequence, or any combination thereof.

[0163] FIG. **7** is a flowchart showing an exemplary process **700** for causing one or more navigational responses based on an analysis of three sets of images, consistent with disclosed embodiments. At step **710**, processing unit **110** may receive a first, second, and third plurality of images via data interface **128**. For instance, cameras included in image acquisition unit **120** (such as image capture devices **122**, **124**, and **126** having fields of view **202**, **204**, and **206**) may capture a first, second, and third plurality of images of an area forward and/or to the side of vehicle **200** and transmit them over a digital connection (e.g., USB, wireless, Bluetooth, etc.) to processing unit



**110**. In some embodiments, processing unit **110** may receive the first, second, and third plurality of images via three or more data interfaces. For example, each of image capture devices **122**, **124**, **126** may have an associated data interface for communicating data to processing unit **110**. The disclosed embodiments are not limited to any particular data interface configurations or protocols. [0164] At step **720**, processing unit **110** may analyze the first, second, and third plurality of images to detect features within the images, such as lane markings, vehicles, pedestrians, road signs, highway exit ramps, traffic lights, road hazards, and the like. The analysis may be performed in a manner similar to the steps described in connection with FIGS. **5A-5D** and **6**, above. For instance, processing unit **110** may perform monocular image analysis (e.g., via execution of monocular image analysis module **402** and based on the steps described in connection with FIGS. **5A-5D**, above) on each of the first, second, and third plurality of images. Alternatively, processing unit **110** may perform stereo image analysis (e.g., via execution of stereo image analysis module **404** and based on the steps described in connection with FIG. **6**, above) on the first and second plurality of images, the second and third plurality of images, and/or the first and third plurality of images. The processed information corresponding to the analysis of the first, second, and/or third plurality of images may be combined. In some embodiments, processing unit **110** may perform a combination of monocular and stereo image analyses. For example, processing unit **110** may perform monocular image analysis (e.g., via execution of monocular image analysis module **402**) on the first plurality of images and stereo image analysis (e.g., via execution of stereo image analysis module **404**) on the second and third plurality of images. The configuration of image capture devices **122**, **124**, and **126**—including their respective locations and fields of view **202**, **204**, and **206**—may influence the types of analyses conducted on the first, second, and third plurality of images. The disclosed embodiments are not limited to a particular configuration of image capture devices **122**, **124**, and **126**, or the types of analyses conducted on the first, second, and third plurality of images.

[0165] In some embodiments, processing unit **110** may perform testing on system **100** based on the images acquired and analyzed at steps **710** and **720**. Such testing may provide an indicator of the overall performance of system **100** for certain configurations of image capture devices **122**, **124**, and **126**. For example, processing unit **110** may determine the proportion of “false hits” (e.g., cases where system **100** incorrectly determined the presence of a vehicle or pedestrian) and “misses.”

[0166] At step **730**, processing unit **110** may cause one or more navigational responses in vehicle **200** based on information derived from two of the first, second, and third plurality of images. Selection of two of the first, second, and third plurality of images may depend on various factors, such as, for example, the number, types, and sizes of objects detected in each of the plurality of images. Processing unit **110** may also make the selection based on image quality and resolution, the effective field of view reflected in the images, the number of captured frames, the extent to which one or more objects of interest actually appear in the frames (e.g., the percentage of frames in which an object appears, the proportion of the object that appears in each such frame, etc.), and the like.

[0167] In some embodiments, processing unit **110** may select information derived from two of the first, second, and third plurality of images by determining the extent to which information derived from one image source is consistent with information derived from other image sources. For example, processing unit **110** may combine the processed information derived from each of image capture devices **122**, **124**, and **126** (whether by monocular analysis, stereo analysis, or any combination of the two) and determine visual indicators (e.g., lane markings, a detected vehicle and its location and/or path, a detected traffic light, etc.) that are consistent across the images captured from each of image capture devices **122**, **124**, and **126**. Processing unit **110** may also exclude information that is inconsistent across the captured images (e.g., a vehicle changing lanes, a lane model indicating a vehicle that is too close to vehicle **200**, etc.). Thus, processing unit **110** may select information derived from two of the first, second, and third plurality of images based on the determinations of consistent and inconsistent information.

[0168] Navigational responses may include, for example, a turn, a lane shift, a change in acceleration, and the like. Processing unit **110** may cause the one or more navigational responses based on the analysis performed at step **720** and the techniques as described above in connection with FIG. **4**. Processing unit **110** may also use data derived from execution of velocity and acceleration module **406** to cause the one or more navigational responses. In some embodiments, processing unit **110** may cause the one or more navigational responses based on a relative position, relative velocity, and/or relative acceleration between vehicle **200** and an object detected within any of the first, second, and third plurality of images. Multiple navigational responses may occur simultaneously, in sequence, or any combination thereof.

#### Sparse Road Model for Autonomous Vehicle Navigation

[0169] In some embodiments, the disclosed systems and methods may use a sparse map for autonomous vehicle navigation. In particular, the sparse map may be for autonomous vehicle navigation along a road segment. For example, the sparse map may provide sufficient information for navigating an autonomous vehicle without storing and/or updating a large quantity of data. As discussed below in further detail, an autonomous vehicle may use the sparse map to navigate one or more roads based on one or more stored trajectories.

#### Sparse Map for Autonomous Vehicle Navigation

[0170] In some embodiments, the disclosed systems and methods may generate a sparse map for autonomous vehicle navigation. For example, the sparse map may provide sufficient information for navigation without requiring excessive data storage or data transfer rates. As discussed below in further detail, a vehicle (which may be an autonomous vehicle) may use the sparse map to navigate one or more roads. For example, in some embodiments, the sparse map may include data related to a road and potentially landmarks along the road that may be sufficient for vehicle navigation, but which also exhibit small data footprints. For example, the sparse data maps described in detail below may require significantly less storage space and data transfer bandwidth as compared with digital maps including detailed map information, such as image data collected along a road.

[0171] For example, rather than storing detailed representations of a road segment, the sparse data map may store three-dimensional polynomial representations of preferred vehicle paths along a road. These paths may require very little data storage space. Further, in the described sparse data maps, landmarks may be identified and included in the sparse map road model to aid in navigation. These landmarks may be located at any spacing suitable for enabling vehicle navigation, but in some cases, such landmarks need not be identified and included in the model at high densities and short spacings. Rather, in some cases, navigation may be possible based on landmarks that are spaced apart by at least 50 meters, at least 100 meters, at least 500 meters, at least 1 kilometer, or at least 2 kilometers. As will be discussed in more detail in other sections, the sparse map may be generated based on data collected or measured by vehicles equipped with various sensors and devices, such as image capture devices, Global Positioning System sensors, motion sensors, etc., as the vehicles travel along roadways. In some cases, the sparse map may be generated based on data collected during multiple drives of one or more vehicles along a particular roadway. Generating a sparse map using multiple drives of one or more vehicles may be referred to as “crowdsourcing” a sparse map.

[0172] Consistent with disclosed embodiments, an autonomous vehicle system may use a sparse map for navigation. For example, the disclosed systems and methods may distribute a sparse map for generating a road navigation model for an autonomous vehicle and may navigate an autonomous vehicle along a road segment using a sparse map and/or a generated road navigation model. Sparse maps consistent with the present disclosure may include one or more three-dimensional contours that may represent predetermined trajectories that autonomous vehicles may traverse as they move along associated road segments.

[0173] Sparse maps consistent with the present disclosure may also include data representing one or more road features. Such road features may include recognized landmarks, road signature

profiles, and any other road-related features useful in navigating a vehicle. Sparse maps consistent with the present disclosure may enable autonomous navigation of a vehicle based on relatively small amounts of data included in the sparse map. For example, rather than including detailed representations of a road, such as road edges, road curvature, images associated with road segments, or data detailing other physical features associated with a road segment, the disclosed embodiments of the sparse map may require relatively little storage space (and relatively little bandwidth when portions of the sparse map are transferred to a vehicle) but may still adequately provide for autonomous vehicle navigation. The small data footprint of the disclosed sparse maps, discussed in further detail below, may be achieved in some embodiments by storing representations of road-related elements that require small amounts of data but still enable autonomous navigation. [0174] For example, rather than storing detailed representations of various aspects of a road, the disclosed sparse maps may store polynomial representations of one or more trajectories that a vehicle may follow along the road. Thus, rather than storing (or having to transfer) details regarding the physical nature of the road to enable navigation along the road, using the disclosed sparse maps, a vehicle may be navigated along a particular road segment without, in some cases, having to interpret physical aspects of the road, but rather, by aligning its path of travel with a trajectory (e.g., a polynomial spline) along the particular road segment. In this way, the vehicle may be navigated based mainly upon the stored trajectory (e.g., a polynomial spline) that may require much less storage space than an approach involving storage of roadway images, road parameters, road layout, etc.

[0175] In addition to the stored polynomial representations of trajectories along a road segment, the disclosed sparse maps may also include small data objects that may represent a road feature. In some embodiments, the small data objects may include digital signatures, which are derived from a digital image (or a digital signal) that was obtained by a sensor (e.g., a camera or other sensor, such as a suspension sensor) onboard a vehicle traveling along the road segment. The digital signature may have a reduced size relative to the signal that was acquired by the sensor. In some embodiments, the digital signature may be created to be compatible with a classifier function that is configured to detect and to identify the road feature from the signal that is acquired by the sensor, for example, during a subsequent drive. In some embodiments, a digital signature may be created such that the digital signature has a footprint that is as small as possible, while retaining the ability to correlate or match the road feature with the stored signature based on an image (or a digital signal generated by a sensor, if the stored signature is not based on an image and/or includes other data) of the road feature that is captured by a camera onboard a vehicle traveling along the same road segment at a subsequent time.

[0176] In some embodiments, a size of the data objects may be further associated with a uniqueness of the road feature. For example, for a road feature that is detectable by a camera onboard a vehicle, and where the camera system onboard the vehicle is coupled to a classifier that is capable of distinguishing the image data corresponding to that road feature as being associated with a particular type of road feature, for example, a road sign, and where such a road sign is locally unique in that area (e.g., there is no identical road sign or road sign of the same type nearby), it may be sufficient to store data indicating the type of the road feature and its location.

[0177] As will be discussed in further detail below, road features (e.g., landmarks along a road segment) may be stored as small data objects that may represent a road feature in relatively few bytes, while at the same time providing sufficient information for recognizing and using such a feature for navigation. In one example, a road sign may be identified as a recognized landmark on which navigation of a vehicle may be based. A representation of the road sign may be stored in the sparse map to include, e.g., a few bytes of data indicating a type of landmark (e.g., a stop sign) and a few bytes of data indicating a location of the landmark (e.g., coordinates). Navigating based on such data-light representations of the landmarks (e.g., using representations sufficient for locating, recognizing, and navigating based upon the landmarks) may provide a desired level of navigational

functionality associated with sparse maps without significantly increasing the data overhead associated with the sparse maps. This lean representation of landmarks (and other road features) may take advantage of the sensors and processors included onboard such vehicles that are configured to detect, identify, and/or classify certain road features.

[0178] When, for example, a sign or even a particular type of a sign is locally unique (e.g., when there is no other sign or no other sign of the same type) in a given area, the sparse map may use data indicating a type of a landmark (a sign or a specific type of sign), and during navigation (e.g., autonomous navigation) when a camera onboard an autonomous vehicle captures an image of the area including a sign (or of a specific type of sign), the processor may process the image, detect the sign (if indeed present in the image), classify the image as a sign (or as a specific type of sign), and correlate the location of the image with the location of the sign as stored in the sparse map.

[0179] The sparse map may include any suitable representation of objects identified along a road segment. In some cases, the objects may be referred to as semantic objects or non-semantic objects. Semantic objects may include, for example, objects associated with a predetermined type classification. This type classification may be useful in reducing the amount of data required to describe the semantic object recognized in an environment, which can be beneficial both in the harvesting phase (e.g., to reduce costs associated with bandwidth use for transferring drive information from a plurality of harvesting vehicles to a server) and during the navigation phase (e.g., reduction of map data can speed transfer of map tiles from a server to a navigating vehicle and can also reduce costs associated with bandwidth use for such transfers). Semantic object classification types may be assigned to any type of objects or features that are expected to be encountered along a roadway.

[0180] Semantic objects may further be divided into two or more logical groups. For example, in some cases, one group of semantic object types may be associated with predetermined dimensions. Such semantic objects may include certain speed limit signs, yield signs, merge signs, stop signs, traffic lights, directional arrows on a roadway, manhole covers, or any other type of object that may be associated with a standardized size. One benefit offered by such semantic objects is that very little data may be needed to represent/fully define the objects. For example, if a standardized size of a speed limit size is known, then a harvesting vehicle may need only identify (through analysis of a captured image) the presence of a speed limit sign (a recognized type) along with an indication of a position of the detected speed limit sign (e.g., a 2D position in the captured image (or, alternatively, a 3D position in real world coordinates) of a center of the sign or a certain corner of the sign) to provide sufficient information for map generation on the server side. Where 2D image positions are transmitted to the server, a position associated with the captured image where the sign was detected may also be transmitted so the server can determine a real-world position of the sign (e.g., through structure in motion techniques using multiple captured images from one or more harvesting vehicles). Even with this limited information (requiring just a few bytes to define each detected object), the server may construct the map including a fully represented speed limit sign based on the type classification (representative of a speed limit sign) received from one or more harvesting vehicles along with the position information for the detected sign.

[0181] Semantic objects may also include other recognized object or feature types that are not associated with certain standardized characteristics. Such objects or features may include potholes, tar seams, light poles, non-standardized signs, curbs, trees, tree branches, or any other type of recognized object type with one or more variable characteristics (e.g., variable dimensions). In such cases, in addition to transmitting to a server an indication of the detected object or feature type (e.g., pothole, pole, etc.) and position information for the detected object or feature, a harvesting vehicle may also transmit an indication of a size of the object or feature. The size may be expressed in 2D image dimensions (e.g., with a bounding box or one or more dimension values) or real-world dimensions (determined through structure in motion calculations, based on LIDAR or RADAR system outputs, based on trained neural network outputs, etc.).

[0182] Non-semantic objects or features may include any detectable objects or features that fall outside of a recognized category or type, but that still may provide valuable information in map generation. In some cases, such non-semantic features may include a detected corner of a building or a corner of a detected window of a building, a unique stone or object near a roadway, a concrete splatter in a roadway shoulder, or any other detectable object or feature. Upon detecting such an object or feature one or more harvesting vehicles may transmit to a map generation server a location of one or more points (2D image points or 3D real world points) associated with the detected object/feature. Additionally, a compressed or simplified image segment (e.g., an image hash) may be generated for a region of the captured image including the detected object or feature. This image hash may be calculated based on a predetermined image processing algorithm and may form an effective signature for the detected non-semantic object or feature. Such a signature may be useful for navigation relative to a sparse map including the non-semantic feature or object, as a vehicle traversing the roadway may apply an algorithm similar to the algorithm used to generate the image hash in order to confirm/verify the presence in a captured image of the mapped non-semantic feature or object. Using this technique, non-semantic features may add to the richness of the sparse maps (e.g., to enhance their usefulness in navigation) without adding significant data overhead.

[0183] As noted, target trajectories may be stored in the sparse map. These target trajectories (e.g., 3D splines) may represent the preferred or recommended paths for each available lane of a roadway, each valid pathway through a junction, for merges and exits, etc. In addition to target trajectories, other road feature may also be detected, harvested, and incorporated in the sparse maps in the form of representative splines. Such features may include, for example, road edges, lane markings, curbs, guardrails, or any other objects or features that extend along a roadway or road segment.

#### Generating a Sparse Map

[0184] In some embodiments, a sparse map may include at least one line representation of a road surface feature extending along a road segment and a plurality of landmarks associated with the road segment. In certain aspects, the sparse map may be generated via “crowdsourcing,” for example, through image analysis of a plurality of images acquired as one or more vehicles traverse the road segment.

[0185] FIG. 8 shows a sparse map **800** that one or more vehicles, e.g., vehicle **200** (which may be an autonomous vehicle), may access for providing autonomous vehicle navigation. Sparse map **800** may be stored in a memory, such as memory **140** or **150**. Such memory devices may include any types of non-transitory storage devices or computer-readable media. For example, in some embodiments, memory **140** or **150** may include hard drives, compact discs, flash memory, magnetic based memory devices, optical based memory devices, etc. In some embodiments, sparse map **800** may be stored in a database (e.g., map database **160**) that may be stored in memory **140** or **150**, or other types of storage devices.

[0186] In some embodiments, sparse map **800** may be stored on a storage device or a non-transitory computer-readable medium provided onboard vehicle **200** (e.g., a storage device included in a navigation system onboard vehicle **200**). A processor (e.g., processing unit **110**) provided on vehicle **200** may access sparse map **800** stored in the storage device or computer-readable medium provided onboard vehicle **200** in order to generate navigational instructions for guiding the autonomous vehicle **200** as the vehicle traverses a road segment.

[0187] Sparse map **800** need not be stored locally with respect to a vehicle, however. In some embodiments, sparse map **800** may be stored on a storage device or computer-readable medium provided on a remote server that communicates with vehicle **200** or a device associated with vehicle **200**. A processor (e.g., processing unit **110**) provided on vehicle **200** may receive data included in sparse map **800** from the remote server and may execute the data for guiding the autonomous driving of vehicle **200**. In such embodiments, the remote server may store all of sparse

map **800** or only a portion thereof. Accordingly, the storage device or computer-readable medium provided onboard vehicle **200** and/or onboard one or more additional vehicles may store the remaining portion(s) of sparse map **800**.

[0188] Furthermore, in such embodiments, sparse map **800** may be made accessible to a plurality of vehicles traversing various road segments (e.g., tens, hundreds, thousands, or millions of vehicles, etc.). It should be noted also that sparse map **800** may include multiple sub-maps. For example, in some embodiments, sparse map **800** may include hundreds, thousands, millions, or more, of sub-maps (e.g., map tiles) that may be used in navigating a vehicle. Such sub-maps may be referred to as local maps or map tiles, and a vehicle traveling along a roadway may access any number of local maps relevant to a location in which the vehicle is traveling. The local map sections of sparse map **800** may be stored with a Global Navigation Satellite System (GNSS) key as an index to the database of sparse map **800**. Thus, while computation of steering angles for navigating a host vehicle in the present system may be performed without reliance upon a GNSS position of the host vehicle, road features, or landmarks, such GNSS information may be used for retrieval of relevant local maps.

[0189] In general, sparse map **800** may be generated based on data (e.g., drive information) collected from one or more vehicles as they travel along roadways. For example, using sensors aboard the one or more vehicles (e.g., cameras, speedometers, GPS, accelerometers, etc.), the trajectories that the one or more vehicles travel along a roadway may be recorded, and the polynomial representation of a preferred trajectory for vehicles making subsequent trips along the roadway may be determined based on the collected trajectories travelled by the one or more vehicles. Similarly, data collected by the one or more vehicles may aid in identifying potential landmarks along a particular roadway. Data collected from traversing vehicles may also be used to identify road profile information, such as road width profiles, road roughness profiles, traffic line spacing profiles, road conditions, etc. Using the collected information, sparse map **800** may be generated and distributed (e.g., for local storage or via on-the-fly data transmission) for use in navigating one or more autonomous vehicles. However, in some embodiments, map generation may not end upon initial generation of the map. As will be discussed in greater detail below, sparse map **800** may be continuously or periodically updated based on data collected from vehicles as those vehicles continue to traverse roadways included in sparse map **800**.

[0190] Data recorded in sparse map **800** may include position information based on Global Positioning System (GPS) data. For example, location information may be included in sparse map **800** for various map elements, including, for example, landmark locations, road profile locations, etc. Locations for map elements included in sparse map **800** may be obtained using GPS data collected from vehicles traversing a roadway. For example, a vehicle passing an identified landmark may determine a location of the identified landmark using GPS position information associated with the vehicle and a determination of a location of the identified landmark relative to the vehicle (e.g., based on image analysis of data collected from one or more cameras on board the vehicle). Such location determinations of an identified landmark (or any other feature included in sparse map **800**) may be repeated as additional vehicles pass the location of the identified landmark. Some or all of the additional location determinations may be used to refine the location information stored in sparse map **800** relative to the identified landmark. For example, in some embodiments, multiple position measurements relative to a particular feature stored in sparse map **800** may be averaged together. Any other mathematical operations, however, may also be used to refine a stored location of a map element based on a plurality of determined locations for the map element.

[0191] In a particular example, harvesting vehicles may traverse a particular road segment. Each harvesting vehicle captures images of their respective environments. The images may be collected at any suitable frame capture rate (e.g., 9 Hz, etc.). Image analysis processor(s) aboard each harvesting vehicle analyze the captured images to detect the presence of semantic and/or non-

semantic features/objects. At a high level, the harvesting vehicles transmit to a mapping-server indications of detections of the semantic and/or non-semantic objects/features along with positions associated with those objects/features. In more detail, type indicators, dimension indicators, etc. may be transmitted together with the position information. The position information may include any suitable information for enabling the mapping server to aggregate the detected objects/features into a sparse map useful in navigation. In some cases, the position information may include one or more 2D image positions (e.g., X-Y pixel locations) in a captured image where the semantic or non-semantic features/objects were detected. Such image positions may correspond to a center of the feature/object, a corner, etc. In this scenario, to aid the mapping server in reconstructing the drive information and aligning the drive information from multiple harvesting vehicles, each harvesting vehicle may also provide the server with a location (e.g., a GPS location) where each image was captured.

[0192] In other cases, the harvesting vehicle may provide to the server one or more 3D real world points associated with the detected objects/features. Such 3D points may be relative to a predetermined origin (such as an origin of a drive segment) and may be determined through any suitable technique. In some cases, a structure in motion technique may be used to determine the 3D real world position of a detected object/feature. For example, a certain object such as a particular speed limit sign may be detected in two or more captured images. Using information such as the known ego motion (speed, trajectory, GPS position, etc.) of the harvesting vehicle between the captured images, along with observed changes of the speed limit sign in the captured images (change in X-Y pixel location, change in size, etc.), the real-world position of one or more points associated with the speed limit sign may be determined and passed along to the mapping server. Such an approach is optional, as it requires more computation on the part of the harvesting vehicle systems. The sparse map of the disclosed embodiments may enable autonomous navigation of a vehicle using relatively small amounts of stored data. In some embodiments, sparse map **800** may have a data density (e.g., including data representing the target trajectories, landmarks, and any other stored road features) of less than 2 MB per kilometer of roads, less than 1 MB per kilometer of roads, less than 500 kB per kilometer of roads, or less than 100 kB per kilometer of roads. In some embodiments, the data density of sparse map **800** may be less than 10 kB per kilometer of roads or even less than 2 kB per kilometer of roads (e.g., 1.6 kB per kilometer), or no more than 10 kB per kilometer of roads, or no more than 20 kB per kilometer of roads. In some embodiments, most, if not all, of the roadways of the United States may be navigated autonomously using a sparse map having a total of 4 GB or less of data. These data density values may represent an average over an entire sparse map **800**, over a local map within sparse map **800**, and/or over a particular road segment within sparse map **800**.

[0193] As noted, sparse map **800** may include representations of a plurality of target trajectories **810** for guiding autonomous driving or navigation along a road segment. Such target trajectories may be stored as three-dimensional splines. The target trajectories stored in sparse map **800** may be determined based on two or more reconstructed trajectories of prior traversals of vehicles along a particular road segment, for example. A road segment may be associated with a single target trajectory or multiple target trajectories. For example, on a two lane road, a first target trajectory may be stored to represent an intended path of travel along the road in a first direction, and a second target trajectory may be stored to represent an intended path of travel along the road in another direction (e.g., opposite to the first direction). Additional target trajectories may be stored with respect to a particular road segment. For example, on a multi-lane road one or more target trajectories may be stored representing intended paths of travel for vehicles in one or more lanes associated with the multi-lane road. In some embodiments, each lane of a multi-lane road may be associated with its own target trajectory. In other embodiments, there may be fewer target trajectories stored than lanes present on a multi-lane road. In such cases, a vehicle navigating the multi-lane road may use any of the stored target trajectories to guide its navigation by taking into

account an amount of lane offset from a lane for which a target trajectory is stored (e.g., if a vehicle is traveling in the left most lane of a three lane highway, and a target trajectory is stored only for the middle lane of the highway, the vehicle may navigate using the target trajectory of the middle lane by accounting for the amount of lane offset between the middle lane and the left-most lane when generating navigational instructions).

[0194] In some embodiments, the target trajectory may represent an ideal path that a vehicle should take as the vehicle travels. The target trajectory may be located, for example, at an approximate center of a lane of travel. In other cases, the target trajectory may be located elsewhere relative to a road segment. For example, a target trajectory may approximately coincide with a center of a road, an edge of a road, or an edge of a lane, etc. In such cases, navigation based on the target trajectory may include a determined amount of offset to be maintained relative to the location of the target trajectory. Moreover, in some embodiments, the determined amount of offset to be maintained relative to the location of the target trajectory may differ based on a type of vehicle (e.g., a passenger vehicle including two axles may have a different offset from a truck including more than two axles along at least a portion of the target trajectory).

[0195] Sparse map **800** may also include data relating to a plurality of predetermined landmarks **820** associated with particular road segments, local maps, etc. As discussed in greater detail below, these landmarks may be used in navigation of the autonomous vehicle. For example, in some embodiments, the landmarks may be used to determine a current position of the vehicle relative to a stored target trajectory. With this position information, the autonomous vehicle may be able to adjust a heading direction to match a direction of the target trajectory at the determined location.

[0196] The plurality of landmarks **820** may be identified and stored in sparse map **800** at any suitable spacing. In some embodiments, landmarks may be stored at relatively high densities (e.g., every few meters or more). In some embodiments, however, significantly larger landmark spacing values may be employed. For example, in sparse map **800**, identified (or recognized) landmarks may be spaced apart by 10 meters, 20 meters, 50 meters, 100 meters, 1 kilometer, or 2 kilometers. In some cases, the identified landmarks may be located at distances of even more than 2 kilometers apart.

[0197] Between landmarks, and therefore between determinations of vehicle position relative to a target trajectory, the vehicle may navigate based on dead reckoning in which the vehicle uses sensors to determine its ego motion and estimate its position relative to the target trajectory. Because errors may accumulate during navigation by dead reckoning, over time the position determinations relative to the target trajectory may become increasingly less accurate. The vehicle may use landmarks occurring in sparse map **800** (and their known locations) to remove the dead reckoning-induced errors in position determination. In this way, the identified landmarks included in sparse map **800** may serve as navigational anchors from which an accurate position of the vehicle relative to a target trajectory may be determined. Because a certain amount of error may be acceptable in position location, an identified landmark need not always be available to an autonomous vehicle. Rather, suitable navigation may be possible even based on landmark spacings, as noted above, of 10 meters, 20 meters, 50 meters, 100 meters, 500 meters, 1 kilometer, 2 kilometers, or more. In some embodiments, a density of 1 identified landmark every 1 km of road may be sufficient to maintain a longitudinal position determination accuracy within 1 m. Thus, not every potential landmark appearing along a road segment need be stored in sparse map **800**.

[0198] Moreover, in some embodiments, lane markings may be used for localization of the vehicle during landmark spacings. By using lane markings during landmark spacings, the accumulation of errors during navigation by dead reckoning may be minimized.

[0199] In addition to target trajectories and identified landmarks, sparse map **800** may include information relating to various other road features. For example, FIG. 9A illustrates a representation of curves along a particular road segment that may be stored in sparse map **800**. In some embodiments, a single lane of a road may be modeled by a three-dimensional polynomial



description of left and right sides of the road. Such polynomials representing left and right sides of a single lane are shown in FIG. 9A. Regardless of how many lanes a road may have, the road may be represented using polynomials in a way similar to that illustrated in FIG. 9A. For example, left and right sides of a multi-lane road may be represented by polynomials similar to those shown in FIG. 9A, and intermediate lane markings included on a multi-lane road (e.g., dashed markings representing lane boundaries, solid yellow lines representing boundaries between lanes traveling in different directions, etc.) may also be represented using polynomials such as those shown in FIG. 9A.

[0200] As shown in FIG. 9A, a lane **900** may be represented using polynomials (e.g., a first order, second order, third order, or any suitable order polynomials). For illustration, lane **900** is shown as a two-dimensional lane and the polynomials are shown as two-dimensional polynomials. As depicted in FIG. 9A, lane **900** includes a left side **910** and a right side **920**. In some embodiments, more than one polynomial may be used to represent a location of each side of the road or lane boundary. For example, each of left side **910** and right side **920** may be represented by a plurality of polynomials of any suitable length. In some cases, the polynomials may have a length of about 100 m, although other lengths greater than or less than 100 m may also be used. Additionally, the polynomials can overlap with one another in order to facilitate seamless transitions in navigating based on subsequently encountered polynomials as a host vehicle travels along a roadway. For example, each of left side **910** and right side **920** may be represented by a plurality of third order polynomials separated into segments of about 100 meters in length (an example of the first predetermined range), and overlapping each other by about 50 meters. The polynomials representing the left side **910** and the right side **920** may or may not have the same order. For example, in some embodiments, some polynomials may be second order polynomials, some may be third order polynomials, and some may be fourth order polynomials.

[0201] In the example shown in FIG. 9A, left side **910** of lane **900** is represented by two groups of third order polynomials. The first group includes polynomial segments **911**, **912**, and **913**. The second group includes polynomial segments **914**, **915**, and **916**. The two groups, while substantially parallel to each other, follow the locations of their respective sides of the road. Polynomial segments **911**, **912**, **913**, **914**, **915**, and **916** have a length of about 100 meters and overlap adjacent segments in the series by about 50 meters. As noted previously, however, polynomials of different lengths and different overlap amounts may also be used. For example, the polynomials may have lengths of 500 m, 1 km, or more, and the overlap amount may vary from 0 to 50 m, 50 m to 100 m, or greater than 100 m. Additionally, while FIG. 9A is shown as representing polynomials extending in 2D space (e.g., on the surface of the paper), it is to be understood that these polynomials may represent curves extending in three dimensions (e.g., including a height component) to represent elevation changes in a road segment in addition to X-Y curvature. In the example shown in FIG. 9A, right side **920** of lane **900** is further represented by a first group having polynomial segments **921**, **922**, and **923** and a second group having polynomial segments **924**, **925**, and **926**.

[0202] Returning to the target trajectories of sparse map **800**, FIG. 9B shows a three-dimensional polynomial representing a target trajectory for a vehicle traveling along a particular road segment. The target trajectory represents not only the X-Y path that a host vehicle should travel along a particular road segment, but also the elevation change that the host vehicle will experience when traveling along the road segment. Thus, each target trajectory in sparse map **800** may be represented by one or more three-dimensional polynomials, like the three-dimensional polynomial **950** shown in FIG. 9B. Sparse map **800** may include a plurality of trajectories (e.g., millions or billions or more to represent trajectories of vehicles along various road segments along roadways throughout the world). In some embodiments, each target trajectory may correspond to a spline connecting three-dimensional polynomial segments.

[0203] Regarding the data footprint of polynomial curves stored in sparse map **800**, in some

embodiments, each third degree polynomial may be represented by four parameters, each requiring four bytes of data. Suitable representations may be obtained with third degree polynomials requiring about 192 bytes of data for every 100 m. This may translate to approximately 200 kB per hour in data usage/transfer requirements for a host vehicle traveling approximately 100 km/hr.

[0204] Sparse map **800** may describe the lanes network using a combination of geometry descriptors and meta-data. The geometry may be described by polynomials or splines as described above. The meta-data may describe the number of lanes, special characteristics (such as a car pool lane), and possibly other sparse labels. The total footprint of such indicators may be negligible.

[0205] Accordingly, a sparse map according to embodiments of the present disclosure may include at least one line representation of a road surface feature extending along the road segment, each line representation representing a path along the road segment substantially corresponding with the road surface feature. In some embodiments, as discussed above, the at least one line representation of the road surface feature may include a spline, a polynomial representation, or a curve.

Furthermore, in some embodiments, the road surface feature may include at least one of a road edge or a lane marking. Moreover, as discussed below with respect to “crowdsourcing,” the road surface feature may be identified through image analysis of a plurality of images acquired as one or more vehicles traverse the road segment.

[0206] As previously noted, sparse map **800** may include a plurality of predetermined landmarks associated with a road segment. Rather than storing actual images of the landmarks and relying, for example, on image recognition analysis based on captured images and stored images, each landmark in sparse map **800** may be represented and recognized using less data than a stored, actual image would require. Data representing landmarks may still include sufficient information for describing or identifying the landmarks along a road. Storing data describing characteristics of landmarks, rather than the actual images of landmarks, may reduce the size of sparse map **800**.

[0207] FIG. **10** illustrates examples of types of landmarks that may be represented in sparse map **800**. The landmarks may include any visible and identifiable objects along a road segment. The landmarks may be selected such that they are fixed and do not change often with respect to their locations and/or content. The landmarks included in sparse map **800** may be useful in determining a location of vehicle **200** with respect to a target trajectory as the vehicle traverses a particular road segment. Examples of landmarks may include traffic signs, directional signs, general signs (e.g., rectangular signs), roadside fixtures (e.g., lampposts, reflectors, etc.), and any other suitable category. In some embodiments, lane marks on the road, may also be included as landmarks in sparse map **800**.

[0208] Examples of landmarks shown in FIG. **10** include traffic signs, directional signs, roadside fixtures, and general signs. Traffic signs may include, for example, speed limit signs (e.g., speed limit sign **1000**), yield signs (e.g., yield sign **1005**), route number signs (e.g., route number sign **1010**), traffic light signs (e.g., traffic light sign **1015**), stop signs (e.g., stop sign **1020**). Directional signs may include a sign that includes one or more arrows indicating one or more directions to different places. For example, directional signs may include a highway sign **1025** having arrows for directing vehicles to different roads or places, an exit sign **1030** having an arrow directing vehicles off a road, etc. Accordingly, at least one of the plurality of landmarks may include a road sign.

[0209] General signs may be unrelated to traffic. For example, general signs may include billboards used for advertisement, or a welcome board adjacent a border between two countries, states, counties, cities, or towns. FIG. **10** shows a general sign **1040** (“Joe’s Restaurant”). Although general sign **1040** may have a rectangular shape, as shown in FIG. **10**, general sign **1040** may have other shapes, such as square, circle, triangle, etc.

[0210] Landmarks may also include roadside fixtures. Roadside fixtures may be objects that are not signs, and may not be related to traffic or directions. For example, roadside fixtures may include lampposts (e.g., lamppost **1035**), power line posts, traffic light posts, etc.

[0211] Landmarks may also include beacons that may be specifically designed for usage in an

autonomous vehicle navigation system. For example, such beacons may include stand-alone structures placed at predetermined intervals to aid in navigating a host vehicle. Such beacons may also include visual/graphical information added to existing road signs (e.g., icons, emblems, bar codes, etc.) that may be identified or recognized by a vehicle traveling along a road segment. Such beacons may also include electronic components. In such embodiments, electronic beacons (e.g., RFID tags, etc.) may be used to transmit non-visual information to a host vehicle. Such information may include, for example, landmark identification and/or landmark location information that a host vehicle may use in determining its position along a target trajectory.

[0212] In some embodiments, the landmarks included in sparse map **800** may be represented by a data object of a predetermined size. The data representing a landmark may include any suitable parameters for identifying a particular landmark. For example, in some embodiments, landmarks stored in sparse map **800** may include parameters such as a physical size of the landmark (e.g., to support estimation of distance to the landmark based on a known size/scale), a distance to a previous landmark, lateral offset, height, a type code (e.g., a landmark type-what type of directional sign, traffic sign, etc.), a GPS coordinate (e.g., to support global localization), and any other suitable parameters. Each parameter may be associated with a data size. For example, a landmark size may be stored using 8 bytes of data. A distance to a previous landmark, a lateral offset, and height may be specified using 12 bytes of data. A type code associated with a landmark such as a directional sign or a traffic sign may require about 2 bytes of data. For general signs, an image signature enabling identification of the general sign may be stored using 50 bytes of data storage. The landmark GPS position may be associated with 16 bytes of data storage. These data sizes for each parameter are examples only, and other data sizes may also be used. Representing landmarks in sparse map **800** in this manner may offer a lean solution for efficiently representing landmarks in the database. In some embodiments, objects may be referred to as standard semantic objects or non-standard semantic objects. A standard semantic object may include any class of object for which there's a standardized set of characteristics (e.g., speed limit signs, warning signs, directional signs, traffic lights, etc. having known dimensions or other characteristics). A non-standard semantic object may include any object that is not associated with a standardized set of characteristics (e.g., general advertising signs, signs identifying business establishments, potholes, trees, etc. that may have variable dimensions). Each non-standard semantic object may be represented with 38 bytes of data (e.g., 8 bytes for size; 12 bytes for distance to previous landmark, lateral offset, and height; 2 bytes for a type code; and 16 bytes for position coordinates). Standard semantic objects may be represented using even less data, as size information may not be needed by the mapping server to fully represent the object in the sparse map.

[0213] Sparse map **800** may use a tag system to represent landmark types. In some cases, each traffic sign or directional sign may be associated with its own tag, which may be stored in the database as part of the landmark identification. For example, the database may include on the order of 1000 different tags to represent various traffic signs and on the order of about 10000 different tags to represent directional signs. Of course, any suitable number of tags may be used, and additional tags may be created as needed. General purpose signs may be represented in some embodiments using less than about 100 bytes (e.g., about 86 bytes including 8 bytes for size; 12 bytes for distance to previous landmark, lateral offset, and height; 50 bytes for an image signature; and 16 bytes for GPS coordinates).

[0214] Thus, for semantic road signs not requiring an image signature, the data density impact to sparse map **800**, even at relatively high landmark densities of about 1 per 50 m, may be on the order of about 760 bytes per kilometer (e.g., 20 landmarks per km $\times$ 38 bytes per landmark=760 bytes). Even for general purpose signs including an image signature component, the data density impact is about 1.72 kB per km (e.g., 20 landmarks per km $\times$ 86 bytes per landmark=1,720 bytes). For semantic road signs, this equates to about 76 kB per hour of data usage for a vehicle traveling 100 km/hr. For general purpose signs, this equates to about 170 kB per hour for a vehicle traveling

100 km/hr. It should be noted that in some environments (e.g., urban environments) there may be a much higher density of detected objects available for inclusion in the sparse map (perhaps more than one per meter). In some embodiments, a generally rectangular object, such as a rectangular sign, may be represented in sparse map **800** by no more than 100 bytes of data. The representation of the generally rectangular object (e.g., general sign **1040**) in sparse map **800** may include a condensed image signature or image hash (e.g., condensed image signature **1045**) associated with the generally rectangular object. This condensed image signature/image hash may be determined using any suitable image hashing algorithm and may be used, for example, to aid in identification of a general purpose sign, for example, as a recognized landmark. Such a condensed image signature (e.g., image information derived from actual image data representing an object) may avoid a need for storage of an actual image of an object or a need for comparative image analysis performed on actual images in order to recognize landmarks.

[0215] Referring to FIG. **10**, sparse map **800** may include or store a condensed image signature **1045** associated with a general sign **1040**, rather than an actual image of general sign **1040**. For example, after an image capture device (e.g., image capture device **122**, **124**, or **126**) captures an image of general sign **1040**, a processor (e.g., image processor **190** or any other processor that can process images either aboard or remotely located relative to a host vehicle) may perform an image analysis to extract/create condensed image signature **1045** that includes a unique signature or pattern associated with general sign **1040**. In one embodiment, condensed image signature **1045** may include a shape, color pattern, a brightness pattern, or any other feature that may be extracted from the image of general sign **1040** for describing general sign **1040**.

[0216] For example, in FIG. **10**, the circles, triangles, and stars shown in condensed image signature **1045** may represent areas of different colors. The pattern represented by the circles, triangles, and stars may be stored in sparse map **800**, e.g., within the 50 bytes designated to include an image signature. Notably, the circles, triangles, and stars are not necessarily meant to indicate that such shapes are stored as part of the image signature. Rather, these shapes are meant to conceptually represent recognizable areas having discernible color differences, textual areas, graphical shapes, or other variations in characteristics that may be associated with a general purpose sign. Such condensed image signatures can be used to identify a landmark in the form of a general sign. For example, the condensed image signature can be used to perform a same-not-same analysis based on a comparison of a stored condensed image signature with image data captured, for example, using a camera onboard an autonomous vehicle.

[0217] Accordingly, the plurality of landmarks may be identified through image analysis of the plurality of images acquired as one or more vehicles traverse the road segment. As explained below with respect to “crowdsourcing,” in some embodiments, the image analysis to identify the plurality of landmarks may include accepting potential landmarks when a ratio of images in which the landmark does appear to images in which the landmark does not appear exceeds a threshold. Furthermore, in some embodiments, the image analysis to identify the plurality of landmarks may include rejecting potential landmarks when a ratio of images in which the landmark does not appear to images in which the landmark does appear exceeds a threshold.

[0218] Returning to the target trajectories a host vehicle may use to navigate a particular road segment, FIG. **11A** shows polynomial representations trajectories capturing during a process of building or maintaining sparse map **800**. A polynomial representation of a target trajectory included in sparse map **800** may be determined based on two or more reconstructed trajectories of prior traversals of vehicles along the same road segment. In some embodiments, the polynomial representation of the target trajectory included in sparse map **800** may be an aggregation of two or more reconstructed trajectories of prior traversals of vehicles along the same road segment. In some embodiments, the polynomial representation of the target trajectory included in sparse map **800** may be an average of the two or more reconstructed trajectories of prior traversals of vehicles along the same road segment. Other mathematical operations may also be used to construct a target

trajectory along a road path based on reconstructed trajectories collected from vehicles traversing along a road segment.

[0219] As shown in FIG. **11A**, a road segment **1100** may be travelled by a number of vehicles **200** at different times. Each vehicle **200** may collect data relating to a path that the vehicle took along the road segment. The path traveled by a particular vehicle may be determined based on camera data, accelerometer information, speed sensor information, and/or GPS information, among other potential sources. Such data may be used to reconstruct trajectories of vehicles traveling along the road segment, and based on these reconstructed trajectories, a target trajectory (or multiple target trajectories) may be determined for the particular road segment. Such target trajectories may represent a preferred path of a host vehicle (e.g., guided by an autonomous navigation system) as the vehicle travels along the road segment.

[0220] In the example shown in FIG. **11A**, a first reconstructed trajectory **1101** may be determined based on data received from a first vehicle traversing road segment **1100** at a first time period (e.g., day 1), a second reconstructed trajectory **1102** may be obtained from a second vehicle traversing road segment **1100** at a second time period (e.g., day 2), and a third reconstructed trajectory **1103** may be obtained from a third vehicle traversing road segment **1100** at a third time period (e.g., day 3). Each trajectory **1101**, **1102**, and **1103** may be represented by a polynomial, such as a three-dimensional polynomial. It should be noted that in some embodiments, any of the reconstructed trajectories may be assembled onboard the vehicles traversing road segment **1100**.

[0221] Additionally, or alternatively, such reconstructed trajectories may be determined on a server side based on information received from vehicles traversing road segment **1100**. For example, in some embodiments, vehicles **200** may transmit data to one or more servers relating to their motion along road segment **1100** (e.g., steering angle, heading, time, position, speed, sensed road geometry, and/or sensed landmarks, among things). The server may reconstruct trajectories for vehicles **200** based on the received data. The server may also generate a target trajectory for guiding navigation of autonomous vehicle that will travel along the same road segment **1100** at a later time based on the first, second, and third trajectories **1101**, **1102**, and **1103**. While a target trajectory may be associated with a single prior traversal of a road segment, in some embodiments, each target trajectory included in sparse map **800** may be determined based on two or more reconstructed trajectories of vehicles traversing the same road segment. In FIG. **11A**, the target trajectory is represented by **1110**. In some embodiments, the target trajectory **1110** may be generated based on an average of the first, second, and third trajectories **1101**, **1102**, and **1103**. In some embodiments, the target trajectory **1110** included in sparse map **800** may be an aggregation (e.g., a weighted combination) of two or more reconstructed trajectories.

[0222] At the mapping server, the server may receive actual trajectories for a particular road segment from multiple harvesting vehicles traversing the road segment. To generate a target trajectory for each valid path along the road segment (e.g., each lane, each drive direction, each path through a junction, etc.), the received actual trajectories may be aligned. The alignment process may include using detected objects/features identified along the road segment along with harvested positions of those detected objects/features to correlate the actual, harvested trajectories with one another. Once aligned, an average or “best fit” target trajectory for each available lane, etc. may be determined based on the aggregated, correlated/aligned actual trajectories.

[0223] FIGS. **11B** and **11C** further illustrate the concept of target trajectories associated with road segments present within a geographic region **1111**. As shown in FIG. **11B**, a first road segment **1120** within geographic region **1111** may include a multilane road, which includes two lanes **1122** designated for vehicle travel in a first direction and two additional lanes **1124** designated for vehicle travel in a second direction opposite to the first direction. Lanes **1122** and lanes **1124** may be separated by a double yellow line **1123**. Geographic region **1111** may also include a branching road segment **1130** that intersects with road segment **1120**. Road segment **1130** may include a two-lane road, each lane being designated for a different direction of travel. Geographic region **1111**

may also include other road features, such as a stop line **1132**, a stop sign **1134**, a speed limit sign **1136**, and a hazard sign **1138**.

[0224] As shown in FIG. **11C**, sparse map **800** may include a local map **1140** including a road model for assisting with autonomous navigation of vehicles within geographic region **1111**. For example, local map **1140** may include target trajectories for one or more lanes associated with road segments **1120** and/or **1130** within geographic region **1111**. For example, local map **1140** may include target trajectories **1141** and/or **1142** that an autonomous vehicle may access or rely upon when traversing lanes **1122**. Similarly, local map **1140** may include target trajectories **1143** and/or **1144** that an autonomous vehicle may access or rely upon when traversing lanes **1124**. Further, local map **1140** may include target trajectories **1145** and/or **1146** that an autonomous vehicle may access or rely upon when traversing road segment **1130**. Target trajectory **1147** represents a preferred path an autonomous vehicle should follow when transitioning from lanes **1120** (and specifically, relative to target trajectory **1141** associated with a right-most lane of lanes **1120**) to road segment **1130** (and specifically, relative to a target trajectory **1145** associated with a first side of road segment **1130**). Similarly, target trajectory **1148** represents a preferred path an autonomous vehicle should follow when transitioning from road segment **1130** (and specifically, relative to target trajectory **1146**) to a portion of road segment **1124** (and specifically, as shown, relative to a target trajectory **1143** associated with a left lane of lanes **1124**).

[0225] Sparse map **800** may also include representations of other road-related features associated with geographic region **1111**. For example, sparse map **800** may also include representations of one or more landmarks identified in geographic region **1111**. Such landmarks may include a first landmark **1150** associated with stop line **1132**, a second landmark **1152** associated with stop sign **1134**, a third landmark associated with speed limit sign **1154**, and a fourth landmark **1156** associated with hazard sign **1138**. Such landmarks may be used, for example, to assist an autonomous vehicle in determining its current location relative to any of the shown target trajectories, such that the vehicle may adjust its heading to match a direction of the target trajectory at the determined location.

[0226] In some embodiments, sparse map **800** may also include road signature profiles. Such road signature profiles may be associated with any discernible/measurable variation in at least one parameter associated with a road. For example, in some cases, such profiles may be associated with variations in road surface information such as variations in surface roughness of a particular road segment, variations in road width over a particular road segment, variations in distances between dashed lines painted along a particular road segment, variations in road curvature along a particular road segment, etc. FIG. **11D** shows an example of a road signature profile **1160**. While profile **1160** may represent any of the parameters mentioned above, or others, in one example, profile **1160** may represent a measure of road surface roughness, as obtained, for example, by monitoring one or more sensors providing outputs indicative of an amount of suspension displacement as a vehicle travels a particular road segment.

[0227] Alternatively or concurrently, profile **1160** may represent variation in road width, as determined based on image data obtained via a camera onboard a vehicle traveling a particular road segment. Such profiles may be useful, for example, in determining a particular location of an autonomous vehicle relative to a particular target trajectory. That is, as it traverses a road segment, an autonomous vehicle may measure a profile associated with one or more parameters associated with the road segment. If the measured profile can be correlated/matched with a predetermined profile that plots the parameter variation with respect to position along the road segment, then the measured and predetermined profiles may be used (e.g., by overlaying corresponding sections of the measured and predetermined profiles) in order to determine a current position along the road segment and, therefore, a current position relative to a target trajectory for the road segment.

[0228] In some embodiments, sparse map **800** may include different trajectories based on different characteristics associated with a user of autonomous vehicles, environmental conditions, and/or

other parameters relating to driving. For example, in some embodiments, different trajectories may be generated based on different user preferences and/or profiles. Sparse map **800** including such different trajectories may be provided to different autonomous vehicles of different users. For example, some users may prefer to avoid toll roads, while others may prefer to take the shortest or fastest routes, regardless of whether there is a toll road on the route. The disclosed systems may generate different sparse maps with different trajectories based on such different user preferences or profiles. As another example, some users may prefer to travel in a fast moving lane, while others may prefer to maintain a position in the central lane at all times.

[0229] Different trajectories may be generated and included in sparse map **800** based on different environmental conditions, such as day and night, snow, rain, fog, etc. Autonomous vehicles driving under different environmental conditions may be provided with sparse map **800** generated based on such different environmental conditions. In some embodiments, cameras provided on autonomous vehicles may detect the environmental conditions, and may provide such information back to a server that generates and provides sparse maps. For example, the server may generate or update an already generated sparse map **800** to include trajectories that may be more suitable or safer for autonomous driving under the detected environmental conditions. The update of sparse map **800** based on environmental conditions may be performed dynamically as the autonomous vehicles are traveling along roads.

[0230] Other different parameters relating to driving may also be used as a basis for generating and providing different sparse maps to different autonomous vehicles. For example, when an autonomous vehicle is traveling at a high speed, turns may be tighter. Trajectories associated with specific lanes, rather than roads, may be included in sparse map **800** such that the autonomous vehicle may maintain within a specific lane as the vehicle follows a specific trajectory. When an image captured by a camera onboard the autonomous vehicle indicates that the vehicle has drifted outside of the lane (e.g., crossed the lane mark), an action may be triggered within the vehicle to bring the vehicle back to the designated lane according to the specific trajectory.

#### Crowdsourcing a Sparse Map

[0231] The disclosed sparse maps may be efficiently (and passively) generated through the power of crowdsourcing. For example, any private or commercial vehicle equipped with a camera (e.g., a simple, low resolution camera regularly included as OEM equipment on today's vehicles) and an appropriate image analysis processor can serve as a harvesting vehicle. No special equipment (e.g., high definition imaging and/or positioning systems) are required. As a result of the disclosed crowdsourcing technique, the generated sparse maps may be extremely accurate and may include extremely refined position information (enabling navigation error limits of 10 cm or less) without requiring any specialized imaging or sensing equipment as input to the map generation process. Crowdsourcing also enables much more rapid (and inexpensive) updates to the generated maps, as new drive information is continuously available to the mapping server system from any roads traversed by private or commercial vehicles minimally equipped to also serve as harvesting vehicles. There is no need for designated vehicles equipped with high-definition imaging and mapping sensors. Therefore, the expense associated with building such specialized vehicles can be avoided. Further, updates to the presently disclosed sparse maps may be made much more rapidly than systems that rely upon dedicated, specialized mapping vehicles (which by virtue of their expense and special equipment are typically limited to a fleet of specialized vehicles of far lower numbers than the number of private or commercial vehicles already available for performing the disclosed harvesting techniques).

[0232] The disclosed sparse maps generated through crowdsourcing may be extremely accurate because they may be generated based on many inputs from multiple (**10s**, hundreds, millions, etc.) of harvesting vehicles that have collected drive information along a particular road segment. For example, every harvesting vehicle that drives along a particular road segment may record its actual trajectory and may determine position information relative to detected objects/features along the

road segment. This information is passed along from multiple harvesting vehicles to a server. The actual trajectories are aggregated to generate a refined, target trajectory for each valid drive path along the road segment. Additionally, the position information collected from the multiple harvesting vehicles for each of the detected objects/features along the road segment (semantic or non-semantic) can also be aggregated. As a result, the mapped position of each detected object/feature may constitute an average of hundreds, thousands, or millions of individually determined positions for each detected object/feature. Such a technique may yield extremely accurate mapped positions for the detected objects/features.

[0233] In some embodiments, the disclosed systems and methods may generate a sparse map for autonomous vehicle navigation. For example, disclosed systems and methods may use crowdsourced data for generation of a sparse map that one or more autonomous vehicles may use to navigate along a system of roads. As used herein, “crowdsourcing” means that data are received from various vehicles (e.g., autonomous vehicles) travelling on a road segment at different times, and such data are used to generate and/or update the road model, including sparse map tiles. The model or any of its sparse map tiles may, in turn, be transmitted to the vehicles or other vehicles later travelling along the road segment for assisting autonomous vehicle navigation. The road model may include a plurality of target trajectories representing preferred trajectories that autonomous vehicles should follow as they traverse a road segment. The target trajectories may be the same as a reconstructed actual trajectory collected from a vehicle traversing a road segment, which may be transmitted from the vehicle to a server. In some embodiments, the target trajectories may be different from actual trajectories that one or more vehicles previously took when traversing a road segment. The target trajectories may be generated based on actual trajectories (e.g., through averaging or any other suitable operation).

[0234] The vehicle trajectory data that a vehicle may upload to a server may correspond with the actual reconstructed trajectory for the vehicle or may correspond to a recommended trajectory, which may be based on or related to the actual reconstructed trajectory of the vehicle, but may differ from the actual reconstructed trajectory. For example, vehicles may modify their actual, reconstructed trajectories and submit (e.g., recommend) to the server the modified actual trajectories. The road model may use the recommended, modified trajectories as target trajectories for autonomous navigation of other vehicles.

[0235] In addition to trajectory information, other information for potential use in building a sparse data map **800** may include information relating to potential landmark candidates. For example, through crowd sourcing of information, the disclosed systems and methods may identify potential landmarks in an environment and refine landmark positions. The landmarks may be used by a navigation system of autonomous vehicles to determine and/or adjust the position of the vehicle along the target trajectories.

[0236] The reconstructed trajectories that a vehicle may generate as the vehicle travels along a road may be obtained by any suitable method. In some embodiments, the reconstructed trajectories may be developed by stitching together segments of motion for the vehicle, using, e.g., ego motion estimation (e.g., three dimensional translation and three dimensional rotation of the camera, and hence the body of the vehicle). The rotation and translation estimation may be determined based on analysis of images captured by one or more image capture devices along with information from other sensors or devices, such as inertial sensors and speed sensors. For example, the inertial sensors may include an accelerometer or other suitable sensors configured to measure changes in translation and/or rotation of the vehicle body. The vehicle may include a speed sensor that measures a speed of the vehicle.

[0237] In some embodiments, the ego motion of the camera (and hence the vehicle body) may be estimated based on an optical flow analysis of the captured images. An optical flow analysis of a sequence of images identifies movement of pixels from the sequence of images, and based on the identified movement, determines motions of the vehicle. The ego motion may be integrated over



time and along the road segment to reconstruct a trajectory associated with the road segment that the vehicle has followed.

[0238] Data (e.g., reconstructed trajectories) collected by multiple vehicles in multiple drives along a road segment at different times may be used to construct the road model (e.g., including the target trajectories, etc.) included in sparse data map **800**. Data collected by multiple vehicles in multiple drives along a road segment at different times may also be averaged to increase an accuracy of the model. In some embodiments, data regarding the road geometry and/or landmarks may be received from multiple vehicles that travel through the common road segment at different times. Such data received from different vehicles may be combined to generate the road model and/or to update the road model.

[0239] The geometry of a reconstructed trajectory (and also a target trajectory) along a road segment may be represented by a curve in three dimensional space, which may be a spline connecting three dimensional polynomials. The reconstructed trajectory curve may be determined from analysis of a video stream or a plurality of images captured by a camera installed on the vehicle. In some embodiments, a location is identified in each frame or image that is a few meters ahead of the current position of the vehicle. This location is where the vehicle is expected to travel to in a predetermined time period. This operation may be repeated frame by frame, and at the same time, the vehicle may compute the camera's ego motion (rotation and translation). At each frame or image, a short range model for the desired path is generated by the vehicle in a reference frame that is attached to the camera. The short range models may be stitched together to obtain a three dimensional model of the road in some coordinate frame, which may be an arbitrary or predetermined coordinate frame. The three dimensional model of the road may then be fitted by a spline, which may include or connect one or more polynomials of suitable orders.

[0240] To conclude the short range road model at each frame, one or more detection modules may be used. For example, a bottom-up lane detection module may be used. The bottom-up lane detection module may be useful when lane marks are drawn on the road. This module may look for edges in the image and assembles them together to form the lane marks. A second module may be used together with the bottom-up lane detection module. The second module is an end-to-end deep neural network, which may be trained to predict the correct short range path from an input image. In both modules, the road model may be detected in the image coordinate frame and transformed to a three dimensional space that may be virtually attached to the camera.

[0241] Although the reconstructed trajectory modeling method may introduce an accumulation of errors due to the integration of ego motion over a long period of time, which may include a noise component, such errors may be inconsequential as the generated model may provide sufficient accuracy for navigation over a local scale. In addition, it is possible to cancel the integrated error by using external sources of information, such as satellite images or geodetic measurements. For example, the disclosed systems and methods may use a GNSS receiver to cancel accumulated errors. However, the GNSS positioning signals may not be always available and accurate. The disclosed systems and methods may enable a steering application that depends weakly on the availability and accuracy of GNSS positioning. In such systems, the usage of the GNSS signals may be limited. For example, in some embodiments, the disclosed systems may use the GNSS signals for database indexing purposes only.

[0242] In some embodiments, the range scale (e.g., local scale) that may be relevant for an autonomous vehicle navigation steering application may be on the order of 50 meters, 100 meters, 200 meters, 300 meters, etc. Such distances may be used, as the geometrical road model is mainly used for two purposes: planning the trajectory ahead and localizing the vehicle on the road model. In some embodiments, the planning task may use the model over a typical range of 40 meters ahead (or any other suitable distance ahead, such as 20 meters, 30 meters, 50 meters), when the control algorithm steers the vehicle according to a target point located 1.3 seconds ahead (or any other time such as 1.5 seconds, 1.7 seconds, 2 seconds, etc.). The localization task uses the road model over a

typical range of 60 meters behind the car (or any other suitable distances, such as 50 meters, 100 meters, 150 meters, etc.), according to a method called “tail alignment” described in more detail in another section. The disclosed systems and methods may generate a geometrical model that has sufficient accuracy over particular range, such as 100 meters, such that a planned trajectory will not deviate by more than, for example, 30 cm from the lane center.

[0243] As explained above, a three dimensional road model may be constructed from detecting short range sections and stitching them together. The stitching may be enabled by computing a six degree ego motion model, using the videos and/or images captured by the camera, data from the inertial sensors that reflect the motions of the vehicle, and the host vehicle velocity signal. The accumulated error may be small enough over some local range scale, such as of the order of 100 meters. All this may be completed in a single drive over a particular road segment.

[0244] In some embodiments, multiple drives may be used to average the resulted model, and to increase its accuracy further. The same car may travel the same route multiple times, or multiple cars may send their collected model data to a central server. In any case, a matching procedure may be performed to identify overlapping models and to enable averaging in order to generate target trajectories. The constructed model (e.g., including the target trajectories) may be used for steering once a convergence criterion is met. Subsequent drives may be used for further model improvements and in order to accommodate infrastructure changes.

[0245] Sharing of driving experience (such as sensed data) between multiple cars becomes feasible if they are connected to a central server. Each vehicle client may store a partial copy of a universal road model, which may be relevant for its current position. A bidirectional update procedure between the vehicles and the server may be performed by the vehicles and the server. The small footprint concept discussed above enables the disclosed systems and methods to perform the bidirectional updates using a very small bandwidth.

[0246] Information relating to potential landmarks may also be determined and forwarded to a central server. For example, the disclosed systems and methods may determine one or more physical properties of a potential landmark based on one or more images that include the landmark. The physical properties may include a physical size (e.g., height, width) of the landmark, a distance from a vehicle to a landmark, a distance between the landmark to a previous landmark, the lateral position of the landmark (e.g., the position of the landmark relative to the lane of travel), the GPS coordinates of the landmark, a type of landmark, identification of text on the landmark, etc. For example, a vehicle may analyze one or more images captured by a camera to detect a potential landmark, such as a speed limit sign.

[0247] The vehicle may determine a distance from the vehicle to the landmark or a position associated with the landmark (e.g., any semantic or non-semantic object or feature along a road segment) based on the analysis of the one or more images. In some embodiments, the distance may be determined based on analysis of images of the landmark using a suitable image analysis method, such as a scaling method and/or an optical flow method. As previously noted, a position of the object/feature may include a 2D image position (e.g., an X-Y pixel position in one or more captured images) of one or more points associated with the object/feature or may include a 3D real-world position of one or more points (e.g., determined through structure in motion/optical flow techniques, LIDAR or RADAR information, etc.). In some embodiments, the disclosed systems and methods may be configured to determine a type or classification of a potential landmark. In case the vehicle determines that a certain potential landmark corresponds to a predetermined type or classification stored in a sparse map, it may be sufficient for the vehicle to communicate to the server an indication of the type or classification of the landmark, along with its location. The server may store such indications. At a later time, during navigation, a navigating vehicle may capture an image that includes a representation of the landmark, process the image (e.g., using a classifier), and compare the result landmark in order to confirm detection of the mapped landmark and to use the mapped landmark in localizing the navigating vehicle relative to the sparse map.

[0248] In some embodiments, multiple autonomous vehicles travelling on a road segment may communicate with a server. The vehicles (or clients) may generate a curve describing its drive (e.g., through ego motion integration) in an arbitrary coordinate frame. The vehicles may detect landmarks and locate them in the same frame. The vehicles may upload the curve and the landmarks to the server. The server may collect data from vehicles over multiple drives, and generate a unified road model. For example, as discussed below with respect to FIG. 19, the server may generate a sparse map having the unified road model using the uploaded curves and landmarks.

[0249] The server may also distribute the model to clients (e.g., vehicles). For example, the server may distribute the sparse map to one or more vehicles. The server may continuously or periodically update the model when receiving new data from the vehicles. For example, the server may process the new data to evaluate whether the data includes information that should trigger an update, or creation of new data on the server. The server may distribute the updated model or the updates to the vehicles for providing autonomous vehicle navigation.

[0250] The server may use one or more criteria for determining whether new data received from the vehicles should trigger an update to the model or trigger creation of new data. For example, when the new data indicates that a previously recognized landmark at a specific location no longer exists, or is replaced by another landmark, the server may determine that the new data should trigger an update to the model. As another example, when the new data indicates that a road segment has been closed, and when this has been corroborated by data received from other vehicles, the server may determine that the new data should trigger an update to the model.

[0251] The server may distribute the updated model (or the updated portion of the model) to one or more vehicles that are traveling on the road segment, with which the updates to the model are associated. The server may also distribute the updated model to vehicles that are about to travel on the road segment, or vehicles whose planned trip includes the road segment, with which the updates to the model are associated. For example, while an autonomous vehicle is traveling along another road segment before reaching the road segment with which an update is associated, the server may distribute the updates or updated model to the autonomous vehicle before the vehicle reaches the road segment.

[0252] In some embodiments, the remote server may collect trajectories and landmarks from multiple clients (e.g., vehicles that travel along a common road segment). The server may match curves using landmarks and create an average road model based on the trajectories collected from the multiple vehicles. The server may also compute a graph of roads and the most probable path at each node or conjunction of the road segment. For example, the remote server may align the trajectories to generate a crowdsourced sparse map from the collected trajectories.

[0253] The server may average landmark properties received from multiple vehicles that travelled along the common road segment, such as the distances between one landmark to another (e.g., a previous one along the road segment) as measured by multiple vehicles, to determine an arc-length parameter and support localization along the path and speed calibration for each client vehicle. The server may average the physical dimensions of a landmark measured by multiple vehicles travelled along the common road segment and recognized the same landmark. The averaged physical dimensions may be used to support distance estimation, such as the distance from the vehicle to the landmark. The server may average lateral positions of a landmark (e.g., position from the lane in which vehicles are travelling in to the landmark) measured by multiple vehicles travelled along the common road segment and recognized the same landmark. The averaged lateral portion may be used to support lane assignment. The server may average the GPS coordinates of the landmark measured by multiple vehicles travelled along the same road segment and recognized the same landmark. The averaged GPS coordinates of the landmark may be used to support global localization or positioning of the landmark in the road model.

[0254] In some embodiments, the server may identify model changes, such as constructions,

detours, new signs, removal of signs, etc., based on data received from the vehicles. The server may continuously or periodically or instantaneously update the model upon receiving new data from the vehicles. The server may distribute updates to the model or the updated model to vehicles for providing autonomous navigation. For example, as discussed further below, the server may use crowdsourced data to filter out “ghost” landmarks detected by vehicles.

[0255] In some embodiments, the server may analyze driver interventions during the autonomous driving. The server may analyze data received from the vehicle at the time and location where intervention occurs, and/or data received prior to the time the intervention occurred. The server may identify certain portions of the data that caused or are closely related to the intervention, for example, data indicating a temporary lane closure setup, data indicating a pedestrian in the road. The server may update the model based on the identified data. For example, the server may modify one or more trajectories stored in the model.

[0256] FIG. 12 is a schematic illustration of a system that uses crowdsourcing to generate a sparse map (as well as distribute and navigate using a crowdsourced sparse map). FIG. 12 shows a road segment 1200 that includes one or more lanes. A plurality of vehicles 1205, 1210, 1215, 1220, and 1225 may travel on road segment 1200 at the same time or at different times (although shown as appearing on road segment 1200 at the same time in FIG. 12). At least one of vehicles 1205, 1210, 1215, 1220, and 1225 may be an autonomous vehicle. For simplicity of the present example, all of the vehicles 1205, 1210, 1215, 1220, and 1225 are presumed to be autonomous vehicles.

[0257] Each vehicle may be similar to vehicles disclosed in other embodiments (e.g., vehicle 200), and may include components or devices included in or associated with vehicles disclosed in other embodiments. Each vehicle may be equipped with an image capture device or camera (e.g., image capture device 122 or camera 122). Each vehicle may communicate with a remote server 1230 via one or more networks (e.g., over a cellular network and/or the Internet, etc.) through wireless communication paths 1235, as indicated by the dashed lines. Each vehicle may transmit data to server 1230 and receive data from server 1230. For example, server 1230 may collect data from multiple vehicles travelling on the road segment 1200 at different times, and may process the collected data to generate an autonomous vehicle road navigation model, or an update to the model. Server 1230 may transmit the autonomous vehicle road navigation model or the update to the model to the vehicles that transmitted data to server 1230. Server 1230 may transmit the autonomous vehicle road navigation model or the update to the model to other vehicles that travel on road segment 1200 at later times.

[0258] As vehicles 1205, 1210, 1215, 1220, and 1225 travel on road segment 1200, navigation information collected (e.g., detected, sensed, or measured) by vehicles 1205, 1210, 1215, 1220, and 1225 may be transmitted to server 1230. In some embodiments, the navigation information may be associated with the common road segment 1200. The navigation information may include a trajectory associated with each of the vehicles 1205, 1210, 1215, 1220, and 1225 as each vehicle travels over road segment 1200. In some embodiments, the trajectory may be reconstructed based on data sensed by various sensors and devices provided on vehicle 1205. For example, the trajectory may be reconstructed based on at least one of accelerometer data, speed data, landmarks data, road geometry or profile data, vehicle positioning data, and ego motion data. In some embodiments, the trajectory may be reconstructed based on data from inertial sensors, such as accelerometer, and the velocity of vehicle 1205 sensed by a speed sensor. In addition, in some embodiments, the trajectory may be determined (e.g., by a processor onboard each of vehicles 1205, 1210, 1215, 1220, and 1225) based on sensed ego motion of the camera, which may indicate three dimensional translation and/or three dimensional rotations (or rotational motions). The ego motion of the camera (and hence the vehicle body) may be determined from analysis of one or more images captured by the camera.

[0259] In some embodiments, the trajectory of vehicle 1205 may be determined by a processor provided aboard vehicle 1205 and transmitted to server 1230. In other embodiments, server 1230

may receive data sensed by the various sensors and devices provided in vehicle **1205**, and determine the trajectory based on the data received from vehicle **1205**.

[0260] In some embodiments, the navigation information transmitted from vehicles **1205**, **1210**, **1215**, **1220**, and **1225** to server **1230** may include data regarding the road surface, the road geometry, or the road profile. The geometry of road segment **1200** may include lane structure and/or landmarks. The lane structure may include the total number of lanes of road segment **1200**, the type of lanes (e.g., one-way lane, two-way lane, driving lane, passing lane, etc.), markings on lanes, width of lanes, etc. In some embodiments, the navigation information may include a lane assignment, e.g., which lane of a plurality of lanes a vehicle is traveling in. For example, the lane assignment may be associated with a numerical value “3” indicating that the vehicle is traveling on the third lane from the left or right. As another example, the lane assignment may be associated with a text value “center lane” indicating the vehicle is traveling on the center lane.

[0261] Server **1230** may store the navigation information on a non-transitory computer-readable medium, such as a hard drive, a compact disc, a tape, a memory, etc. Server **1230** may generate (e.g., through a processor included in server **1230**) at least a portion of an autonomous vehicle road navigation model for the common road segment **1200** based on the navigation information received from the plurality of vehicles **1205**, **1210**, **1215**, **1220**, and **1225** and may store the model as a portion of a sparse map. Server **1230** may determine a trajectory associated with each lane based on crowdsourced data (e.g., navigation information) received from multiple vehicles (e.g., **1205**, **1210**, **1215**, **1220**, and **1225**) that travel on a lane of road segment at different times. Server **1230** may generate the autonomous vehicle road navigation model or a portion of the model (e.g., an updated portion) based on a plurality of trajectories determined based on the crowd sourced navigation data. Server **1230** may transmit the model or the updated portion of the model to one or more of autonomous vehicles **1205**, **1210**, **1215**, **1220**, and **1225** traveling on road segment **1200** or any other autonomous vehicles that travel on road segment at a later time for updating an existing autonomous vehicle road navigation model provided in a navigation system of the vehicles. The autonomous vehicle road navigation model may be used by the autonomous vehicles in autonomously navigating along the common road segment **1200**.

[0262] As explained above, the autonomous vehicle road navigation model may be included in a sparse map (e.g., sparse map **800** depicted in FIG. **8**). Sparse map **800** may include sparse recording of data related to road geometry and/or landmarks along a road, which may provide sufficient information for guiding autonomous navigation of an autonomous vehicle, yet does not require excessive data storage. In some embodiments, the autonomous vehicle road navigation model may be stored separately from sparse map **800**, and may use map data from sparse map **800** when the model is executed for navigation. In some embodiments, the autonomous vehicle road navigation model may use map data included in sparse map **800** for determining target trajectories along road segment **1200** for guiding autonomous navigation of autonomous vehicles **1205**, **1210**, **1215**, **1220**, and **1225** or other vehicles that later travel along road segment **1200**. For example, when the autonomous vehicle road navigation model is executed by a processor included in a navigation system of vehicle **1205**, the model may cause the processor to compare the trajectories determined based on the navigation information received from vehicle **1205** with predetermined trajectories included in sparse map **800** to validate and/or correct the current traveling course of vehicle **1205**.

[0263] In the autonomous vehicle road navigation model, the geometry of a road feature or target trajectory may be encoded by a curve in a three-dimensional space. In one embodiment, the curve may be a three dimensional spline including one or more connecting three dimensional polynomials. As one of skill in the art would understand, a spline may be a numerical function that is piece-wise defined by a series of polynomials for fitting data. A spline for fitting the three dimensional geometry data of the road may include a linear spline (first order), a quadratic spline (second order), a cubic spline (third order), or any other splines (other orders), or a combination

thereof. The spline may include one or more three dimensional polynomials of different orders connecting (e.g., fitting) data points of the three dimensional geometry data of the road. In some embodiments, the autonomous vehicle road navigation model may include a three dimensional spline corresponding to a target trajectory along a common road segment (e.g., road segment **1200**) or a lane of the road segment **1200**.

[0264] As explained above, the autonomous vehicle road navigation model included in the sparse map may include other information, such as identification of at least one landmark along road segment **1200**. The landmark may be visible within a field of view of a camera (e.g., camera **122**) installed on each of vehicles **1205**, **1210**, **1215**, **1220**, and **1225**. In some embodiments, camera **122** may capture an image of a landmark. A processor (e.g., processor **180**, **190**, or processing unit **110**) provided on vehicle **1205** may process the image of the landmark to extract identification information for the landmark. The landmark identification information, rather than an actual image of the landmark, may be stored in sparse map **800**. The landmark identification information may require much less storage space than an actual image. Other sensors or systems (e.g., GPS system) may also provide certain identification information of the landmark (e.g., position of landmark). The landmark may include at least one of a traffic sign, an arrow marking, a lane marking, a dashed lane marking, a traffic light, a stop line, a directional sign (e.g., a highway exit sign with an arrow indicating a direction, a highway sign with arrows pointing to different directions or places), a landmark beacon, or a lamppost. A landmark beacon refers to a device (e.g., an RFID device) installed along a road segment that transmits or reflects a signal to a receiver installed on a vehicle, such that when the vehicle passes by the device, the beacon received by the vehicle and the location of the device (e.g., determined from GPS location of the device) may be used as a landmark to be included in the autonomous vehicle road navigation model and/or the sparse map **800**.

[0265] The identification of at least one landmark may include a position of the at least one landmark. The position of the landmark may be determined based on position measurements performed using sensor systems (e.g., Global Positioning Systems, inertial based positioning systems, landmark beacon, etc.) associated with the plurality of vehicles **1205**, **1210**, **1215**, **1220**, and **1225**. In some embodiments, the position of the landmark may be determined by averaging the position measurements detected, collected, or received by sensor systems on different vehicles **1205**, **1210**, **1215**, **1220**, and **1225** through multiple drives. For example, vehicles **1205**, **1210**, **1215**, **1220**, and **1225** may transmit position measurements data to server **1230**, which may average the position measurements and use the averaged position measurement as the position of the landmark. The position of the landmark may be continuously refined by measurements received from vehicles in subsequent drives.

[0266] The identification of the landmark may include a size of the landmark. The processor provided on a vehicle (e.g., **1205**) may estimate the physical size of the landmark based on the analysis of the images. Server **1230** may receive multiple estimates of the physical size of the same landmark from different vehicles over different drives. Server **1230** may average the different estimates to arrive at a physical size for the landmark, and store that landmark size in the road model. The physical size estimate may be used to further determine or estimate a distance from the vehicle to the landmark. The distance to the landmark may be estimated based on the current speed of the vehicle and a scale of expansion based on the position of the landmark appearing in the images relative to the focus of expansion of the camera. For example, the distance to landmark may be estimated by  $Z=V*dt*R/D$ , where  $V$  is the speed of vehicle,  $R$  is the distance in the image from the landmark at time  $t1$  to the focus of expansion, and  $D$  is the change in distance for the landmark in the image from  $t1$  to  $t2$ .  $dt$  represents the  $(t2-t1)$ . For example, the distance to landmark may be estimated by  $Z=V*dt*R/D$ , where  $V$  is the speed of vehicle,  $R$  is the distance in the image between the landmark and the focus of expansion,  $dt$  is a time interval, and  $D$  is the image displacement of the landmark along the epipolar line. Other equations equivalent to the above equation, such as  $Z=V*\omega/\Delta\omega$ , may be used for estimating the distance to the landmark. Here,  $V$  is the vehicle speed,

$\omega$  is an image length (like the object width), and  $\Delta\omega$  is the change of that image length in a unit of time.

[0267] When the physical size of the landmark is known, the distance to the landmark may also be determined based on the following equation:  $Z=f*W/\omega$ , where  $f$  is the focal length,  $W$  is the size of the landmark (e.g., height or width),  $\omega$  is the number of pixels when the landmark leaves the image. From the above equation, a change in distance  $Z$  may be calculated using  $\Delta Z=f*W*\Delta\omega/\omega^2+f*\Delta W/\omega$ , where  $\Delta W$  decays to zero by averaging, and where  $\Delta\omega$  is the number of pixels representing a bounding box accuracy in the image. A value estimating the physical size of the landmark may be calculated by averaging multiple observations at the server side. The resulting error in distance estimation may be very small. There are two sources of error that may occur when using the formula above, namely  $\Delta W$  and  $\Delta\omega$ . Their contribution to the distance error is given by  $\Delta Z=f*W*\Delta\omega/\omega^2+f*\Delta W/\omega$ . However,  $\Delta W$  decays to zero by averaging; hence  $\Delta Z$  is determined by  $\Delta\omega$  (e.g., the inaccuracy of the bounding box in the image).

[0268] For landmarks of unknown dimensions, the distance to the landmark may be estimated by tracking feature points on the landmark between successive frames. For example, certain features appearing on a speed limit sign may be tracked between two or more image frames. Based on these tracked features, a distance distribution per feature point may be generated. The distance estimate may be extracted from the distance distribution. For example, the most frequent distance appearing in the distance distribution may be used as the distance estimate. As another example, the average of the distance distribution may be used as the distance estimate.

[0269] FIG. 13 illustrates an example autonomous vehicle road navigation model represented by a plurality of three dimensional splines **1301**, **1302**, and **1303**. The curves **1301**, **1302**, and **1303** shown in FIG. 13 are for illustration purpose only. Each spline may include one or more three dimensional polynomials connecting a plurality of data points **1310**. Each polynomial may be a first order polynomial, a second order polynomial, a third order polynomial, or a combination of any suitable polynomials having different orders. Each data point **1310** may be associated with the navigation information received from vehicles **1205**, **1210**, **1215**, **1220**, and **1225**. In some embodiments, each data point **1310** may be associated with data related to landmarks (e.g., size, location, and identification information of landmarks) and/or road signature profiles (e.g., road geometry, road roughness profile, road curvature profile, road width profile). In some embodiments, some data points **1310** may be associated with data related to landmarks, and others may be associated with data related to road signature profiles.

[0270] FIG. 14 illustrates raw location data **1410** (e.g., GPS data) received from five separate drives. One drive may be separate from another drive if it was traversed by separate vehicles at the same time, by the same vehicle at separate times, or by separate vehicles at separate times. To account for errors in the location data **1410** and for differing locations of vehicles within the same lane (e.g., one vehicle may drive closer to the left of a lane than another), server **1230** may generate a map skeleton **1420** using one or more statistical techniques to determine whether variations in the raw location data **1410** represent actual divergences or statistical errors. Each path within skeleton **1420** may be linked back to the raw data **1410** that formed the path. For example, the path between A and B within skeleton **1420** is linked to raw data **1410** from drives 2, 3, 4, and 5 but not from drive 1. Skeleton **1420** may not be detailed enough to be used to navigate a vehicle (e.g., because it combines drives from multiple lanes on the same road unlike the splines described above) but may provide useful topological information and may be used to define intersections.

[0271] FIG. 15 illustrates an example by which additional detail may be generated for a sparse map within a segment of a map skeleton (e.g., segment A to B within skeleton **1420**). As depicted in FIG. 15, the data (e.g., ego-motion data, road markings data, and the like) may be shown as a function of position  $S$  (or  $S_1$  or  $S_2$ ) along the drive. Server **1230** may identify landmarks for the sparse map by identifying unique matches between landmarks **1501**, **1503**, and **1505** of drive **1510** and landmarks **1507** and **1509** of drive **1520**. Such a matching algorithm may result in

identification of landmarks **1511**, **1513**, and **1515**. One skilled in the art would recognize, however, that other matching algorithms may be used. For example, probability optimization may be used in lieu of or in combination with unique matching. Server **1230** may longitudinally align the drives to align the matched landmarks. For example, server **1230** may select one drive (e.g., drive **1520**) as a reference drive and then shift and/or elastically stretch the other drive(s) (e.g., drive **1510**) for alignment.

[0272] FIG. **16** shows an example of aligned landmark data for use in a sparse map. In the example of FIG. **16**, landmark **1610** comprises a road sign. The example of FIG. **16** further depicts data from a plurality of drives **1601**, **1603**, **1605**, **1607**, **1609**, **1611**, and **1613**. In the example of FIG. **16**, the data from drive **1613** consists of a “ghost” landmark, and the server **1230** may identify it as such because none of drives **1601**, **1603**, **1605**, **1607**, **1609**, and **1611** include an identification of a landmark in the vicinity of the identified landmark in drive **1613**. Accordingly, server **1230** may accept potential landmarks when a ratio of images in which the landmark does appear to images in which the landmark does not appear exceeds a threshold and/or may reject potential landmarks when a ratio of images in which the landmark does not appear to images in which the landmark does appear exceeds a threshold.

[0273] FIG. **17** depicts a system **1700** for generating drive data, which may be used to crowdsource a sparse map. As depicted in FIG. **17**, system **1700** may include a camera **1701** and a locating device **1703** (e.g., a GPS locator). Camera **1701** and locating device **1703** may be mounted on a vehicle (e.g., one of vehicles **1205**, **1210**, **1215**, **1220**, and **1225**). Camera **1701** may produce a plurality of data of multiple types, e.g., ego motion data, traffic sign data, road data, or the like. The camera data and location data may be segmented into drive segments **1705**. For example, drive segments **1705** may each have camera data and location data from less than 1 km of driving.

[0274] In some embodiments, system **1700** may remove redundancies in drive segments **1705**. For example, if a landmark appears in multiple images from camera **1701**, system **1700** may strip the redundant data such that the drive segments **1705** only contain one copy of the location of and any metadata relating to the landmark. By way of further example, if a lane marking appears in multiple images from camera **1701**, system **1700** may strip the redundant data such that the drive segments **1705** only contain one copy of the location of and any metadata relating to the lane marking.

[0275] System **1700** also includes a server (e.g., server **1230**). Server **1230** may receive drive segments **1705** from the vehicle and recombine the drive segments **1705** into a single drive **1707**. Such an arrangement may allow for reduced bandwidth requirements when transferring data between the vehicle and the server while also allowing for the server to store data relating to an entire drive.

[0276] FIG. **18** depicts system **1700** of FIG. **17** further configured for crowdsourcing a sparse map. As in FIG. **17**, system **1700** includes vehicle **1810**, which captures drive data using, for example, a camera (which produces, e.g., ego motion data, traffic sign data, road data, or the like) and a locating device (e.g., a GPS locator). As in FIG. **17**, vehicle **1810** segments the collected data into drive segments (depicted as “DS1 1,” “DS2 1,” “DSN 1” in FIG. **18**). Server **1230** then receives the drive segments and reconstructs a drive (depicted as “Drive 1” in FIG. **18**) from the received segments.

[0277] As further depicted in FIG. **18**, system **1700** also receives data from additional vehicles. For example, vehicle **1820** also captures drive data using, for example, a camera (which produces, e.g., ego motion data, traffic sign data, road data, or the like) and a locating device (e.g., a GPS locator). Similar to vehicle **1810**, vehicle **1820** segments the collected data into drive segments (depicted as “DS1 2,” “DS2 2,” “DSN 2” in FIG. **18**). Server **1230** then receives the drive segments and reconstructs a drive (depicted as “Drive 2” in FIG. **18**) from the received segments. Any number of additional vehicles may be used. For example, FIG. **18** also includes “CAR N” that captures drive data, segments it into drive segments (depicted as “DS1 N,” “DS2 N,” “DSN N” in FIG. **18**), and sends it to server **1230** for reconstruction into a drive (depicted as “Drive N” in FIG. **18**).



[0278] As depicted in FIG. 18, server 1230 may construct a sparse map (depicted as “MAP”) using the reconstructed drives (e.g., “Drive 1,” “Drive 2,” and “Drive N”) collected from a plurality of vehicles (e.g., “CAR 1” (also labeled vehicle 1810), “CAR 2” (also labeled vehicle 1820), and “CAR N”).

[0279] FIG. 19 is a flowchart showing an example process 1900 for generating a sparse map for autonomous vehicle navigation along a road segment. Process 1900 may be performed by one or more processing devices included in server 1230.

[0280] Process 1900 may include receiving a plurality of images acquired as one or more vehicles traverse the road segment (step 1905). Server 1230 may receive images from cameras included within one or more of vehicles 1205, 1210, 1215, 1220, and 1225. For example, camera 122 may capture one or more images of the environment surrounding vehicle 1205 as vehicle 1205 travels along road segment 1200. In some embodiments, server 1230 may also receive stripped down image data that has had redundancies removed by a processor on vehicle 1205, as discussed above with respect to FIG. 17.

[0281] Process 1900 may further include identifying, based on the plurality of images, at least one line representation of a road surface feature extending along the road segment (step 1910). Each line representation may represent a path along the road segment substantially corresponding with the road surface feature. For example, server 1230 may analyze the environmental images received from camera 122 to identify a road edge or a lane marking and determine a trajectory of travel along road segment 1200 associated with the road edge or lane marking. In some embodiments, the trajectory (or line representation) may include a spline, a polynomial representation, or a curve. Server 1230 may determine the trajectory of travel of vehicle 1205 based on camera ego motions (e.g., three dimensional translation and/or three dimensional rotational motions) received at step 1905.

[0282] Process 1900 may also include identifying, based on the plurality of images, a plurality of landmarks associated with the road segment (step 1910). For example, server 1230 may analyze the environmental images received from camera 122 to identify one or more landmarks, such as road sign along road segment 1200. Server 1230 may identify the landmarks using analysis of the plurality of images acquired as one or more vehicles traverse the road segment. To enable crowdsourcing, the analysis may include rules regarding accepting and rejecting possible landmarks associated with the road segment. For example, the analysis may include accepting potential landmarks when a ratio of images in which the landmark does appear to images in which the landmark does not appear exceeds a threshold and/or rejecting potential landmarks when a ratio of images in which the landmark does not appear to images in which the landmark does appear exceeds a threshold.

[0283] Process 1900 may include other operations or steps performed by server 1230. For example, the navigation information may include a target trajectory for vehicles to travel along a road segment, and process 1900 may include clustering, by server 1230, vehicle trajectories related to multiple vehicles travelling on the road segment and determining the target trajectory based on the clustered vehicle trajectories, as discussed in further detail below. Clustering vehicle trajectories may include clustering, by server 1230, the multiple trajectories related to the vehicles travelling on the road segment into a plurality of clusters based on at least one of the absolute heading of vehicles or lane assignment of the vehicles. Generating the target trajectory may include averaging, by server 1230, the clustered trajectories. By way of further example, process 1900 may include aligning data received in step 1905. Other processes or steps performed by server 1230, as described above, may also be included in process 1900.

[0284] The disclosed systems and methods may include other features. For example, the disclosed systems may use local coordinates, rather than global coordinates. For autonomous driving, some systems may present data in world coordinates. For example, longitude and latitude coordinates on the earth surface may be used. In order to use the map for steering, the host vehicle may determine

its position and orientation relative to the map. It seems natural to use a GPS device on board, in order to position the vehicle on the map and in order to find the rotation transformation between the body reference frame and the world reference frame (e.g., North, East and Down). Once the body reference frame is aligned with the map reference frame, then the desired route may be expressed in the body reference frame and the steering commands may be computed or generated.

[0285] The disclosed systems and methods may enable autonomous vehicle navigation (e.g., steering control) with low footprint models, which may be collected by the autonomous vehicles themselves without the aid of expensive surveying equipment. To support the autonomous navigation (e.g., steering applications), the road model may include a sparse map having the geometry of the road, its lane structure, and landmarks that may be used to determine the location or position of vehicles along a trajectory included in the model. As discussed above, generation of the sparse map may be performed by a remote server that communicates with vehicles travelling on the road and that receives data from the vehicles. The data may include sensed data, trajectories reconstructed based on the sensed data, and/or recommended trajectories that may represent modified reconstructed trajectories. As discussed below, the server may transmit the model back to the vehicles or other vehicles that later travel on the road to aid in autonomous navigation.

[0286] FIG. 20 illustrates a block diagram of server 1230. Server 1230 may include a communication unit 2005, which may include both hardware components (e.g., communication control circuits, switches, and antenna), and software components (e.g., communication protocols, computer codes). For example, communication unit 2005 may include at least one network interface. Server 1230 may communicate with vehicles 1205, 1210, 1215, 1220, and 1225 through communication unit 2005. For example, server 1230 may receive, through communication unit 2005, navigation information transmitted from vehicles 1205, 1210, 1215, 1220, and 1225. Server 1230 may distribute, through communication unit 2005, the autonomous vehicle road navigation model to one or more autonomous vehicles.

[0287] Server 1230 may include at least one non-transitory storage medium 2010, such as a hard drive, a compact disc, a tape, etc. Storage device 1410 may be configured to store data, such as navigation information received from vehicles 1205, 1210, 1215, 1220, and 1225 and/or the autonomous vehicle road navigation model that server 1230 generates based on the navigation information. Storage device 2010 may be configured to store any other information, such as a sparse map (e.g., sparse map 800 discussed above with respect to FIG. 8).

[0288] In addition to or in place of storage device 2010, server 1230 may include a memory 2015. Memory 2015 may be similar to or different from memory 140 or 150. Memory 2015 may be a non-transitory memory, such as a flash memory, a random access memory, etc. Memory 2015 may be configured to store data, such as computer codes or instructions executable by a processor (e.g., processor 2020), map data (e.g., data of sparse map 800), the autonomous vehicle road navigation model, and/or navigation information received from vehicles 1205, 1210, 1215, 1220, and 1225.

[0289] Server 1230 may include at least one processing device 2020 configured to execute computer codes or instructions stored in memory 2015 to perform various functions. For example, processing device 2020 may analyze the navigation information received from vehicles 1205, 1210, 1215, 1220, and 1225, and generate the autonomous vehicle road navigation model based on the analysis. Processing device 2020 may control communication unit 1405 to distribute the autonomous vehicle road navigation model to one or more autonomous vehicles (e.g., one or more of vehicles 1205, 1210, 1215, 1220, and 1225 or any vehicle that travels on road segment 1200 at a later time). Processing device 2020 may be similar to or different from processor 180, 190, or processing unit 110.

[0290] FIG. 21 illustrates a block diagram of memory 2015, which may store computer code or instructions for performing one or more operations for generating a road navigation model for use in autonomous vehicle navigation. As shown in FIG. 21, memory 2015 may store one or more modules for performing the operations for processing vehicle navigation information. For example,

memory **2015** may include a model generating module **2105** and a model distributing module **2110**. Processor **2020** may execute the instructions stored in any of modules **2105** and **2110** included in memory **2015**.

[0291] Model generating module **2105** may store instructions which, when executed by processor **2020**, may generate at least a portion of an autonomous vehicle road navigation model for a common road segment (e.g., road segment **1200**) based on navigation information received from vehicles **1205**, **1210**, **1215**, **1220**, and **1225**. For example, in generating the autonomous vehicle road navigation model, processor **2020** may cluster vehicle trajectories along the common road segment **1200** into different clusters. Processor **2020** may determine a target trajectory along the common road segment **1200** based on the clustered vehicle trajectories for each of the different clusters. Such an operation may include finding a mean or average trajectory of the clustered vehicle trajectories (e.g., by averaging data representing the clustered vehicle trajectories) in each cluster. In some embodiments, the target trajectory may be associated with a single lane of the common road segment **1200**.

[0292] The road model and/or sparse map may store trajectories associated with a road segment. These trajectories may be referred to as target trajectories, which are provided to autonomous vehicles for autonomous navigation. The target trajectories may be received from multiple vehicles, or may be generated based on actual trajectories or recommended trajectories (actual trajectories with some modifications) received from multiple vehicles. The target trajectories included in the road model or sparse map may be continuously updated (e.g., averaged) with new trajectories received from other vehicles.

[0293] Vehicles travelling on a road segment may collect data by various sensors. The data may include landmarks, road signature profile, vehicle motion (e.g., accelerometer data, speed data), vehicle position (e.g., GPS data), and may either reconstruct the actual trajectories themselves, or transmit the data to a server, which will reconstruct the actual trajectories for the vehicles. In some embodiments, the vehicles may transmit data relating to a trajectory (e.g., a curve in an arbitrary reference frame), landmarks data, and lane assignment along traveling path to server **1230**. Various vehicles travelling along the same road segment at multiple drives may have different trajectories. Server **1230** may identify routes or trajectories associated with each lane from the trajectories received from vehicles through a clustering process.

[0294] FIG. 22 illustrates a process of clustering vehicle trajectories associated with vehicles **1205**, **1210**, **1215**, **1220**, and **1225** for determining a target trajectory for the common road segment (e.g., road segment **1200**). The target trajectory or a plurality of target trajectories determined from the clustering process may be included in the autonomous vehicle road navigation model or sparse map **800**. In some embodiments, vehicles **1205**, **1210**, **1215**, **1220**, and **1225** traveling along road segment **1200** may transmit a plurality of trajectories **2200** to server **1230**. In some embodiments, server **1230** may generate trajectories based on landmark, road geometry, and vehicle motion information received from vehicles **1205**, **1210**, **1215**, **1220**, and **1225**. To generate the autonomous vehicle road navigation model, server **1230** may cluster vehicle trajectories **1600** into a plurality of clusters **2205**, **2210**, **2215**, **2220**, **2225**, and **2230**, as shown in FIG. 22.

[0295] Clustering may be performed using various criteria. In some embodiments, all drives in a cluster may be similar with respect to the absolute heading along the road segment **1200**. The absolute heading may be obtained from GPS signals received by vehicles **1205**, **1210**, **1215**, **1220**, and **1225**. In some embodiments, the absolute heading may be obtained using dead reckoning. Dead reckoning, as one of skill in the art would understand, may be used to determine the current position and hence heading of vehicles **1205**, **1210**, **1215**, **1220**, and **1225** by using previously determined position, estimated speed, etc. Trajectories clustered by absolute heading may be useful for identifying routes along the roadways.

[0296] In some embodiments, all the drives in a cluster may be similar with respect to the lane assignment (e.g., in the same lane before and after a junction) along the drive on road segment

**1200.** Trajectories clustered by lane assignment may be useful for identifying lanes along the roadways. In some embodiments, both criteria (e.g., absolute heading and lane assignment) may be used for clustering.

[0297] In each cluster **2205**, **2210**, **2215**, **2220**, **2225**, and **2230**, trajectories may be averaged to obtain a target trajectory associated with the specific cluster. For example, the trajectories from multiple drives associated with the same lane cluster may be averaged. The averaged trajectory may be a target trajectory associated with a specific lane. To average a cluster of trajectories, server **1230** may select a reference frame of an arbitrary trajectory  $C_0$ . For all other trajectories ( $C_1, \dots, C_n$ ), server **1230** may find a rigid transformation that maps  $C_i$  to  $C_0$ , where  $i=1, 2, \dots, n$ , where  $n$  is a positive integer number, corresponding to the total number of trajectories included in the cluster. Server **1230** may compute a mean curve or trajectory in the  $C_0$  reference frame.

[0298] In some embodiments, the landmarks may define an arc length matching between different drives, which may be used for alignment of trajectories with lanes. In some embodiments, lane marks before and after a junction may be used for alignment of trajectories with lanes.

[0299] To assemble lanes from the trajectories, server **1230** may select a reference frame of an arbitrary lane. Server **1230** may map partially overlapping lanes to the selected reference frame. Server **1230** may continue mapping until all lanes are in the same reference frame. Lanes that are next to each other may be aligned as if they were the same lane, and later they may be shifted laterally.

[0300] Landmarks recognized along the road segment may be mapped to the common reference frame, first at the lane level, then at the junction level. For example, the same landmarks may be recognized multiple times by multiple vehicles in multiple drives. The data regarding the same landmarks received in different drives may be slightly different. Such data may be averaged and mapped to the same reference frame, such as the  $C_0$  reference frame. Additionally or alternatively, the variance of the data of the same landmark received in multiple drives may be calculated.

[0301] In some embodiments, each lane of road segment **120** may be associated with a target trajectory and certain landmarks. The target trajectory or a plurality of such target trajectories may be included in the autonomous vehicle road navigation model, which may be used later by other autonomous vehicles travelling along the same road segment **1200**. Landmarks identified by vehicles **1205**, **1210**, **1215**, **1220**, and **1225** while the vehicles travel along road segment **1200** may be recorded in association with the target trajectory. The data of the target trajectories and landmarks may be continuously or periodically updated with new data received from other vehicles in subsequent drives.

[0302] For localization of an autonomous vehicle, the disclosed systems and methods may use an Extended Kalman Filter. The location of the vehicle may be determined based on three dimensional position data and/or three dimensional orientation data, prediction of future location ahead of vehicle's current location by integration of ego motion. The localization of vehicle may be corrected or adjusted by image observations of landmarks. For example, when vehicle detects a landmark within an image captured by the camera, the landmark may be compared to a known landmark stored within the road model or sparse map **800**. The known landmark may have a known location (e.g., GPS data) along a target trajectory stored in the road model and/or sparse map **800**. Based on the current speed and images of the landmark, the distance from the vehicle to the landmark may be estimated. The location of the vehicle along a target trajectory may be adjusted based on the distance to the landmark and the landmark's known location (stored in the road model or sparse map **800**). The landmark's position/location data (e.g., mean values from multiple drives) stored in the road model and/or sparse map **800** may be presumed to be accurate.

[0303] In some embodiments, the disclosed system may form a closed loop subsystem, in which estimation of the vehicle six degrees of freedom location (e.g., three dimensional position data plus three dimensional orientation data) may be used for navigating (e.g., steering the wheel of) the autonomous vehicle to reach a desired point (e.g., 1.3 second ahead in the stored). In turn, data

measured from the steering and actual navigation may be used to estimate the six degrees of freedom location.

[0304] In some embodiments, poles along a road, such as lampposts and power or cable line poles may be used as landmarks for localizing the vehicles. Other landmarks such as traffic signs, traffic lights, arrows on the road, stop lines, as well as static features or signatures of an object along the road segment may also be used as landmarks for localizing the vehicle. When poles are used for localization, the x observation of the poles (i.e., the viewing angle from the vehicle) may be used, rather than the y observation (i.e., the distance to the pole) since the bottoms of the poles may be occluded and sometimes they are not on the road plane.

[0305] FIG. 23 illustrates a navigation system for a vehicle, which may be used for autonomous navigation using a crowdsourced sparse map. For illustration, the vehicle is referenced as vehicle 1205. The vehicle shown in FIG. 23 may be any other vehicle disclosed herein, including, for example, vehicles 1210, 1215, 1220, and 1225, as well as vehicle 200 shown in other embodiments. As shown in FIG. 12, vehicle 1205 may communicate with server 1230. Vehicle 1205 may include an image capture device 122 (e.g., camera 122). Vehicle 1205 may include a navigation system 2300 configured for providing navigation guidance for vehicle 1205 to travel on a road (e.g., road segment 1200). Vehicle 1205 may also include other sensors, such as a speed sensor 2320 and an accelerometer 2325. Speed sensor 2320 may be configured to detect the speed of vehicle 1205. Accelerometer 2325 may be configured to detect an acceleration or deceleration of vehicle 1205. Vehicle 1205 shown in FIG. 23 may be an autonomous vehicle, and the navigation system 2300 may be used for providing navigation guidance for autonomous driving. Alternatively, vehicle 1205 may also be a non-autonomous, human-controlled vehicle, and navigation system 2300 may still be used for providing navigation guidance.

[0306] Navigation system 2300 may include a communication unit 2305 configured to communicate with server 1230 through communication path 1235. Navigation system 2300 may also include a GPS unit 2310 configured to receive and process GPS signals. Navigation system 2300 may further include at least one processor 2315 configured to process data, such as GPS signals, map data from sparse map 800 (which may be stored on a storage device provided onboard vehicle 1205 and/or received from server 1230), road geometry sensed by a road profile sensor 2330, images captured by camera 122, and/or autonomous vehicle road navigation model received from server 1230. The road profile sensor 2330 may include different types of devices for measuring different types of road profile, such as road surface roughness, road width, road elevation, road curvature, etc. For example, the road profile sensor 2330 may include a device that measures the motion of a suspension of vehicle 2305 to derive the road roughness profile. In some embodiments, the road profile sensor 2330 may include radar sensors to measure the distance from vehicle 1205 to road sides (e.g., barrier on the road sides), thereby measuring the width of the road. In some embodiments, the road profile sensor 2330 may include a device configured for measuring the up and down elevation of the road. In some embodiment, the road profile sensor 2330 may include a device configured to measure the road curvature. For example, a camera (e.g., camera 122 or another camera) may be used to capture images of the road showing road curvatures. Vehicle 1205 may use such images to detect road curvatures.

[0307] The at least one processor 2315 may be programmed to receive, from camera 122, at least one environmental image associated with vehicle 1205. The at least one processor 2315 may analyze the at least one environmental image to determine navigation information related to the vehicle 1205. The navigation information may include a trajectory related to the travel of vehicle 1205 along road segment 1200. The at least one processor 2315 may determine the trajectory based on motions of camera 122 (and hence the vehicle), such as three dimensional translation and three dimensional rotational motions. In some embodiments, the at least one processor 2315 may determine the translation and rotational motions of camera 122 based on analysis of a plurality of images acquired by camera 122. In some embodiments, the navigation information may include

lane assignment information (e.g., in which lane vehicle **1205** is travelling along road segment **1200**). The navigation information transmitted from vehicle **1205** to server **1230** may be used by server **1230** to generate and/or update an autonomous vehicle road navigation model, which may be transmitted back from server **1230** to vehicle **1205** for providing autonomous navigation guidance for vehicle **1205**.

[0308] The at least one processor **2315** may also be programmed to transmit the navigation information from vehicle **1205** to server **1230**. In some embodiments, the navigation information may be transmitted to server **1230** along with road information. The road location information may include at least one of the GPS signal received by the GPS unit **2310**, landmark information, road geometry, lane information, etc. The at least one processor **2315** may receive, from server **1230**, the autonomous vehicle road navigation model or a portion of the model. The autonomous vehicle road navigation model received from server **1230** may include at least one update based on the navigation information transmitted from vehicle **1205** to server **1230**. The portion of the model transmitted from server **1230** to vehicle **1205** may include an updated portion of the model. The at least one processor **2315** may cause at least one navigational maneuver (e.g., steering such as making a turn, braking, accelerating, passing another vehicle, etc.) by vehicle **1205** based on the received autonomous vehicle road navigation model or the updated portion of the model.

[0309] The at least one processor **2315** may be configured to communicate with various sensors and components included in vehicle **1205**, including communication unit **2305**, GPS unit **2310**, camera **122**, speed sensor **2320**, accelerometer **2325**, and road profile sensor **2330**. The at least one processor **2315** may collect information or data from various sensors and components and transmit the information or data to server **1230** through communication unit **2305**. Alternatively or additionally, various sensors or components of vehicle **1205** may also communicate with server **1230** and transmit data or information collected by the sensors or components to server **1230**.

[0310] In some embodiments, vehicles **1205**, **1210**, **1215**, **1220**, and **1225** may communicate with each other, and may share navigation information with each other, such that at least one of the vehicles **1205**, **1210**, **1215**, **1220**, and **1225** may generate the autonomous vehicle road navigation model using crowdsourcing, e.g., based on information shared by other vehicles. In some embodiments, vehicles **1205**, **1210**, **1215**, **1220**, and **1225** may share navigation information with each other and each vehicle may update its own the autonomous vehicle road navigation model provided in the vehicle. In some embodiments, at least one of the vehicles **1205**, **1210**, **1215**, **1220**, and **1225** (e.g., vehicle **1205**) may function as a hub vehicle. The at least one processor **2315** of the hub vehicle (e.g., vehicle **1205**) may perform some or all of the functions performed by server **1230**. For example, the at least one processor **2315** of the hub vehicle may communicate with other vehicles and receive navigation information from other vehicles. The at least one processor **2315** of the hub vehicle may generate the autonomous vehicle road navigation model or an update to the model based on the shared information received from other vehicles. The at least one processor **2315** of the hub vehicle may transmit the autonomous vehicle road navigation model or the update to the model to other vehicles for providing autonomous navigation guidance.

#### Navigation Based on Sparse Maps

[0311] As previously discussed, the autonomous vehicle road navigation model including sparse map **800** may include a plurality of mapped lane marks and a plurality of mapped objects/features associated with a road segment. As discussed in greater detail below, these mapped lane marks, objects, and features may be used when the autonomous vehicle navigates. For example, in some embodiments, the mapped objects and features may be used to localized a host vehicle relative to the map (e.g., relative to a mapped target trajectory). The mapped lane marks may be used (e.g., as a check) to determine a lateral position and/or orientation relative to a planned or target trajectory. With this position information, the autonomous vehicle may be able to adjust a heading direction to match a direction of a target trajectory at the determined position.

[0312] Vehicle **200** may be configured to detect lane marks in a given road segment. The road

segment may include any markings on a road for guiding vehicle traffic on a roadway. For example, the lane marks may be continuous or dashed lines demarking the edge of a lane of travel. The lane marks may also include double lines, such as a double continuous lines, double dashed lines or a combination of continuous and dashed lines indicating, for example, whether passing is permitted in an adjacent lane. The lane marks may also include freeway entrance and exit markings indicating, for example, a deceleration lane for an exit ramp or dotted lines indicating that a lane is turn-only or that the lane is ending. The markings may further indicate a work zone, a temporary lane shift, a path of travel through an intersection, a median, a special purpose lane (e.g., a bike lane, HOV lane, etc.), or other miscellaneous markings (e.g., crosswalk, a speed hump, a railway crossing, a stop line, etc.).

[0313] Vehicle **200** may use cameras, such as image capture devices **122** and **124** included in image acquisition unit **120**, to capture images of the surrounding lane marks. Vehicle **200** may analyze the images to detect point locations associated with the lane marks based on features identified within one or more of the captured images. These point locations may be uploaded to a server to represent the lane marks in sparse map **800**. Depending on the position and field of view of the camera, lane marks may be detected for both sides of the vehicle simultaneously from a single image. In other embodiments, different cameras may be used to capture images on multiple sides of the vehicle. Rather than uploading actual images of the lane marks, the marks may be stored in sparse map **800** as a spline or a series of points, thus reducing the size of sparse map **800** and/or the data that must be uploaded remotely by the vehicle.

[0314] FIGS. **24A-24D** illustrate exemplary point locations that may be detected by vehicle **200** to represent particular lane marks. Similar to the landmarks described above, vehicle **200** may use various image recognition algorithms or software to identify point locations within a captured image. For example, vehicle **200** may recognize a series of edge points, corner points or various other point locations associated with a particular lane mark. FIG. **24A** shows a continuous lane mark **2410** that may be detected by vehicle **200**. Lane mark **2410** may represent the outside edge of a roadway, represented by a continuous white line. As shown in FIG. **24A**, vehicle **200** may be configured to detect a plurality of edge location points **2411** along the lane mark. Location points **2411** may be collected to represent the lane mark at any intervals sufficient to create a mapped lane mark in the sparse map. For example, the lane mark may be represented by one point per meter of the detected edge, one point per every five meters of the detected edge, or at other suitable spacings. In some embodiments, the spacing may be determined by other factors, rather than at set intervals such as, for example, based on points where vehicle **200** has a highest confidence ranking of the location of the detected points. Although FIG. **24A** shows edge location points on an interior edge of lane mark **2410**, points may be collected on the outside edge of the line or along both edges. Further, while a single line is shown in FIG. **24A**, similar edge points may be detected for a double continuous line. For example, points **2411** may be detected along an edge of one or both of the continuous lines.

[0315] Vehicle **200** may also represent lane marks differently depending on the type or shape of lane mark. FIG. **24B** shows an exemplary dashed lane mark **2420** that may be detected by vehicle **200**. Rather than identifying edge points, as in FIG. **24A**, vehicle may detect a series of corner points **2421** representing corners of the lane dashes to define the full boundary of the dash. While FIG. **24B** shows each corner of a given dash marking being located, vehicle **200** may detect or upload a subset of the points shown in the figure. For example, vehicle **200** may detect the leading edge or leading corner of a given dash mark, or may detect the two corner points nearest the interior of the lane. Further, not every dash mark may be captured, for example, vehicle **200** may capture and/or record points representing a sample of dash marks (e.g., every other, every third, every fifth, etc.) or dash marks at a predefined spacing (e.g., every meter, every five meters, every 10 meters, etc.) Corner points may also be detected for similar lane marks, such as markings showing a lane is for an exit ramp, that a particular lane is ending, or other various lane marks that

may have detectable corner points. Corner points may also be detected for lane marks consisting of double dashed lines or a combination of continuous and dashed lines.

[0316] In some embodiments, the points uploaded to the server to generate the mapped lane marks may represent other points besides the detected edge points or corner points. FIG. 24C illustrates a series of points that may represent a centerline of a given lane mark. For example, continuous lane **2410** may be represented by centerline points **2441** along a centerline **2440** of the lane mark. In some embodiments, vehicle **200** may be configured to detect these center points using various image recognition techniques, such as convolutional neural networks (CNN), scale-invariant feature transform (SIFT), histogram of oriented gradients (HOG) features, or other techniques. Alternatively, vehicle **200** may detect other points, such as edge points **2411** shown in FIG. 24A, and may calculate centerline points **2441**, for example, by detecting points along each edge and determining a midpoint between the edge points. Similarly, dashed lane mark **2420** may be represented by centerline points **2451** along a centerline **2450** of the lane mark. The centerline points may be located at the edge of a dash, as shown in FIG. 24C, or at various other locations along the centerline. For example, each dash may be represented by a single point in the geometric center of the dash. The points may also be spaced at a predetermined interval along the centerline (e.g., every meter, 5 meters, 10 meters, etc.). The centerline points **2451** may be detected directly by vehicle **200**, or may be calculated based on other detected reference points, such as corner points **2421**, as shown in FIG. 24B. A centerline may also be used to represent other lane mark types, such as a double line, using similar techniques as above.

[0317] In some embodiments, vehicle **200** may identify points representing other features, such as a vertex between two intersecting lane marks. FIG. 24D shows exemplary points representing an intersection between two lane marks **2460** and **2465**. Vehicle **200** may calculate a vertex point **2466** representing an intersection between the two lane marks. For example, one of lane marks **2460** or **2465** may represent a train crossing area or other crossing area in the road segment. While lane marks **2460** and **2465** are shown as crossing each other perpendicularly, various other configurations may be detected. For example, the lane marks **2460** and **2465** may cross at other angles, or one or both of the lane marks may terminate at the vertex point **2466**. Similar techniques may also be applied for intersections between dashed or other lane mark types. In addition to vertex point **2466**, various other points **2467** may also be detected, providing further information about the orientation of lane marks **2460** and **2465**.

[0318] Vehicle **200** may associate real-world coordinates with each detected point of the lane mark. For example, location identifiers may be generated, including coordinate for each point, to upload to a server for mapping the lane mark. The location identifiers may further include other identifying information about the points, including whether the point represents a corner point, an edge point, center point, etc. Vehicle **200** may therefore be configured to determine a real-world position of each point based on analysis of the images. For example, vehicle **200** may detect other features in the image, such as the various landmarks described above, to locate the real-world position of the lane marks. This may involve determining the location of the lane marks in the image relative to the detected landmark or determining the position of the vehicle based on the detected landmark and then determining a distance from the vehicle (or target trajectory of the vehicle) to the lane mark. When a landmark is not available, the location of the lane mark points may be determined relative to a position of the vehicle determined based on dead reckoning. The real-world coordinates included in the location identifiers may be represented as absolute coordinates (e.g., latitude/longitude coordinates), or may be relative to other features, such as based on a longitudinal position along a target trajectory and a lateral distance from the target trajectory. The location identifiers may then be uploaded to a server for generation of the mapped lane marks in the navigation model (such as sparse map **800**). In some embodiments, the server may construct a spline representing the lane marks of a road segment. Alternatively, vehicle **200** may generate the spline and upload it to the server to be recorded in the navigational model.



[0319] FIG. 24E shows an exemplary navigation model or sparse map for a corresponding road segment that includes mapped lane marks. The sparse map may include a target trajectory **2475** for a vehicle to follow along a road segment. As described above, target trajectory **2475** may represent an ideal path for a vehicle to take as it travels the corresponding road segment, or may be located elsewhere on the road (e.g., a centerline of the road, etc.). Target trajectory **2475** may be calculated in the various methods described above, for example, based on an aggregation (e.g., a weighted combination) of two or more reconstructed trajectories of vehicles traversing the same road segment.

[0320] In some embodiments, the target trajectory may be generated equally for all vehicle types and for all road, vehicle, and/or environment conditions. In other embodiments, however, various other factors or variables may also be considered in generating the target trajectory. A different target trajectory may be generated for different types of vehicles (e.g., a private car, a light truck, and a full trailer). For example, a target trajectory with relatively tighter turning radii may be generated for a small private car than a larger semi-trailer truck. In some embodiments, road, vehicle and environmental conditions may be considered as well. For example, a different target trajectory may be generated for different road conditions (e.g., wet, snowy, icy, dry, etc.), vehicle conditions (e.g., tire condition or estimated tire condition, brake condition or estimated brake condition, amount of fuel remaining, etc.) or environmental factors (e.g., time of day, visibility, weather, etc.). The target trajectory may also depend on one or more aspects or features of a particular road segment (e.g., speed limit, frequency and size of turns, grade, etc.). In some embodiments, various user settings may also be used to determine the target trajectory, such as a set driving mode (e.g., desired driving aggressiveness, economy mode, etc.).

[0321] The sparse map may also include mapped lane marks **2470** and **2480** representing lane marks along the road segment. The mapped lane marks may be represented by a plurality of location identifiers **2471** and **2481**. As described above, the location identifiers may include locations in real world coordinates of points associated with a detected lane mark. Similar to the target trajectory in the model, the lane marks may also include elevation data and may be represented as a curve in three-dimensional space. For example, the curve may be a spline connecting three dimensional polynomials of suitable order the curve may be calculated based on the location identifiers. The mapped lane marks may also include other information or metadata about the lane mark, such as an identifier of the type of lane mark (e.g., between two lanes with the same direction of travel, between two lanes of opposite direction of travel, edge of a roadway, etc.) and/or other characteristics of the lane mark (e.g., continuous, dashed, single line, double line, yellow, white, etc.). In some embodiments, the mapped lane marks may be continuously updated within the model, for example, using crowdsourcing techniques. The same vehicle may upload location identifiers during multiple occasions of travelling the same road segment or data may be selected from a plurality of vehicles (such as **1205**, **1210**, **1215**, **1220**, and **1225**) travelling the road segment at different times. Sparse map **800** may then be updated or refined based on subsequent location identifiers received from the vehicles and stored in the system. As the mapped lane marks are updated and refined, the updated road navigation model and/or sparse map may be distributed to a plurality of autonomous vehicles.

[0322] Generating the mapped lane marks in the sparse map may also include detecting and/or mitigating errors based on anomalies in the images or in the actual lane marks themselves. FIG. 24F shows an exemplary anomaly **2495** associated with detecting a lane mark **2490**. Anomaly **2495** may appear in the image captured by vehicle **200**, for example, from an object obstructing the camera's view of the lane mark, debris on the lens, etc. In some instances, the anomaly may be due to the lane mark itself, which may be damaged or worn away, or partially covered, for example, by dirt, debris, water, snow or other materials on the road. Anomaly **2495** may result in an erroneous point **2491** being detected by vehicle **200**. Sparse map **800** may provide the correct the mapped lane mark and exclude the error. In some embodiments, vehicle **200** may detect erroneous point

**2491** for example, by detecting anomaly **2495** in the image, or by identifying the error based on detected lane mark points before and after the anomaly. Based on detecting the anomaly, the vehicle may omit point **2491** or may adjust it to be in line with other detected points. In other embodiments, the error may be corrected after the point has been uploaded, for example, by determining the point is outside of an expected threshold based on other points uploaded during the same trip, or based on an aggregation of data from previous trips along the same road segment. [0323] The mapped lane marks in the navigation model and/or sparse map may also be used for navigation by an autonomous vehicle traversing the corresponding roadway. For example, a vehicle navigating along a target trajectory may periodically use the mapped lane marks in the sparse map to align itself with the target trajectory. As mentioned above, between landmarks the vehicle may navigate based on dead reckoning in which the vehicle uses sensors to determine its ego motion and estimate its position relative to the target trajectory. Errors may accumulate over time and vehicle's position determinations relative to the target trajectory may become increasingly less accurate. Accordingly, the vehicle may use lane marks occurring in sparse map **800** (and their known locations) to reduce the dead reckoning-induced errors in position determination. In this way, the identified lane marks included in sparse map **800** may serve as navigational anchors from which an accurate position of the vehicle relative to a target trajectory may be determined.

[0324] FIG. **25A** shows an exemplary image **2500** of a vehicle's surrounding environment that may be used for navigation based on the mapped lane marks. Image **2500** may be captured, for example, by vehicle **200** through image capture devices **122** and **124** included in image acquisition unit **120**. Image **2500** may include an image of at least one lane mark **2510**, as shown in FIG. **25A**. Image **2500** may also include one or more landmarks **2521**, such as road sign, used for navigation as described above. Some elements shown in FIG. **25A**, such as elements **2511**, **2530**, and **2520** which do not appear in the captured image **2500** but are detected and/or determined by vehicle **200** are also shown for reference.

[0325] Using the various techniques described above with respect to FIGS. **24A-D** and **24F**, a vehicle may analyze image **2500** to identify lane mark **2510**. Various points **2511** may be detected corresponding to features of the lane mark in the image. Points **2511**, for example, may correspond to an edge of the lane mark, a corner of the lane mark, a midpoint of the lane mark, a vertex between two intersecting lane marks, or various other features or locations. Points **2511** may be detected to correspond to a location of points stored in a navigation model received from a server. For example, if a sparse map is received containing points that represent a centerline of a mapped lane mark, points **2511** may also be detected based on a centerline of lane mark **2510**.

[0326] The vehicle may also determine a longitudinal position represented by element **2520** and located along a target trajectory. Longitudinal position **2520** may be determined from image **2500**, for example, by detecting landmark **2521** within image **2500** and comparing a measured location to a known landmark location stored in the road model or sparse map **800**. The location of the vehicle along a target trajectory may then be determined based on the distance to the landmark and the landmark's known location. The longitudinal position **2520** may also be determined from images other than those used to determine the position of a lane mark. For example, longitudinal position **2520** may be determined by detecting landmarks in images from other cameras within image acquisition unit **120** taken simultaneously or near simultaneously to image **2500**. In some instances, the vehicle may not be near any landmarks or other reference points for determining longitudinal position **2520**. In such instances, the vehicle may be navigating based on dead reckoning and thus may use sensors to determine its ego motion and estimate a longitudinal position **2520** relative to the target trajectory. The vehicle may also determine a distance **2530** representing the actual distance between the vehicle and lane mark **2510** observed in the captured image(s). The camera angle, the speed of the vehicle, the width of the vehicle, or various other factors may be accounted for in determining distance **2530**.

[0327] FIG. **25B** illustrates a lateral localization correction of the vehicle based on the mapped lane

marks in a road navigation model. As described above, vehicle **200** may determine a distance **2530** between vehicle **200** and a lane mark **2510** using one or more images captured by vehicle **200**. Vehicle **200** may also have access to a road navigation model, such as sparse map **800**, which may include a mapped lane mark **2550** and a target trajectory **2555**. Mapped lane mark **2550** may be modeled using the techniques described above, for example using crowdsourced location identifiers captured by a plurality of vehicles. Target trajectory **2555** may also be generated using the various techniques described previously. Vehicle **200** may also determine or estimate a longitudinal position **2520** along target trajectory **2555** as described above with respect to FIG. 25A. Vehicle **200** may then determine an expected distance **2540** based on a lateral distance between target trajectory **2555** and mapped lane mark **2550** corresponding to longitudinal position **2520**. The lateral localization of vehicle **200** may be corrected or adjusted by comparing the actual distance **2530**, measured using the captured image(s), with the expected distance **2540** from the model.

[0328] FIGS. 25C and 25D provide illustrations associated with another example for localizing a host vehicle during navigation based on mapped landmarks/objects/features in a sparse map. FIG. 25C conceptually represents a series of images captured from a vehicle navigating along a road segment **2560**. In this example, road segment **2560** includes a straight section of a two-lane divided highway delineated by road edges **2561** and **2562** and center lane marking **2563**. As shown, the host vehicle is navigating along a lane **2564**, which is associated with a mapped target trajectory **2565**. Thus, in an ideal situation (and without influencers such as the presence of target vehicles or objects in the roadway, etc.) the host vehicle should closely track the mapped target trajectory **2565** as it navigates along lane **2564** of road segment **2560**. In reality, the host vehicle may experience drift as it navigates along mapped target trajectory **2565**. For effective and safe navigation, this drift should be maintained within acceptable limits (e.g.,  $\pm 10$  cm of lateral displacement from target trajectory **2565** or any other suitable threshold). To periodically account for drift and to make any needed course corrections to ensure that the host vehicle follows target trajectory **2565**, the disclosed navigation systems may be able to localize the host vehicle along the target trajectory **2565** (e.g., determine a lateral and longitudinal position of the host vehicle relative to the target trajectory **2565**) using one or more mapped features/objects included in the sparse map.

[0329] As a simple example, FIG. 25C shows a speed limit sign **2566** as it may appear in five different, sequentially captured images as the host vehicle navigates along road segment **2560**. For example, at a first time,  $t_0$ , sign **2566** may appear in a captured image near the horizon. As the host vehicle approaches sign **2566**, in subsequently captured images at times  $t_1$ ,  $t_2$ ,  $t_3$ , and  $t_4$ , sign **2566** will appear at different 2D X-Y pixel locations of the captured images. For example, in the captured image space, sign **2566** will move downward and to the right along curve **2567** (e.g., a curve extending through the center of the sign in each of the five captured image frames). Sign **2566** will also appear to increase in size as it is approached by the host vehicle (i.e., it will occupy a great number of pixels in subsequently captured images).

[0330] These changes in the image space representations of an object, such as sign **2566**, may be exploited to determine a localized position of the host vehicle along a target trajectory. For example, as described in the present disclosure, any detectable object or feature, such as a semantic feature like sign **2566** or a detectable non-semantic feature, may be identified by one or more harvesting vehicles that previously traversed a road segment (e.g., road segment **2560**). A mapping server may collect the harvested drive information from a plurality of vehicles, aggregate and correlate that information, and generate a sparse map including, for example, a target trajectory **2565** for lane **2564** of road segment **2560**. The sparse map may also store a location of sign **2566** (along with type information, etc.). During navigation (e.g., prior to entering road segment **2560**), a host vehicle may be supplied with a map tile including a sparse map for road segment **2560**. To navigate in lane **2564** of road segment **2560**, the host vehicle may follow mapped target trajectory **2565**.

[0331] The mapped representation of sign **2566** may be used by the host vehicle to localize itself

relative to the target trajectory. For example, a camera on the host vehicle will capture an image **2570** of the environment of the host vehicle, and that captured image **2570** may include an image representation of sign **2566** having a certain size and a certain X-Y image location, as shown in FIG. **25D**. This size and X-Y image location can be used to determine the host vehicle's position relative to target trajectory **2565**. For example, based on the sparse map including a representation of sign **2566**, a navigation processor of the host vehicle can determine that in response to the host vehicle traveling along target trajectory **2565**, a representation of sign **2566** should appear in captured images such that a center of sign **2566** will move (in image space) along line **2567**. If a captured image, such as image **2570**, shows the center (or other reference point) displaced from line **2567** (e.g., the expected image space trajectory), then the host vehicle navigation system can determine that at the time of the captured image it was not located on target trajectory **2565**. From the image, however, the navigation processor can determine an appropriate navigational correction to return the host vehicle to the target trajectory **2565**. For example, if analysis shows an image location of sign **2566** that is displaced in the image by a distance **2572** to the left of the expected image space location on line **2567**, then the navigation processor may cause a heading change by the host vehicle (e.g., change the steering angle of the wheels) to move the host vehicle leftward by a distance **2573**. In this way, each captured image can be used as part of a feedback loop process such that a difference between an observed image position of sign **2566** and expected image trajectory **2567** may be minimized to ensure that the host vehicle continues along target trajectory **2565** with little to no deviation. Of course, the more mapped objects that are available, the more often the described localization technique may be employed, which can reduce or eliminate drift-induced deviations from target trajectory **2565**.

[0332] The process described above may be useful for detecting a lateral orientation or displacement of the host vehicle relative to a target trajectory. Localization of the host vehicle relative to target trajectory **2565** may also include a determination of a longitudinal location of the target vehicle along the target trajectory. For example, captured image **2570** includes a representation of sign **2566** as having a certain image size (e.g., 2D X-Y pixel area). This size can be compared to an expected image size of mapped sign **2566** as it travels through image space along line **2567** (e.g., as the size of the sign progressively increases, as shown in FIG. **25C**). Based on the image size of sign **2566** in image **2570**, and based on the expected size progression in image space relative to mapped target trajectory **2565**, the host vehicle can determine its longitudinal position (at the time when image **2570** was captured) relative to target trajectory **2565**. This longitudinal position coupled with any lateral displacement relative to target trajectory **2565**, as described above, allows for full localization of the host vehicle relative to target trajectory **2565**, as the host vehicle navigates along road **2560**.

[0333] FIGS. **25C** and **25D** provide just one example of the disclosed localization technique using a single mapped object and a single target trajectory. In other examples, there may be many more target trajectories (e.g., one target trajectory for each viable lane of a multi-lane highway, urban street, complex junction, etc.) and there may be many more mapped available for localization. For example, a sparse map representative of an urban environment may include many objects per meter available for localization.

[0334] FIG. **26A** is a flowchart showing an exemplary process **2600A** for mapping a lane mark for use in autonomous vehicle navigation, consistent with disclosed embodiments. At step **2610**, process **2600A** may include receiving two or more location identifiers associated with a detected lane mark. For example, step **2610** may be performed by server **1230** or one or more processors associated with the server. The location identifiers may include locations in real-world coordinates of points associated with the detected lane mark, as described above with respect to FIG. **24E**. In some embodiments, the location identifiers may also contain other data, such as additional information about the road segment or the lane mark. Additional data may also be received during step **2610**, such as accelerometer data, speed data, landmarks data, road geometry or profile data,

vehicle positioning data, ego motion data, or various other forms of data described above. The location identifiers may be generated by a vehicle, such as vehicles **1205**, **1210**, **1215**, **1220**, and **1225**, based on images captured by the vehicle. For example, the identifiers may be determined based on acquisition, from a camera associated with a host vehicle, of at least one image representative of an environment of the host vehicle, analysis of the at least one image to detect the lane mark in the environment of the host vehicle, and analysis of the at least one image to determine a position of the detected lane mark relative to a location associated with the host vehicle. As described above, the lane mark may include a variety of different marking types, and the location identifiers may correspond to a variety of points relative to the lane mark. For example, where the detected lane mark is part of a dashed line marking a lane boundary, the points may correspond to detected corners of the lane mark. Where the detected lane mark is part of a continuous line marking a lane boundary, the points may correspond to a detected edge of the lane mark, with various spacings as described above. In some embodiments, the points may correspond to the centerline of the detected lane mark, as shown in FIG. **24C**, or may correspond to a vertex between two intersecting lane marks and at least one two other points associated with the intersecting lane marks, as shown in FIG. **24D**.

[0335] At step **2612**, process **2600A** may include associating the detected lane mark with a corresponding road segment. For example, server **1230** may analyze the real-world coordinates, or other information received during step **2610**, and compare the coordinates or other information to location information stored in an autonomous vehicle road navigation model. Server **1230** may determine a road segment in the model that corresponds to the real-world road segment where the lane mark was detected.

[0336] At step **2614**, process **2600A** may include updating an autonomous vehicle road navigation model relative to the corresponding road segment based on the two or more location identifiers associated with the detected lane mark. For example, the autonomous road navigation model may be sparse map **800**, and server **1230** may update the sparse map to include or adjust a mapped lane mark in the model. Server **1230** may update the model based on the various methods or processes described above with respect to FIG. **24E**. In some embodiments, updating the autonomous vehicle road navigation model may include storing one or more indicators of position in real world coordinates of the detected lane mark. The autonomous vehicle road navigation model may also include a at least one target trajectory for a vehicle to follow along the corresponding road segment, as shown in FIG. **24E**.

[0337] At step **2616**, process **2600A** may include distributing the updated autonomous vehicle road navigation model to a plurality of autonomous vehicles. For example, server **1230** may distribute the updated autonomous vehicle road navigation model to vehicles **1205**, **1210**, **1215**, **1220**, and **1225**, which may use the model for navigation. The autonomous vehicle road navigation model may be distributed via one or more networks (e.g., over a cellular network and/or the Internet, etc.), through wireless communication paths **1235**, as shown in FIG. **12**.

[0338] In some embodiments, the lane marks may be mapped using data received from a plurality of vehicles, such as through a crowdsourcing technique, as described above with respect to FIG. **24E**. For example, process **2600A** may include receiving a first communication from a first host vehicle, including location identifiers associated with a detected lane mark, and receiving a second communication from a second host vehicle, including additional location identifiers associated with the detected lane mark. For example, the second communication may be received from a subsequent vehicle travelling on the same road segment, or from the same vehicle on a subsequent trip along the same road segment. Process **2600A** may further include refining a determination of at least one position associated with the detected lane mark based on the location identifiers received in the first communication and based on the additional location identifiers received in the second communication. This may include using an average of the multiple location identifiers and/or filtering out “ghost” identifiers that may not reflect the real-world position of the lane mark.

[0339] FIG. 26B is a flowchart showing an exemplary process 2600B for autonomously navigating a host vehicle along a road segment using mapped lane marks. Process 2600B may be performed, for example, by processing unit 110 of autonomous vehicle 200. At step 2620, process 2600B may include receiving from a server-based system an autonomous vehicle road navigation model. In some embodiments, the autonomous vehicle road navigation model may include a target trajectory for the host vehicle along the road segment and location identifiers associated with one or more lane marks associated with the road segment. For example, vehicle 200 may receive sparse map 800 or another road navigation model developed using process 2600A. In some embodiments, the target trajectory may be represented as a three-dimensional spline, for example, as shown in FIG. 9B. As described above with respect to FIGS. 24A-F, the location identifiers may include locations in real world coordinates of points associated with the lane mark (e.g., corner points of a dashed lane mark, edge points of a continuous lane mark, a vertex between two intersecting lane marks and other points associated with the intersecting lane marks, a centerline associated with the lane mark, etc.).

[0340] At step 2621, process 2600B may include receiving at least one image representative of an environment of the vehicle. The image may be received from an image capture device of the vehicle, such as through image capture devices 122 and 124 included in image acquisition unit 120. The image may include an image of one or more lane marks, similar to image 2500 described above.

[0341] At step 2622, process 2600B may include determining a longitudinal position of the host vehicle along the target trajectory. As described above with respect to FIG. 25A, this may be based on other information in the captured image (e.g., landmarks, etc.) or by dead reckoning of the vehicle between detected landmarks.

[0342] At step 2623, process 2600B may include determining an expected lateral distance to the lane mark based on the determined longitudinal position of the host vehicle along the target trajectory and based on the two or more location identifiers associated with the at least one lane mark. For example, vehicle 200 may use sparse map 800 to determine an expected lateral distance to the lane mark. As shown in FIG. 25B, longitudinal position 2520 along a target trajectory 2555 may be determined in step 2622. Using sparse map 800, vehicle 200 may determine an expected distance 2540 to mapped lane mark 2550 corresponding to longitudinal position 2520.

[0343] At step 2624, process 2600B may include analyzing the at least one image to identify the at least one lane mark. Vehicle 200, for example, may use various image recognition techniques or algorithms to identify the lane mark within the image, as described above. For example, lane mark 2510 may be detected through image analysis of image 2500, as shown in FIG. 25A.

[0344] At step 2625, process 2600B may include determining an actual lateral distance to the at least one lane mark based on analysis of the at least one image. For example, the vehicle may determine a distance 2530, as shown in FIG. 25A, representing the actual distance between the vehicle and lane mark 2510. The camera angle, the speed of the vehicle, the width of the vehicle, the position of the camera relative to the vehicle, or various other factors may be accounted for in determining distance 2530.

[0345] At step 2626, process 2600B may include determining an autonomous steering action for the host vehicle based on a difference between the expected lateral distance to the at least one lane mark and the determined actual lateral distance to the at least one lane mark. For example, as described above with respect to FIG. 25B, vehicle 200 may compare actual distance 2530 with an expected distance 2540. The difference between the actual and expected distance may indicate an error (and its magnitude) between the vehicle's actual position and the target trajectory to be followed by the vehicle. Accordingly, the vehicle may determine an autonomous steering action or other autonomous action based on the difference. For example, if actual distance 2530 is less than expected distance 2540, as shown in FIG. 25B, the vehicle may determine an autonomous steering action to direct the vehicle left, away from lane mark 2510. Thus, the vehicle's position relative to

the target trajectory may be corrected. Process **2600B** may be used, for example, to improve navigation of the vehicle between landmarks.

[0346] Processes **2600A** and **2600B** provide examples only of techniques that may be used for navigating a host vehicle using the disclosed sparse maps. In other examples, processes consistent with those described relative to FIGS. **25C** and **25D** may also be employed.

[0347] In some embodiments, the disclosed systems, methods, and non-transitory computer-readable media may use one or more AV maps. An autonomous vehicle (AV) map (or AV map) may include information to support and/or implement one or more autonomous vehicle (AV) functions of a vehicle in manner such that the vehicle operates in a safe manner and/or navigates in an accurate manner. The AV functions supported and/or implemented by an autonomous vehicle (or a semi-autonomous vehicle) may include one or more autonomously controlled functions (e.g., functions determined, selected, and/or implemented based on instructions executed by at least one processor) such as steering, accelerating, and/or braking the vehicle. The AV functions may be part of a driving policy, such as RSS, for example, which was developed and is implemented by Mobileye, of Jerusalem, Israel. The information for operating the vehicle in a safe manner may include, but is not limited to, information pertaining to one or more regulations applicable to a location of the vehicle or a jurisdiction (e.g., a right-hand traffic or left-hand traffic jurisdiction), information related to an environment in which the vehicle is located (e.g., information related to a drivable path, stop sign, traffic light, speed limit, lane markings, landmarks, free space, virtual or physical stop lines, traffic light relevancy information, etc.), and/or information related to adjusting and/or adapting navigation of the vehicle to account for one or more objects (e.g., other vehicles, pedestrians, objects, obstacles, occlusions, hazards, road work zones, traffic cones, etc.) in an environment of the vehicle. The vehicle may sense objects in the environment of the vehicle using one or more sensors (e.g., cameras, radar, lidar), as discussed herein. The AV map may serve as a redundant source of information for the sensed information and, in some cases, may also supplement the sensed information (e.g., provide a location of a virtual stop line, where no stop line is marked on the road). In some embodiments, operating the vehicle in a safe manner may further include operating the vehicle to maintain a comfort level for one or more passengers in the vehicle. The comfort level may include one or more predetermined criteria (e.g., related to speed, acceleration, and/or turning) selected to cause the vehicle to operate in a manner that meets or exceeds a designated or selected level of passenger comfort. The level of comfort may be formalized and expressed in appropriate mathematical formulae. For example, mathematical formulae that limit the extent of jerk or acceleration in different directions that is applied to a passenger may be used. The information for operating the vehicle in an accurate manner may include, but is not limited to, information pertaining to a planned or designated navigational path (e.g., a trajectory, as discussed herein) or a route (e.g., from a particular location, such as a starting location, to a destination). In some embodiments, the information included in an AV map may further include information that supports and/or implements one or more AV functions in an efficient manner. The information for operating the vehicle in an efficient manner may include information pertaining to speed, acceleration, lane changes, and/or lane positioning, and/or traveling and/or selecting a path or route based on traffic conditions (e.g., traveling a route that is a greater distance than a shorter route experiencing traffic conditions) and/or other factors or attributes related to potential routes (e.g., weather conditions, road conditions, or other characteristics of routes, such as traveling a highway without traffic lights instead of a road with traffic lights). In some embodiments, the AV map may further include at least some information from a high-definition (HD) map. In some embodiments, the AV map may be a sparse map, as described above. In some embodiments, the AV map may be crowdsourced, as discussed herein. In this disclosure, the terms “AV map” and “sparse map” are used interchangeable.

#### Object Road Edges

[0348] When navigating, an autonomous or semi-autonomous vehicle may need to detect one or

more objects positioned along the edges of a road segment and/or a boundary associated with a region of free space. The region of free space may correspond to a navigable area in which the vehicle may travel without incident, and the boundary may designate at least a portion of an edge of the region of free space. In some instances, the boundary may correspond to an edge of a road (e.g., a curb or an edge of road pavement). However, in some cases, the boundary of the region of free space may correspond to an edge of one or more objects positioned along the edges of a road segment. For example, in the case of parked vehicles lining the side of a road segment (e.g., parallel parked vehicles), a host vehicle (e.g., an autonomous or semi-autonomous vehicle) navigating the road segment may need to detect the edges of the parked cars, curbs, barriers, etc. to determine a boundary of a free space region in which the host vehicle may safely travel.

[0349] The disclosed systems and methods may identify a boundary associated with a plurality of objects positioned along the edges of a road segment. The objects may include parked vehicles, fences, barriers, trash containers, construction markers, or any combination of objects. In some embodiments, the disclosed systems and methods may determine one boundary line (or edge) for all objects sensed along a road segment rather than determining individual boundary lines or edges for each individual object.

[0350] FIG. 27 is a flowchart showing an exemplary process 2700 for navigating a host vehicle relative to a road segment, consistent with the disclosed embodiments. Process 2700 may be performed by at least one processor or processing device, such as processing unit 110 included in system 100 embarked in host vehicle 200, or various other devices described herein. For example, processing unit 110 executing process 2700 may be configured to navigate host vehicle 200 relative to a road segment. It is to be understood that above and throughout the present description, the term “processor” is used as a shorthand for “at least one processor.” In other words, a processor may include one or more structures (e.g., circuitry) that perform logic operations whether such structures are collocated, connected, or scattered. In some embodiments, a non-transitory computer-readable medium may contain instructions that when executed by a processor cause the processor to perform process 2700. Further, process 2700 is not necessarily limited to the steps shown in FIG. 27 and any steps or processes of the various embodiments described throughout the present disclosure may also be included in process 2700.

[0351] Additionally, some disclosed embodiments may involve communicating with at least one trained system (e.g., a trained model or a trained neural network). Communicating with at least one trained system may include inputting one or more different data entries to the trained system as well as reading or extracting one or more outputs from the trained system. In some embodiments, the at least one trained system may be local to the system. For example, referring to FIG. 1, a trained system may be directly implemented within processing unit 110. Alternatively, in some embodiments, the at least one trained model may be external to the system, and the at least one processor may communicate with the at least one trained model using any known means for sending and/or receiving data. For example, referring to FIG. 1, processing unit 110 may use wireless transceiver 172 to interact and communicate with an external trained model.

[0352] At step 2702, processing unit 110 receives at least one image captured by at least one camera from an environment of the host vehicle. The at least one image may include representations of at least two objects with offset edges in the environment of the host vehicle. The captured image may be acquired by at least one camera onboard the host vehicle. For instance, a camera included in image acquisition unit 120 (such as image capture device 122, 124, or 126 having, respectively, fields of view 202, 204, and 206) may capture the image representing a scene occurring in an area forward of vehicle 200 (or to the sides or rear of a vehicle, for example). Additionally, in some embodiments, the at least one camera may send the captured image to the at least one processor. For example, image capture device 122 may transmit the first image frame over a data connection (e.g., digital, wired, USB, wireless, Bluetooth, etc.) to processing unit 110 via data interface 128.



[0353] As used herein, an object present in an environment of the host vehicle refers to any physical entity that may be represented in a captured image acquired by a camera onboard the host vehicle and/or detected by one or more of the host vehicle's sensors. Examples of such objects include but are not limited to, other vehicles (e.g., cars, trucks, motorcycles, bicycles, etc.), pedestrians, infrastructure (e.g., traffic signs, signals, road markings, barriers, and buildings), obstacles (e.g., debris, animals, or any unexpected items on the road), or environmental features (e.g., trees, curbs, and other natural or man-made structures). Consistent with the disclosed embodiments, the at least two objects in the environment of the host vehicle may have offset edges. In this context, “offset edges” refers to the positioning of the edges of two or more objects within the vehicle's environment that are not aligned in a straight or parallel manner, but instead are slightly shifted or displaced relative to one another. This may occur in various real-world scenarios, such as when objects (e.g., curbs, road signs, or parked vehicles) are not perfectly aligned along the same plane or axis. For example, when the host vehicle evolves in its environment, it may encounter objects with edges that are offset due to variations in their position or orientation. This offset may result from the objects being placed at different distances from the host vehicle, or due to the terrain or road conditions causing the objects to appear misaligned in the host vehicle's visual field. It would be appreciated that an edge as used herein relates to a line or a surface, which generally demarcates the respective object along the path of the vehicle (e.g., along a lane in which the vehicle is traveling). In reality, this line or surface may be an approximation and may not necessarily correspond accurately to the surface or edge of the object.

[0354] In some embodiments, the at least two objects may be parked vehicles. Additionally, in some embodiments, the parked vehicles may be positioned parallel to an edge of a road segment. In some other embodiments, the at least two objects in the environment of the host vehicle may include a variety of objects such as construction barriers, construction bins, or road debris. It is to be appreciated that the two or more objects in the environment may differ in nature, such as vehicles and road barriers, for example. In some embodiments, the at least two objects are not necessarily static and may include moving objects such as a pedestrian or bicycle that are traveling along the road segment.

[0355] FIG. 28A illustrates an exemplary image, **2800a** captured by an onboard camera (e.g., image capture device **122**) in a host vehicle (e.g., host vehicle **200**) traveling on road segment **2805**. Consistent with the disclosed embodiments, image **2800a** depicts the environment surrounding host vehicle **200** and includes representations of various objects or features within that environment. In the example shown in FIG. 28A, four vehicles—**2810-1**, **2810-2**, **2810-3**, and **2810-4**—are parked parallel to one edge of road segment **2805**. Parked vehicles **2810-1**, **2810-2**, **2810-3**, and **2810-4** have different offset edges **2820-1**, **2820-2**, **2820-3**, and **2820-4** shown as dashed lines.

[0356] In some embodiment, processing unit **110** may be further configured to detect the at least two objects by: analyzing the at least one image, using an output of a RADAR system, using an output of a LIDAR system, using an output of a trained system, or a combination thereof. In other words, the at least two objects may be detected and identified by leveraging multiple sensing and processing techniques, either individually or in combination, to enhance accuracy and robustness. One approach may involve analyzing the at least one captured image, where computer vision algorithms may process visual data to identify and/or classify objects based on characteristics such as shape, color, texture, and contrast. Another approach may rely on RADAR system output, which may detect objects by transmitting radio waves and analyzing their reflections to determine distance, speed, and relative position. RADAR may be particularly useful in poor visibility conditions such as fog, rain, or darkness, where traditional cameras may struggle. Yet another approach may use LIDAR system output, which may employ laser pulses to scan the environment and generate high-resolution 3D point clouds, allowing for precise object detection and depth perception. Furthermore, a trained system, such as an AI-driven neural network or a VIDAR

(Visual Detection and Ranging) system, may be employed to recognize and classify objects based on vast amounts of pre-learned data, improving detection in complex environments with overlapping or partially obscured objects. VIDAR may combine elements of both computer vision and LIDAR, utilizing multiple high-resolution cameras to construct a 3D representation of the environment. Unlike LIDAR, which relies on laser reflections, VIDAR may leverage image-based triangulation techniques, offering a passive and cost-effective alternative for depth perception and object localization. VIDAR may be particularly effective for detecting objects at greater distances while reducing the reliance on active laser scanning.

[0357] To maximize detection reliability, processing unit **110** may also combine these approaches, integrating multiple sensor outputs to compensate for individual limitations, ensuring more comprehensive object identification and tracking. This multi-modal approach may enhance the perception of the surrounding environment beneficial for safe and efficient operation. It is to be appreciated that both a LIDAR and RADAR system may be internal to the host vehicle while the computer vision algorithm and or the trained system may be either internal or external to the host vehicle. In a situation where the computer vision algorithms and/or trained system are external to the host vehicle, processing unit **110** may use wireless transceiver **172** to interact and communicate with the external computer vision algorithms and/or trained system. Referring to FIG. **28A**, any of the parked vehicles (**2810-1** through **2810-4**) may be detected using at least one of the aforementioned techniques or any combination thereof. Additionally, the detection process may include identifying and deriving the associated edges (**2820-1** through **2820-4**) of the parked vehicles, which help define their boundaries within the environment of road segment **2805**.

[0358] Furthermore, in some embodiments, processing unit **110** may be configured to determine whether one or more of the at least two objects are moving. For example, referring to FIG. **28A**, processing unit **110** may determine whether any of the parked vehicles **2810-1** through **2810-4** are exiting their parking spots. To make this determination, processing unit **110** may receive and analyze one or more additional images beyond the initial image received for example at step **2702** of process **2700**. By comparing and tracking the position of the detected objects across these successive images, processing unit **110** may apply computer vision algorithms to track the movement and identify if any of the objects are shifting, such as a parked vehicle beginning to move.

[0359] Additionally, in some embodiments, processing unit **110** can employ other techniques, such as RADAR, LIDAR, or a trained system (e.g., VIDAR system), either individually or in combination, to detect the motion of the objects. For instance, RADAR may measure doppler (from which speed can be derived) and changes in distance over time, while LIDAR may provide high-resolution 3D mapping (and advanced Lidar systems, such as FMCW Lidars can also provide doppler measurements), allowing the system to monitor object positions and detect movement. VIDAR, which integrates vision and ranging capabilities, may also be used to track the movement of objects using both visual and depth data. These techniques, particularly when combined, may enhance detection accuracy and ensure that even subtle movements are recognized, such as a vehicle slightly shifting within its parking space or a pedestrian walking along the road segment. By leveraging multiple sensor modalities, processing unit **110** may more reliably determine if the at least two objects are moving.

[0360] At step **2704**, processing unit **110** analyzes the at least one image to determine a free space boundary relative to the at least two objects. The free space boundary may represent an edge of at least one region of free space and may follow a path different from a path represented by the offset edges of the at least two objects. In the context of the present disclosure, a region of free space refers to an open area that is free from obstacles, where movement or safe navigation by the host vehicle can occur without interference. The region of free space may be part of an environment that is accessible to the host vehicle based on the detected positions of objects. The free space boundary may represent the dividing line between the region of free and areas occupied or obstructed by the

two or more objects. The free space boundary may therefore mark the limits of where the host vehicle can move freely. In some embodiments, the free space boundary may be a continuous line. The free space continuous boundary line may extend from a first one of the detected objects to a last one of the detected objects (e.g., such as a line of parked vehicles). Additionally, in some embodiments, the free boundary space may be substantially straight. Unlike offset edges, which provide a buffer around obstacles, the free space boundary may represent the true extent of navigable space, ensuring efficient and safe motion planning.

[0361] Determining the free space boundary (e.g., a continuous boundary line or edge) for objects detected along a road segment may offer several advantages compared to identifying individual boundary lines for each object separately. First, it may reduce computational load and minimize processing resources by simplifying the determination of the relevant free space boundary, leading to more efficient navigation. Second, this approach may be significantly faster than individually measuring each vehicle or object along the road. Third, it may enhance detection accuracy in scenarios where one object partially or fully obscures another, ensuring a more reliable representation of the environment. Fourth, a continuous boundary may help facilitate smoother steering for the host vehicle by maintaining a consistent distance from the boundary rather than adjusting position when passing gaps between objects. Finally, a smoother and more predictable driving experience may enhance passenger comfort, reducing unnecessary acceleration, braking, or lateral movements.

[0362] Referring to the exemplary image **2800a** shown in FIG. **28A**, processing unit **110** may analyze image **2800a** and determine free space boundary **2830a** which mark the limit between a region of free space (on the left side of free space boundary **2830a**), and a region occupied by the parked vehicles **2810-1** through **2810-4**. Consistent with the disclosed embodiments, free space boundary **2830a** follows a path different than a path represented by the offset edges **2820-1** through **2820-4** of the parked vehicles **2810-1** through **2810-4**. In some embodiments, the free space boundary may extend along an edge of the at least two objects. For example, referring to FIG. **28A**, free space boundary **2830a** extends along edges **2820-1** through **2820-4** of parked vehicles **2810-1** through **2810-4**. Additionally, or alternatively, in some embodiments, the free space boundary may be substantially parallel to an edge of a road or a curb. For example, free space boundary **2830a** is substantially parallel to the right edge of road segment **2805**. Additionally, or alternatively, in some embodiments the free space boundary may be positioned at an offset relative to a surface of an object that is at the edge of a road. The offset can be different for different object or for different circumstances associated with the object. For example, in the example of parked cars, the offset can be created to accommodate a door opening scenario, which may effectively narrow the free space of a vehicle at the respective location of the offset. In this regard, the edge generation can include obtaining a characteristic of the object at the edge of the road or path of the vehicle and using the characteristic of the object at the edge of the road or path in the edge generation process.

[0363] In some embodiments, analyzing the at least one image to determine the free space boundary associated with at least two objects may include determining a plurality of intersections between a plurality of free space search lines and the offset edges of the at least two objects. Within the context of this disclosure, free space search lines refer to 2D lines used in a vision system to detect obstructions (e.g., caused by the at least two objects) and determine the boundaries of navigable areas. Free space search lines may be substantially perpendicular to a direction of a road segment, an optical axis of a camera, and/or a heading direction of a host vehicles. Each search line may be processed separately. When data from multiple cameras is received, the free space search lines may vary between cameras and are processed individually to identify obstructions at the edges of the free space. Each free space search line may be divided into a left side and right side, and finding obstruction or intersections with offset edges may begin at the center of the free space search line to the outside of the free space search line. It is to be appreciated that the number of free space search lines and the distance between them may depend on one or more factors, including for

example, camera properties, host vehicles dynamics, road conditions, weather conditions, lighting conditions, traffic density, or processing power.

[0364] Once intersections between the free space search lines and the offset edges of the objects are identified, the free space boundary may be determined. For example, the positions of the different intersections may be interpolated, outliers may be removed, and a smoothing process or other preprocessing techniques may be applied. Based on the output of these processes, the free space boundary line may be derived. It is to be appreciated that the free space boundary line does not necessarily cross or directly connect the intersections. For example, there may be an offset distance between the intersections and the free space boundary line. As mentioned earlier, offset edges (e.g., **2820-1** through **2820-4**) may be identified and derived through a detection process involving techniques such as image analysis, RADAR, LIDAR systems, and trained systems (e.g., VIDAR), either individually or in combination. Referring to the exemplary image **2800a** shown in FIG. **28A**, processing unit **110** may determine intersections between free space search lines **2850** (e.g., right side of the lines) and the offset edges **2820-1** through **2840-1** of the parked vehicles to determine free space boundary **2830a**. As further explained below, in some embodiments, the at least two objects may be separated by a distance or gap, in which case at least one free space search line may not intersect an edge. In such situations, if the distance is below a certain threshold, data from neighboring intersections may be used to determine a virtual intersection for the at least one free space search line.

[0365] In some embodiments, analyzing the at least one image to determine the free space boundary associated with at least two objects may include identifying a potential area of free space based on at least one map and determining the free space boundary relative to the at least two objects based on the potential area of free space and the at least one image. Initially, a region of free space may be estimated using a map-based baseline, which may include Autonomous Vehicle (AV) maps designed to support safe and precise vehicle navigation. AV maps may contain information, i.e., geographic and spatial data such as road layouts, intersections, traffic signs, lane markings, speed limits, or landmarks, that aids autonomous or semi-autonomous vehicle functions, such as steering, acceleration, and braking, by providing pre-mapped data of the driving environment. One example of such a map is the Road Experience Management (REM) map developed by Mobileye™ of Jerusalem, Israel, which includes representations of longitudinal road features such as 3D splines, target trajectories, drivable paths, road edges, barriers, and lane markings. It is to be understood that above and throughout the present description, the term “map” refers equally to a digital map and/or a sparse map, i.e., a map that may provide sufficient information for navigating a host vehicle without storing and/or updating large quantities of data (e.g., image data, complete road topography representations of various features, etc.). To determine the free space boundary, map features (e.g., road edges) may first be used as a reference, then the free space region may be adjusted based on real-time detection of objects along the road, such as parked vehicles. The refinement process may involve analyzing image data from vehicle-mounted cameras, as well as data from RADAR, LIDAR, trained systems, VIDAR sensors, or a combination of these sources. By processing this multi-modal data, a free boundary line that extends across multiple detected objects, such as a series of parked vehicles, may be established. For example, in some embodiments, data obtained from image analysis—such as points indicating edges of objects in an image frame—may be combined with map features (e.g., point indicating road edges) to determine object-free space. Object-free space may refer to areas where no obstacles are detected, enabling safe navigation. Such as free space may be delineated via points marking edges of the object free space. VIDAR data (e.g., point clouds corresponding to the object in an image) may then be analyzed in relation to the determined object-free space. Specifically, VIDAR points may be compared with points defining the object-free space, and any VIDAR points located beneath these object-free space points (in image space) may be identified. If these identified VIDAR points are near the road edge (in world space), they may be used as inputs for refining the free-space

boundary. As more evidence accumulates (e.g., over multiple frames and/or multiple analysis), the free-space boundary may be adjusted and finalized based on the identified VIDAR points. This approach may ensure a more accurate and robust determination of navigable areas by integrating multi-modal sensor data. The determined boundary line may enclose a portion of the free space region initially estimated from the map, ensuring that navigation decisions are based on both static map data and real-time environmental observations.

[0366] Referring to FIG. **28A**, processing unit **110** may initially determine the free space boundary **2830a** using a potential area of free space, which is based on a map including information about road segment **2805** stored either in internal or external memory accessible to processing unit **110**. Afterward, processing unit **110** may refine this boundary by analyzing image **2800a** using the representations of parked vehicles **2810-1** through **2810-4** within image **2800a**.

[0367] In situations where one or more of the at least two objects are moving, processing unit **110** may be further configured to dynamically adjust the position of the free space boundary to accommodate the perceived changes in the environment. For example, if a parked vehicle begins to exit its parking space, processing unit **110** may alter the boundary to ensure that the space remains accurately defined and navigable. This adjustment could involve recalculating the available free space based on the new positions of the moving objects, ensuring that the host vehicle maintains safe clearance and optimal route planning. Processing unit **110** may also rely on continuous data from vision-based sensors, RADAR, LIDAR, or VIDAR systems to track the movement and make real-time modifications to the free space boundary, enabling the host vehicle to navigate around dynamic obstacles and adapt to shifting conditions in the environment. By continuously updating the boundary based on object movement, the host vehicle's ability to react to real-world changes may be enhanced, ensuring safer and more efficient navigation.

[0368] At step **2706**, processing unit **110** determines at least one navigational action based on the free space boundary. For example, processing unit **110** may utilize a navigation module or system to determine the appropriate action required to safely and efficiently navigate the host vehicle. This process may involve analyzing the insights provided by the free space boundary, which represents the available space for movement within the environment. The navigation system may consider factors such as the proximity of obstacles, the movement of dynamic objects, and/or the shape of the free space region. The navigation system may obtain this data and compute one or more actions for the vehicle. In some embodiments, the at least one navigational action may include steering, braking, or accelerating the host vehicle. For instance, if the free space boundary indicates an open path, the navigation system may decide to accelerate or continue on a current trajectory.

Conversely, if the free space boundary indicates a narrowing path or the presence of obstacles (e.g., parked cars, pedestrians, or road debris), the navigation system may slow down or adjust the steering angle to avoid collisions and maintain safe navigation. Referring to FIG. **28A**, since none of the parked vehicles **2810-1** through **2810-4** are moving and the free space boundary **2830a** delineates a clear region of free space, the navigation system associated with the host vehicle may decide to maintain the current speed or accelerate the host vehicle.

[0369] Additionally, or alternatively, in some embodiments, the at least one navigational action may include altering a planned trajectory of the host vehicle based on the free space boundary. A planned trajectory may be based on a map. The planned trajectory, which may be pre-determined based on map data and road conditions, may need to be adjusted in real-time to account for dynamic changes in the environment. For instance, if the free space boundary reveals an obstacle or restricted space that was not accounted for in the initial trajectory (such as a parked vehicle, construction barrier, or a pedestrian entering the roadway), the system may adjust the vehicle's path to avoid potential collisions or hazardous situations.

[0370] The navigation system may also integrate other factors such as vehicle dynamics, traffic conditions, and road geometry (e.g., curves, intersections) to optimize the decision-making process. This dynamic and real-time analysis ensures that the vehicle's movements are not only safe but also

smooth and efficient, contributing to overall comfort and safety for both passengers and other road users.

[0371] At step **2708**, processing unit **110** causes the host vehicle to implement the at least one navigational action. For example, processing unit **110** may cause the activation of one or more actuators of the host vehicle (e.g., host vehicle **200**) associated with a steering system (e.g., maintaining or changing a current heading direction), a braking system (e.g., reducing a current speed), or a drive system of vehicle **200** (e.g., accelerating, deaccelerating, reducing a current speed).

[0372] In some embodiments, processing unit **110** may be further configured to determine whether at least one moving object is crossing the free space boundary. Additionally, in some embodiments, the at least one moving object may be a pedestrian, a vehicle, or any other dynamic entity within the environment. The at least one moving object may either be one of the at least two objects with offset edges or an additional object. For example, referring to FIG. **28A**, the moving object could be one of the parked vehicles **2810-1** through **2810-4** exiting its parking spot and crossing the free space boundary **2830a**, or a pedestrian crossing the road segment **2805** and thus crossing the free space boundary **2830a** from a position between the parked vehicles **2810-1** through **2810-4**.

Processing unit **110** may determine whether a moving object is crossing the free space boundary using various methods previously mentioned, such as image analysis, RADAR, LIDAR, or a trained system (e.g., VIDAR), either individually or in combination. Furthermore, in some embodiments, determining the at least one navigational action may be further based on whether the at least one moving object is crossing the free space boundary. For instance, referring to FIG. **28A**, if a moving object (e.g., one of the parked vehicles exiting or a pedestrian crossing) is detected crossing free space boundary **2830a**, the navigational action may involve decelerating, braking, stopping, or taking evasive action to avoid a potential collision or incident. This real-time response may ensure that the host vehicle remains safe and responsive to dynamic changes in its environment, minimizing the risk of accidents.

[0373] In some embodiments, the at least two objects may be spaced apart from one another by a distance. For example, in some embodiments, the distance may be less than 1 meter, 5 meters, or 10 meters. In other words, one or more gaps may exist between objects lining the side of a road segment. For example, returning to the case shown in FIG. **28A** of parallel parked vehicles **2810-1** through **2810-4** on a side of road segment **2805**, one or more gaps may exist between the parked vehicles **2810-1** through **2810-4** (e.g., gaps to enable parked cars to exit a parking space, an empty parallel parking spot, multiple empty parking spaces, etc.).

[0374] In some embodiments, the distance may be greater than a threshold distance enabling the host vehicle to either partially or fully enter an area of free space located between the at least two objects. In the example of parked vehicles, some gaps, such as gaps of less than one car length are not possible for a vehicle to enter. Other gaps, such as gaps constituting an empty parking spot between two parked vehicles, may be sufficiently large for a vehicle to enter (e.g., during a parking maneuver), but may be impractical or impossible for a vehicle to travel within. For example, FIG. **28B** represents a captured image **2800b** that illustrates a situation similar to the one shown in FIG. **28A**, where a space of less than one vehicle exists between parked vehicle **2810-1** and parked vehicle **2810-3**. In contrast, FIG. **28C** represents a captured image **2800c** that illustrates a situation where a space of more than one vehicle exists between parked vehicle **2810-1** and parked vehicle **2810-3**. In some embodiments, when the distance is greater than the threshold distance, the at least one navigational action may include steering the host vehicle to either partially or fully enter the area of free space located between the at least two objects. Alternatively, in some embodiments, the at least one navigational action may include parking the host vehicle within the area of free space located between the at least two objects. In other words, in cases wherein a more substantial distance (e.g., ten meters, 50 meters, 100 meters, or more) exists between objects lining the side of a road segment, it may be appropriate for a host vehicle to navigate in the area of free

space located between the at least two objects. Such navigational actions may include a vehicle entering a gap to clear a lane of travel along a road segment (e.g., along a two-way road with rows of cars parked along opposite sides of the road where insufficient free space exists between the rows of parked cars for two navigating vehicles to pass simultaneously). Such navigational actions may also include a navigating vehicle partially entering a free space gap along a side of a road in order, e.g., to increase a buffer distance relative to other navigating vehicles, pedestrians, objects, etc. In another scenario, such navigational actions may include parking the vehicle. Accordingly, for autonomous or semi-autonomous vehicle navigation, the disclosed system may identify boundaries of regions of free space bordered by objects and determine whether gaps between the objects are sufficient for a host vehicle to travel within. In the situation shown in FIG. 28C, the gap between parked vehicle **2810-1** and parked vehicle **2810-4** may be sufficient to enable the host vehicle to enter either partially or fully within the gap or to park within the gap.

[0375] In some embodiments, the at least one navigational action may be determined based on the free space boundary, the distance between the at least two objects, a speed of the host vehicle, a planned angle of entry into the area of free space, or a combination thereof. For example, the determined navigation action may involve fully entering a gap between parked vehicles if the vehicle is traveling at a first speed. However, if the vehicle is moving at a second speed higher than the first, the vehicle may continue to travel on its path instead, as a larger gap would be needed to accommodate the vehicle's movement. At a higher speed, the vehicle may cover more distance in a shorter time, leaving less room for error. Other factors may also influence the determined navigational action such as the host vehicle's size, terrain type, environmental conditions (e.g., weather), time of day, or road topography (e.g., lane widths, whether a road segment is one-way or two-way, number of lanes, proximity to pedestrians, proximity to moving vehicles, etc.).

[0376] In some embodiments, the distance is greater than a predetermined distance at least a portion of the free space boundary corresponds to a road edge. In other words, over a sufficiently long distance, the free space boundary may indicate that a section of the road segment is clear of any objects. In such cases, the free space boundary may be determined solely based on mapped information. For example, referring to FIG. 28C, the distance between parked vehicles **2810-1** and **2810-3** may exceed a predetermined threshold, and a portion of the free space boundary **2830c** may correspond to the right edge of road segment **2805**.

[0377] In some embodiments, processing unit **110** may be further configured to refine the free space boundary based on a driving policy associated with the host vehicle. A driving policy refers to a decision-making system that guides the vehicle's navigational actions based on sensed conditions. This policy may include a set of rules that trigger specific navigational responses, such as slowing down in preparation for an upcoming stop sign. In other cases, the driving policy may be broader, setting constraints that limit navigational actions in response to sensed conditions, such as ensuring a minimum safe distance or restricting acceleration beyond a certain rate. For example, Responsibility Sensitive Safety (RSS) technology, developed by Mobileye™, is an example of a driving policy framework. RSS is used herein and throughout this description as an example of a Safety Driving Model (SDM) that is a model that is implemented by a host vehicle and designed to provide a safety guarantee regarding the host's vehicle autonomous driving. In the case of RSS, the safety guarantee is provided by a set of rigorous mathematical formulae that guarantee, using a set of assumptions related to reasonable behavior of actors in the environment of the vehicle, that the host vehicle will not cause an accident that is the host vehicle's fault (as defined by the mathematical formulae). Once the continuous boundary line for a portion of the free space region is determined, the host vehicle's navigation system may apply the driving policy to regulate how the vehicle navigates, which may influence the free space boundary. For instance, based on the driving policy, the host vehicle might continue along a straight (or nearly straight) path even if there is a gap between parked vehicles (or other objects), with the free space boundary extending along the parked vehicles, ignoring the gap. This is shown in FIG. 28B, where the free space boundary **2830b**

remains unaffected by the gap between parked vehicles **2810-1** and **2810-3**. However, in other embodiments, if the gap between the parked vehicles is large enough, the driving policy may allow the host vehicle to move fully or partially into the gap. This situation is shown in FIG. **28C**, where free space boundary **2830c** includes a portion that corresponds to the road edge of road segment **2805** and thus indicates the potential entry of the host vehicle within the gap between parked vehicles **2810-1** and **2810-3**.

[0378] In some embodiments, analyzing the at least one image to determine the free space boundary relative to the at least two objects may further comprise receiving an output provided by a trained system. The output may include a height estimate for a plurality of pixels in the at least one image associated with one of the at least two objects. Generating such per-pixel height data from captured images may involve several techniques to infer the three-dimensional structure of the scene. One approach is stereo vision analysis, which leverages images from multiple cameras positioned at different angles to determine height by measuring disparities between the images. For example, to achieve this, a host vehicle may be equipped with a set of cameras, including a center camera, a left surround camera, and a right surround camera. The center camera's field of view (FOV) may partially overlap with the FOVs of both the left and right surround cameras. The center camera may be configured to capture images (referred to as center images) of a portion of the vehicle's environment within its FOV. Similarly, the left and right surround cameras may capture images (left surround images and right surround images, respectively) of the surrounding environment within their respective FOVs. The host vehicle or a processing unit with the host vehicle may receive these images and transmit them to an analysis module—potentially incorporating one or more trained neural networks—that processes the images to generate depth and/or height estimates for individual pixels. The resulting output may include per-pixel depth and/or height information for at least one region of the center image or another captured image.

[0379] Another method, Structure from Motion (SfM), may reconstruct a 3D representation of the scene using a sequence of images taken from different viewpoints as the vehicle moves. Additionally, RADAR and/or LIDAR sensors integrated into the host vehicle may provide distance measurements, which, when fused with camera images, may create detailed 3D maps of the environment. These methods may require capturing and preprocessing multiple images, detecting and matching features, estimating height, postprocessing the 3D maps, and integrating data from other sensors.

[0380] Alternatively, for example, if only a single image from a single camera is available, machine learning, deep learning, or neural network models—trained on large datasets—may be used to predict height. These models may analyze image features and patterns to infer spatial characteristics of objects and surfaces, enabling accurate height estimations from a single image.

[0381] In some embodiments, the output may further include a range estimate for a plurality of pixels in the at least one image associated with one of the at least two objects. Range information may be determined using any of the above-mentioned approaches. Furthermore, in some embodiments, the generation of the range information may be based, at least in part, upon previously generated height information. Conversely, in other embodiments, the generation of height information may be based, at least partially, upon previously generated range information. This reciprocal relationship suggests that insights gleaned from one aspect of the scene (either height or range) may be used to refine estimation regarding the other aspect. For example, by incorporating height information, the distance to objects in the scene may be better estimated, aiding in the generation of accurate range information. Similarly, using range information may enhance the ability to determine the vertical position of objects, thereby improving the generation of height information.

[0382] Consistent with the disclosed embodiments, the plurality of pixels may be associated with one of the at least two objects, i.e., areas of interest in the captured image for which height and/or range information is relevant. Determination of such a plurality of pixels may involve the use of a



pre-segmentation algorithm. These algorithms may scan the image to identify and segment different features or objects, such as vehicles, pedestrians, or obstacles. Based on the output of such pre-segmentation, processing unit **110** may decide which features or objects require height and/or range information. The choice of pixels may be based on the need for height and/or range information to understand the 3D structure and distances in the host vehicle's environment. In some situations, the number of objects or features within the image may be sufficiently important to trigger the analysis of the entire image. In some other scenarios, only specific areas of interest may be identified. By focusing on specific areas of interest, processing unit **110** may more accurately interpret valuable information needed for navigation. This targeted approach may help in avoiding unnecessary processing of irrelevant parts of the image, thus optimizing the vehicle's decision-making process and ensuring safe and efficient operation. In some embodiments, the height estimate and/or range estimate may be associated with the offsets edges of the at least two objects. [0383] In some embodiments, analyzing the at least one image to determine the free space boundary relative to the at least two objects may further comprise using the height estimate and/or the range estimate to determine the free space boundary. In other words, the disclosed systems and methods may further analyze the height and/or range of one or more road edge objects when determining a free speed region and thus a navigable free space for a host vehicle. FIG. **28D** represented a captured image **2800d** that illustrates a situation similar to the one shown in FIG. **28A**, further including estimated height and/or range for parked vehicles **2810-1** through **2810-4**. Specifically, in this example, estimated height and/or range may be represented by bounding boxes **2840-1** through **2840-4**, i.e., geometric shapes that can be used to represent the spatial extent of objects in three-dimensional space. By determining bounding boxes for objects, processing unit **110** may delineate the precise area within the captured image that each object occupies and further use this information to determine the free space boundary. For example, bounding boxes **2840-1** through **2840-4** may be used to determine free space boundary **2840d**. It is to be appreciated that these bounding boxes may be further to determine the position of parked vehicles **2810-1** through **2810-4** edges.

[0384] In some embodiments, determining the at least one navigational action may be further based on the height and/or range estimate. Beyond just using the free space boundary, the navigation system may use height and/or distance estimates to refine its decision-making process regarding vehicle movement. For example, the navigation system may adjust the host vehicle's speed and trajectory based on the size and proximity of surrounding objects. If an object with significant height—such as another vehicle, a pedestrian, or a large obstacle—is detected within a predefined safety zone (e.g., one meter or less), the system may slow down or alter the vehicle's path to maintain a safe distance. Conversely, if the detected object is smaller, such as a curb, a road edge, or a minor obstruction, the system may allow the vehicle to pass by at a relatively higher speed while still ensuring safe navigation. Additionally, the navigation system may incorporate predictive modeling to assess potential risks associated with different types of objects. For example, a tall and stationary object, such as a parked truck, may require a wider clearance to avoid potential side collisions, whereas a low-lying object, such as road debris, may only necessitate a minor speed adjustment. By integrating height and/or range data into the decision-making process, the navigation system may improve the precision of its navigational actions, ensuring smoother and safer autonomous operation while maintaining efficiency in diverse driving scenarios.

[0385] In some embodiments, at least one of the at least two objects may be an object overhanging a road surface of the road and the height and/or range estimate may be relative to at least a portion of the object overhanging the road surface. In some embodiments, the object overhanging the road surface may include at least one of a lamppost, gates, barriers, tree branches, or a traffic sign. Any of these objects may include non-vertical side walls and/or overhanging features that may cause interference with a navigating vehicle depending on the height and extent of overhang relative to a road surface. In some embodiments, the disclosed systems and methods may classify an object

exhibiting one or more overhanging features at a height less than about two meters in height from the road's surface as an obstacle to a host vehicle. However, objects that have overhanging features higher above a road surface (e.g., objects with overhanging features at five or ten meters or more above the road's surface) may not constitute an obstacle to the host vehicle. For example, certain overhanging trees, traffic lights, posts, etc., may not present an issue for host vehicle navigation. Additionally, in some embodiments, the disclosed systems and methods may use a shadow of an object as a guide for edge detection, especially relative to edges of overhanging features extending above a road surface (e.g., a shadow cast by a non-vertical wall of a trapezoidal object).

[0386] FIG. 29 illustrates an exemplary image, 2900 captured by an onboard camera (e.g., image capture device 122) in a host vehicle (e.g., host vehicle 200) traveling on road segment 2905.

Consistent with the disclosed embodiments, image 2900 depicts the environment surrounding host vehicle 200 and includes representations of various objects or features within that environment. In the example shown in FIG. 29, two objects, namely a road barrier 2910-1 and a parked vehicle 21910-2 are represented. Road barrier 2910-1 includes a portion overhanging the surface of road segment 2905. Height and/or range estimates are in this example, illustrated as points 2920.

Consistent with the disclosed embodiments, processing unit 110 may analyze image 2900 to determine a free space boundary 2930. Given the height and range information obtained for road barrier 2910-1, the system may define the free space boundary 2930 to terminate before road barrier 2910-1, as it presents an obstruction. As a result, the navigational action determined in this scenario may be to stop the vehicle before reaching the end of the free space boundary 2930, preventing a potential collision with the overhanging portion of road barrier 2910-1.

[0387] In some embodiments, the at least two objects may be located on one side of the road segment. For instance, in FIGS. 28A-D, parked vehicles 2810-1 through 2810-4 are positioned along the right side of road segment 2905. In contrast, FIG. 29 depicts both road barrier 2910-1 and parked vehicle 2910-2 situated on the left side of road segment 2905. However, in some scenarios, objects may be present on both sides or along the edges of a road segment. For example, in some embodiments, the at least one image (e.g., received at step 2702) may further include representations of at least two additional objects with offset edges in the environment of the host vehicle and located on another side of the road segment. In this situation, processing unit 110 may be further configured to analyze the at least one image to determine an additional free space boundary relative to the at least two additional objects. The additional free space boundary may represent an additional edge of the at least one region of free space and may follow a path different from a path represented by the offset edges of the at least two additional objects. Any of the previously mentioned embodiments related to the free space boundary also apply to the additional free space boundary. In particular, any of the methods or techniques previously described for determining and refining a free space boundary remain applicable to this additional free space boundary. Determining the additional free space boundary may provide several advantages. It may enhance the host vehicle's ability to map and navigate complex environments, especially in narrow roadways or urban settings where obstacles may be present on both sides. For example, in some embodiments, determining the at least one navigational action may be further based on the additional free space boundary. By considering multiple boundaries, the navigation system can optimize path planning, avoid potential collisions, and ensure smooth maneuvering, whether in autonomous or driver-assist modes. Moreover, the ability to refine both primary and additional free space boundaries may improve overall situational awareness, allowing for dynamic adjustments in response to changing road conditions or unexpected obstacles.

[0388] The foregoing description has been presented for purposes of illustration. It is not exhaustive and is not limited to the precise forms or embodiments disclosed. Modifications and adaptations will be apparent to those skilled in the art from consideration of the specification and practice of the disclosed embodiments. Additionally, although aspects of the disclosed embodiments are described as being stored in memory, one skilled in the art will appreciate that

these aspects can also be stored on other types of computer-readable media, such as secondary storage devices, for example, hard disks or CD ROM, or other forms of RAM or ROM, USB media, DVD, Blu-ray, 4K Ultra HD Blu-ray, or other optical drive media.

[0389] Computer programs based on the written description and disclosed methods are within the skill of an experienced developer. The various programs or program modules can be created using any of the techniques known to one skilled in the art or can be designed in connection with existing software. For example, program sections or program modules can be designed in or by means of .Net Framework, .Net Compact Framework (and related languages, such as Visual Basic, C, etc.), Java, C++, Objective-C, HTML, HTML/AJAX combinations, XML, or HTML with included Java applets.

[0390] Moreover, while illustrative embodiments have been described herein, the scope of any and all embodiments having equivalent elements, modifications, omissions, combinations (e.g., of aspects across various embodiments), adaptations, and/or alterations as would be appreciated by those skilled in the art based on the present disclosure. The limitations in the claims are to be interpreted broadly based on the language employed in the claims and not limited to examples described in the present specification or during the prosecution of the application. The examples are to be construed as non-exclusive. Furthermore, the steps of the disclosed methods may be modified in any manner, including by reordering steps and/or inserting or deleting steps. It is intended, therefore, that the specification and examples be considered as illustrative only, with a true scope and spirit being indicated by the following claims and their full scope of equivalents.

## Claims

1. A non-transitory computer-readable medium storing instructions executable by at least one processor to perform a method for navigating a host vehicle relative to a road segment, the method comprising: receiving at least one image captured by at least one camera from an environment of the host vehicle, wherein the at least one image includes representations of at least two objects with offset edges in the environment of the host vehicle; analyzing the at least one image to determine a free space boundary relative to the at least two objects, wherein the free space boundary represents an edge of at least one region of free space and follows a path different from a path represented by the offset edges of the at least two objects; determining at least one navigational action based on the free space boundary; and causing the host vehicle to implement the at least one navigational action.
2. The non-transitory computer-readable medium of claim 1, wherein analyzing the at least one image to determine the free space boundary associated with at least two objects includes identifying a potential area of free space based on at least one map and determining the free space boundary relative to the at least two objects based on the potential area of free space and the at least one image.
3. The non-transitory computer-readable medium of claim 1, wherein the at least two objects are parked vehicles.
4. The non-transitory computer-readable medium of claim 3, wherein the parked vehicles are positioned parallel to an edge of a road segment.
5. The non-transitory computer-readable medium of claim 1, wherein the at least two objects are spaced apart from one another by a distance.
6. The non-transitory computer-readable medium of claim 5, wherein the distance is less than 1 meter.
7. The non-transitory computer-readable medium of claim 5, wherein the distance is less than 5 meters.
8. The non-transitory computer-readable medium of claim 5, wherein the distance is less than 10 meters.

9. The non-transitory computer-readable medium of claim 5, wherein the distance is greater than a threshold distance enabling the host vehicle to either partially or fully enter an area of free space located between the at least two objects.
10. The non-transitory computer-readable medium of claim 9, wherein the at least one navigational action includes steering the host vehicle to either partially or fully enter the area of free space located between the at least two objects.
11. The non-transitory computer-readable medium of claim 9, wherein the at least one navigational action includes parking the host vehicle within the area of free space located between the at least two objects.
12. The non-transitory computer-readable medium of claim 9, wherein the at least one navigational action is determined based on the free space boundary, the distance between the at least two objects, a speed of the host vehicle, a planned angle of entry into the area of free space, or a combination thereof.
13. The non-transitory computer-readable medium of claim 5, wherein the distance is greater than a predetermined distance at least a portion of the free space boundary corresponds to a road edge.
14. The non-transitory computer-readable medium of claim 1, wherein the free space boundary extends along an edge of the at least two objects.
15. The non-transitory computer-readable medium of claim 1, wherein the free space boundary is substantially parallel to an edge of a road or a curb.
16. The non-transitory computer-readable medium of claim 1, wherein the at least one navigational action includes steering, braking, or accelerating the host vehicle.
17. The non-transitory computer-readable medium of claim 1, wherein analyzing the at least one image to determine the free space boundary relative to the at least two objects further comprises receiving an output provided by a trained system, wherein the output includes a height estimate for a plurality of pixels in the at least one image associated with one of the at least two objects.
18. The non-transitory computer-readable medium of claim 17, wherein analyzing the at least one image to determine the free space boundary relative to the at least two objects further comprises using the height estimate to determine the free space boundary.
19. The non-transitory computer-readable medium of claim 17, wherein determining the at least one navigational action is further based on the height estimate.
20. The non-transitory computer-readable medium of claim 17, wherein at least one of the at least two objects is an object overhanging a road surface of the road and the height estimate is relative to at least a portion of the object overhanging the road surface.
21. The non-transitory computer-readable medium of claim 20, wherein the object overhanging the road surface includes at least one of a lamppost, gates, barriers, tree branches, or a traffic sign.
22. The non-transitory computer-readable medium of claim 1, wherein the method further comprises detecting the at least two objects by: analyzing the at least one image, using an output of RADAR system, using an output of a LIDAR system, using an output of a trained system, or a combination thereof.
23. The non-transitory computer-readable medium of claim 1, wherein the method further comprises determining whether at least one moving object is crossing the free space boundary.
24. The non-transitory computer-readable medium of claim 23, wherein the at least one moving object is a pedestrian or a vehicle.
25. The non-transitory computer-readable medium of claim 23, wherein determining the at least one navigational action is further based on whether the at least one moving object is crossing the free space boundary.
26. The non-transitory computer-readable medium of claim 1, wherein the method further comprises refining the free space boundary based on a driving policy associated with the host vehicle.
27. The non-transitory computer-readable medium of claim 1, wherein the method further

comprises determining whether one or more of the at least two objects are moving.

**28.** The non-transitory computer-readable medium of claim 1, wherein the at least one navigational action includes altering a planned trajectory of the host vehicle based on the free space boundary.

**29.** The non-transitory computer-readable medium of claim 1, wherein the at least two objects are located on one side of the road segment.

**30.** The non-transitory computer-readable medium of claim 29, wherein the at least one image further includes representations of at least two additional objects with offset edges in the environment of the host vehicle and located on another side of the road segment, and wherein the method further comprises: analyzing the at least one image to determine an additional free space boundary relative to the at least two additional objects, wherein the additional free space boundary represents an additional edge of the at least one region of free space and follows a path different from a path represented by the offset edges of the at least two additional objects.

**31.** The non-transitory computer-readable medium of claim 30, wherein determining the at least one navigational action is further based on the additional free space boundary.

**32.** A method for navigating a host vehicle relative to a road segment, the method comprising: receiving at least one image captured by at least one camera from an environment of the host vehicle, wherein the at least one image includes representations of at least two objects with offset edges in the environment of the host vehicle and located on one side of the road segment; analyzing the at least one image to determine a free space boundary relative to the at least two objects, wherein the free space boundary represents an edge of at least one region of free space and follows a path different from a path represented by the offset edges of the at least two objects; determining at least one navigational action based on the free space boundary; and causing the host vehicle to implement the at least one navigational action.

**33.** The method of claim 32, wherein analyzing the at least one image to determine the free space boundary associated with at least two objects includes identifying a potential area of free space based on at least one map and determining the free space boundary relative to the at least two objects based on the potential area of free space and the at least one image.

**34.** The method of claim 32, wherein the at least two objects are spaced apart from one another by a distance.

**35.** The method of claim 32, wherein analyzing the at least one image to determine the free space boundary relative to the at least two objects further comprises receiving an output provided by a trained system, wherein the output includes a height estimate for a plurality of pixels in the at least one image associated with one of the at least two objects.

**36.** The method of claim 35, wherein analyzing the at least one image to determine the free space boundary relative to the at least two objects further comprises using the height estimate to determine the free space boundary.

**37.** The method of claim 32, further comprising determining whether at least one moving object is crossing the free space boundary.

**38.** The method of claim 32, further comprising refining the free space boundary based on a driving policy associated with the host vehicle.

**39.** A system for navigating a host vehicle relative to a road segment, the system comprising at least one processor comprising circuitry and a memory, wherein the memory includes instructions that when executed by the circuitry cause the at least one processor to: receive at least one image captured by at least one camera from an environment of the host vehicle, wherein the at least one image includes representations of at least two objects with offset edges in the environment of the host vehicle and located on one side of the road segment; analyze the at least one image to determine a free space boundary relative to the at least two objects, wherein the free space boundary represents an edge of at least one region of free space and follows a path different from a path represented by the offset edges of the at least two objects; determine at least one navigational action based on the free space boundary; and cause the host vehicle to implement the at least one

navigational action.

**40.** The system of claim 39, wherein analyzing the at least one image to determine the free space boundary associated with at least two objects includes identifying a potential area of free space based on at least one map and determining the free space boundary relative to the at least two objects based on the potential area of free space and the at least one image.

**41.** The system of claim 39, wherein the at least two objects are spaced apart from one another by a distance.

**42.** The system of claim 39, wherein analyzing the at least one image to determine the free space boundary relative to the at least two objects further comprises receiving an output provided by a trained system, wherein the output includes a height estimate for a plurality of pixels in the at least one image associated with one of the at least two objects.

**43.** The system of claim 42, wherein analyzing the at least one image to determine the free space boundary relative to the at least two objects further comprises using the height estimate to determine the free space boundary.

**44.** The system of claim 39, wherein the at least one processor is further configured to determine whether at least one moving object is crossing the free space boundary.

**45.** The system of claim 39, wherein the at least one processor is further configured to refine the free space boundary based on a driving policy associated with the host vehicle.

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